Sound-change half, alpha 01

Generated from `Germanic/docs/sound_changes/change_reports/report_manifest.tsv` *and* `Germanic/docs/sound_changes/change_reports/sound_change_half_scaffold.tsv`*. All 70 ordinary sound changes now have pilot/full production prose in the assembled half.*

Coverage summary

- Ordinary chronology-card sound changes represented: 81/81.
- Covered by pilot/full production reports: 81.
- Covered by scaffold placeholders: 0.
- Grouped into multi-change units: 48 changes across 20 units.
- Still needing literature dossiers: 0.
- Still needing human judgement or promotion decisions: 0.
- Negative/boundary-only chronology cards: 12.
- Broad/far/contextual chronology cards: 38.

Unit register

Unit	Change IDs	Status	Chronology status	Literature status	Final treatment
003	SC003	full	one-sided singleton with boundary-only earlier side at the left edge of the tested early-rule chain and a real but broad/far `SC003 < SC044` boundary via `learn`, `learn (3sg)`, `learn (iptv.2sg)`, and `meed`	source-backed with a cautious post-PWGmc West Germanic historical label; keep the CAPR rule name distinct from the historical interpretation	keep as a short singleton full report; treat the SC044 relation as a rightward cross-reference only and keep final `*z` deletion distinct

Unit	Change IDs	Status	Chronology status	Literature status	Final treatment
004	SC004	full	one-sided singleton with boundary-only earlier side at the left edge of the tested expanded-PWGmc chain and a real but broad/far `SC004 < SC036` boundary via `soul`	source-backed with explicit scope caution: the historical support is strongest for the unstressed and word-final side, while CAPR's broader nonfinal `*ai > *ā` packaging remains more explicit than the current handbook prose	keep as a short singleton full note; treat the SC036 relation as a rightward cross-reference only and keep the broader packaging caution explicit
005	SC005	full	one-sided singleton with boundary-only earlier side at the left edge of the tested expanded-PWGmc chain and a real but broad/far `SC005 < SC017` boundary via `shoulder`	source-backed by inflectional and morpho-phonological pre-`m` raising; keep the stage label cautious and do not make the case rest on the compact trace witness alone	keep as a short singleton full note; treat the SC017 relation as a rightward cross-reference only and keep the inflectional setting central

Unit	Change IDs	Status	Chronology status	Literature status	Final treatment
006	SC006	full	one-sided singleton with boundary-only earlier side at the left edge of the tested expanded-PWGmc chain and a real but broad/far `SC006 < SC034` boundary via `show (3sg)`	source-backed by the suffixal anti-umlaut argument and the `geoguþ` pathway; keep that center of gravity clearer than the wider trace spread	keep as a short singleton full note; treat the SC034 relation as a rightward cross-reference only and keep the suffixal evidence central
007	SC007	full	one-sided singleton with boundary-only earlier side at the left edge of the tested expanded-PWGmc chain and a real but broad/far `SC007 < SC043` boundary via `water`	source-backed by the `four` and `water` evidence; keep the final/pre-final environment explicit and do not inflate the rule into a broad long-vowel chapter	keep as a short singleton full note; treat the SC043 relation as a rightward cross-reference only and keep the narrow witness set visible
008	SC008	full	one-sided singleton with boundary-only earlier side at the left edge of the tested expanded-PWGmc chain and a real but broad/far `SC008 < SC031` boundary via `four`	source-backed by `four` plus plural-pronominal material; keep the small witness set explicit and do not overread the broad formal rule	keep as a short singleton full note; treat the SC031 relation as a rightward cross-reference only and keep the pronominal support visible

Unit	Change IDs	Status	Chronology status	Literature status	Final treatment
009	SC009	full	one-sided singleton with boundary-only earlier side at the left edge of the tested expanded-PWGmc chain and a real but broad/far `SC009 < SC032` boundary via `friend`	source-backed by the exceptional `friend` sequence; keep the lexical narrowness explicit and do not enlarge the rule into a broadly productive contraction chapter	keep as a short lexical singleton note; treat the SC032 relation as a rightward cross-reference only and keep the uniqueness of the `friend` family visible
010	SC010	full	one-sided singleton with boundary-only earlier side at the left edge of the tested expanded-PWGmc chain and a tight local reciprocal `SC010 < SC011` boundary via `net`	source-backed by Fulk's strong j-gemination statement and example set; keep the light-syllable plus `*j` environment explicit and do not widen the rule into a general consonant-doubling chapter	keep as a short singleton full note; treat the SC011 relation as an immediate rightward cross-reference or local seam without forcing a grouped chapter
011	SC011	full	one-sided singleton with a tight local reciprocal earlier boundary at `SC010` via `net` and a boundary-only later side that reaches the SC087 search limit	source-backed by Ringe and Taylor's direct statement about post-consonantal `*j > *i`, but keep the trace-light status explicit because the current compact trace gives no direct SC011 hits	keep as a short singleton full note; treat the SC010 relation as a leftward local seam and do not enlarge the rule into a broader high-vowel chapter

Unit	Change IDs	Status	Chronology status	Literature status	Final treatment
012	SC012	full	boundary-limited on both sides; earlier search stops at the left edge of the tested expanded-PWGmc chain and later search reaches the SC087 boundary with no real break; do not turn either side into positive chronology	source-backed by the `lþ > ld` development with a cautious northern-West-Germanic scope; keep the stage-label caution explicit and do not overstate the chronology	keep as a short finished singleton note; state plainly that the order test supplies no positive local boundary and keep the scope caution visible
013	SC013	full	boundary-limited on both sides; earlier search stops at the left edge of the tested expanded-PWGmc chain and later search reaches the SC087 boundary with no real break; do not turn either side into positive chronology	source-backed by Ringe and Taylor's direct systemic statement that PWGmc *d became a stop in all positions; the lexical dossier is compact but sufficient for a short note	keep as a short finished singleton note; treat the tested-chain left edge and the SC087 search boundary as methodological limits rather than chapter architecture

Unit	Change IDs	Status	Chronology status	Literature status	Final treatment
014-015	SC014; SC015	full	short asymmetric opening bridge: SC014 remains negative and boundary-limited; SC015 is the stronger member with broad/far `SC015 < SC036` via `world`; both earlier sides are runner-bounded at `PWGmc-Changes`	literature and book dossiers drafted	keep as a short cautious full opening prelude; SC014 stays brief, SC015 carries the prose, and the SC016-SC020 / SC036 links remain cross-references only
016-020	SC016; SC017; SC019; SC020	pilot	strong local corridor with forward links	substantial pilot source material	keep as pilot multi-change production report
018	SC018	full	boundary-limited on both sides; earlier search stops at bundled `PWGmc-Changes`, later search reaches the SC087 boundary with no real break; do not turn either side into positive chronology	review and book dossiers drafted	keep as a short finished singleton note; keep bundled `PWGmc-Changes` and the SC087 search boundary as methodological limits rather than historical chronology, and do not merge the rule into the pilot corridor or SC021

Unit	Change IDs	Status	Chronology status	Literature status	Final treatment
021	SC021	full	one-sided singleton with runner-bounded earlier side and real but broad/far `SC021 < SC040` via `heaven` (`heofun` vs. `heofon`); bundled `PWGmc-Changes` remains a runner limitation, not a historical boundary	review and book dossiers drafted	keep as a short singleton full note; keep promoted SC039-SC040 as a rightward cross-reference only and do not build a non-contiguous chapter
022	SC022	full	boundary-limited on both sides; earlier search stops at bundled `PWGmc-Changes`, later search reaches the SC087 boundary with no real break; source support is descriptive but sufficient to keep the rule visible	review and book dossiers drafted	keep as a short finished singleton note; keep bundled `PWGmc-Changes` and the SC087 search boundary as methodological limits rather than historical chronology, and do not merge the rule into SC021 or SC023

Unit	Change IDs	Status	Chronology status	Literature status	Final treatment
023	SC023	full	one-sided singleton with runner-bounded earlier side and real but broad/far `SC023 < SC047` via do, where delaying the rule collapses the derivation to no output; bundled `PWGmc-Changes` remains a runner limitation, not a historical boundary	review and book dossiers drafted	keep as a short singleton full note; keep promoted SC046-SC048 as a rightward cross-reference only and do not build a non-contiguous chapter
024	SC024	full	one-sided singleton with runner-bounded earlier side and real but broad/far `SC024 < SC056` via `sheep` and `year` (`sċīep` / `ġīer` vs. `sċēap` / `ġēar`); bundled `PWGmc-Changes` remains a runner limitation, not a historical boundary	review and book dossiers drafted	keep as a short singleton full note; keep promoted SC055-SC056 as a rightward cross-reference only and do not build a non-contiguous chapter

Unit	Change IDs	Status	Chronology status	Literature status	Final treatment
025	SC025	full	boundary-limited on both sides; earlier search stops at bundled `PWGmc-Changes`, later search reaches the SC087 boundary with no real break; keep the long-ē-before-nasal background visible without turning either side into positive chronology	review and book dossiers drafted	keep as a short finished singleton note; keep bundled `PWGmc-Changes` and the SC087 search boundary as methodological limits rather than historical chronology, and do not merge the rule into SC024 or SC026-SC027
026-027	SC026; SC027	full	strong local reciprocal pair	literature and book dossiers drafted	keep as paired full production report
028	SC028	full	historically legible pre-consonantal x-loss note with no positive first-break boundary on either side; earlier search stops at bundled `PWGmc-Changes`, later search reaches the SC087 boundary with no real break	literature and book dossiers drafted	keep as a short finished singleton note immediately before the promoted SC029-SC030 core; keep bundled `PWGmc-Changes` and the SC087 search boundary as methodological limits rather than historical chronology, and do not merge it into SC026-SC027 or SC029-SC030

Unit	Change IDs	Status	Chronology status	Literature status	Final treatment
029-030	SC029; SC030	full	tight local glide/fronting core: reciprocal `SC029 < SC030` via `hay` / `strew`, with SC029's earlier side runner-bounded at `PWGmc-Changes` and SC030 handing forward to SC032 through a broader no-output failure set	literature and book dossiers drafted	keep as a narrow adjacent full report; keep SC028 as a separate left note and treat SC032 as a rightward cross-reference rather than chapter architecture
031-034	SC031; SC032; SC033; SC034	full	strong local core with weaker flanks	literature and book dossiers drafted	keep as cautious four-change full production report
035-037	SC035; SC036; SC037	full	cautious adjacent bridge with explicit hierarchy: SC035 prefix flank, SC036 source-backed center (`SC019 < SC036 < SC040`), SC037 compound/technical flank with only technical-marker SC038 on its own card	literature and book dossiers drafted	keep as grouped full bridge report for now; future narrowing around SC036 remains optional but is not needed now

Unit	Change IDs	Status	Chronology status	Literature status	Final treatment
039-040	SC039; SC040	full	tight local medial-vowel core: reciprocal `SC039 < SC040` via `widow`, with SC040 carrying the broader `world`/`youth` context and a rightward SC072 cross-reference	literature and book dossiers drafted	keep as a narrow adjacent full report; keep bundled `PWGmc-Changes` and SC072 as cross-references rather than chapter architecture
041	SC041	full	historically real final-loss hinge with broad/far `SC020 < SC041 < SC046` chronology and outward cross-references rather than a tight adjacent reciprocal pair	literature and book dossiers drafted	keep as a short singleton note; keep the SC020 and SC046 links as left-ward/rightward cross-references and do not merge it into SC039-SC040, the pilot corridor, or SC046-SC048
042	SC042	full	narrow two-sided `rest` window: SC020 < SC042 < SC043	regional review and book dossiers drafted	keep as a short singleton full context note between SC041 and SC043; keep SC020 as a leftward cross-reference and do not bundle SC042 non-contiguously with SC044-SC045
043	SC043	full	strong local standalone report	substantial source material exists	keep as singleton full production report

Unit	Change IDs	Status	Chronology status	Literature status	Final treatment
044-045	SC044; SC045	full	strong adjacent pair: SC043 < SC044 < SC045, with later SC060 relation kept as cross-reference	regional review and book dossiers drafted	keep as paired full production report immediately right of SC043
046-048	SC046; SC047; SC048	full	cautious adjacent corridor: SC043 < SC046 < SC048 and SC034 < SC047 < SC048 < SC059, with a broad reciprocal SC047-SC048 core	regional review and book dossiers drafted	keep as cautious full corridor report; later narrowing remains optional but no split is needed now
049-050	SC049; SC050	full	narrow allophonic SC049 with right-facing SC050 bridge	literature and book dossiers drafted	keep as short adjacent chronological bridge report
051	SC051	full	strong local standalone report	literature and book dossiers drafted	keep as narrow singleton full production report
052	SC052	full	strong local hinge with two-sided card evidence	literature and book dossiers drafted	keep as standalone chronological full production report
053-054	SC053; SC054	full	negative edge plus narrow bridge evidence	literature and book dossiers drafted	keep as short adjacent chronological bridge report
055-056	SC055; SC056	full	strong local pair with right-edge caution	literature and book dossiers drafted	keep as paired full production report

Unit	Change IDs	Status	Chronology status	Literature status	Final treatment
057	SC057	full	one-sided singleton with real `SC052 < SC057` via bow, `follow`, `hedge`, `seek`, and `singe`, but no later real break before the current runner boundary	dedicated literature dossier drafted	keep as a short singleton note; keep SC052 as a leftward cross-reference only and do not rewrite the runner-bounded later side as a positive historical boundary
058	SC058	full	boundary-limited on both sides; earlier search stops at bundled `PWGmc-Changes`, later search reaches the SC087 boundary with no real break; do not turn either side into positive chronology	dedicated literature dossier drafted	keep as a short residual/context note; keep bundled `PWGmc-Changes` and the SC087 search boundary as methodological limits rather than historical chronology, and do not inflate the rule into a larger grouped report

Unit	Change IDs	Status	Chronology status	Literature status	Final treatment
059	SC059	full	source-backed singleton with real two-sided chronology `SC048 < SC059 < SC078`; the later `SC059 < SC078` edge is broad/far and remains a rightward cross-reference only	literature and book dossiers drafted	keep as a short singleton full note; keep SC046-SC048 and the split late-tail scaffold region, especially SC078, as left-ward/rightward cross-references only and do not recreate the old grouped bridge
060	SC060	full	one-sided singleton with real `SC055 < SC060` via `might` and `night`, but no later real break before the current runner boundary	literature and book dossiers drafted	keep as a short finished singleton note; keep promoted SC055-SC056 as a leftward cross-reference only and do not rewrite the runner-bounded later side as a positive historical boundary
061	SC061	full	one-sided singleton with real `SC023 < SC061` via do, but no later real break before the current runner boundary	literature and book dossiers drafted	keep as a short finished singleton note; keep promoted SC023 as a leftward cross-reference only and do not rewrite the runner-bounded later side as a positive historical boundary

Unit	Change IDs	Status	Chronology status	Literature status	Final treatment
063	SC063	full	strong local standalone report	substantial source material exists	keep as singleton full production report
064-065	SC064; SC065	full	narrow positive SC064 with card-negative SC065	literature and book dossiers drafted	keep as short adjacent chronological bridge report
066-068	SC066; SC067; SC068	full	strong local corridor with one weaker bridge member	literature and book dossiers drafted	keep as narrow full production report
069	SC069	full	broad/far opener: `SC023 < SC069` is real but non-local, and the later side remains runner-bounded through SC087	literature and book dossiers drafted	keep as a short finished late-tail opener/context note; keep SC023 as a distant cross-reference only and do not turn the runner-bounded later side into positive local chapter architecture
070-071	SC070; SC071	full	early local bridge: `SC052 < SC070 < SC071`, while SC071's later side remains runner-bounded	literature and book dossiers drafted	keep as a cautious early late-tail bridge full report; keep SC052 as a leftward cross-reference only, do not rewrite SC071's runner-bounded later side as positive chronology, and do not merge the pair into SC069 or SC072-SC073

Unit	Change IDs	Status	Chronology status	Literature status	Final treatment
072-073	SC072; SC073	full	strongest internal core: `SC064 < SC072 < SC073`, with SC073 also pointing forward to SC085	literature and book dossiers drafted	keep as a compact adjacent full report; keep SC064 and SC085 as outward cross-references only, and do not expand into SC070-SC071, SC074-SC075, SC076, or SC078
074-075	SC074; SC075	full	narrow local bridge: `SC072 < SC074 < SC075`, while SC075's later side remains runner-bounded	literature and book dossiers drafted	keep as a narrow witness-limited medial unstressed-i lowering bridge full report; keep SC072 as a leftward cross-reference only, do not rewrite SC075's runner-bounded later side as positive chronology, and do not merge the pair into SC072-SC073, SC076, or SC078

Unit	Change IDs	Status	Chronology status	Literature status	Final treatment
076	SC076	full	source-backed but chronology-negative singleton: no positive first-break boundary in either direction, with the earlier search blocked by bundled `PWGmc-Changes` and the later side runner-bounded through SC087	literature and book dossiers drafted	keep as a short finished late-tail residual/context note; keep bundled `PWGmc-Changes` and the SC087 search boundary as methodological limits rather than historical chronology, and do not force SC076 into a grouped production unit
078	SC078	full	right-edge hinge with broad earlier `SC070 < SC078` computational limit and narrower later `SC078 < SC086` historical seam	literature and book dossiers drafted	keep as a singleton right-edge full report; keep the rightward SC086 relation as a cross-reference only to promoted SC085-SC086, and do not create a non-contiguous SC078-SC086 chapter

Unit	Change IDs	Status	Chronology status	Literature status	Final treatment
079-080	SC079; SC080	full	opening bridge: `SC055 < SC079 < SC080`, with a broad/far left edge and a narrow local right edge, while SC080 remains one-sided on the right	literature and book dossiers drafted	keep as a modest opening closing-bridge full report; keep the broad/far SC055 relation as a cross-reference only, do not rewrite SC080’s runner-bounded later side as positive chronology, and do not merge the pair into SC078 or SC081-SC083
081-083	SC081; SC082; SC083	full	technical middle bridge: `SC081 < SC082 < SC083`, with SC082 as the strongest center, SC081’s left edge broad/far to SC055, and SC083’s later side runner-bounded	literature and book dossiers drafted	keep as a cautious technical full bridge with SC082 as the center; keep SC055 and SC085 as outward cross-references only, do not rewrite SC083’s runner-bounded later side as positive chronology, and do not merge into SC085-SC086

Unit	Change IDs	Status	Chronology status	Literature status	Final treatment
085-086	SC085; SC086	full	strongest compact closing core: `SC073 < SC085 < SC086`, with SC086 also receiving the SC078 seam as a cross-reference	literature and book dossiers drafted	keep as a focused full closing report; keep SC073 and SC078 as leftward cross-references only, do not claim a positive SC086 < SC087 boundary, and do not expand into SC081-SC083 or SC087
087	SC087	full	terminal singleton with broad/far `SC044 < SC087` and a runner-bounded later side	literature and book dossiers drafted	keep as a short terminal singleton full report; keep SC044-SC045 as a distant leftward cross-reference only, do not claim a positive SC086 < SC087 relation, and do not absorb SC078, SC079-SC080, or SC081-SC083

West Germanic rhotacism

Sound-change report

Historical formulation SC003 *EAFRhotacism* appears here as a singleton consonant report for the change of medial *z* to *r*. Historically, the change is better treated as a post-PWGmc West Germanic rhotacism than as a Proto-Germanic innovation proper. Ringe and Taylor place rhotacism in the post-PWGmc sound-change layer and stress that it was not uniform within WGmc (Ringe and Taylor 2014, 98, 102), while Crist argues that it cannot be inherited from Proto-Northwest Germanic and must follow earlier WGmc *z*-deletion rules (Crist 2001, 104–6; 2002, 1, 4).

That does not make the CAPR rule name unusable. It does mean the backend apparatus should distinguish the implementation label *EAFRhotacism* from the source-supported historical stage label.

Source tradition The source support for the phenomenon itself is strong. Hogg states succinctly that Germanic `/z/` yielded `/r/` in intervocalic position in Old English, though it was generally lost finally (Hogg 1992, 37). Ringe and Taylor go farther: they first note that the change occurred independently in Norse and WGmc and was not uniform even in the latter (Ringe and Taylor 2014, 52), and later advise assigning the process to the post-PWGmc period (Ringe and Taylor 2014, 98, 102). Crist's analysis aligns with that caution by arguing that rhotacism is not inherited from Proto-Northwest Germanic and must follow earlier WGmc *z*-deletion rules (Crist 2001, 104–6; 2002, 1, 4).

This is enough source support not only for a backend report but for a cautious historical label. The clearest sources support the change itself strongly and support locating it in the West Germanic area after PWGmc rather than at Proto-Germanic proper.

CAPR implementation CAPR models the development as one explicit rewrite in vocalic medial environments:

```
define EAFRhotacism [
    {*z} -> {*r} || EnglishStarVocalic _ ?
];
```

The implementation is broader than a strictly intervocalic *V_V* rule because the current model also needs to retain medial *VzC* environments. That choice matches the local FST comment, but it should still be kept distinct from the narrower formulations in the historical literature.

For implementation continuity, CAPR keeps the exact rule name *EAFRhotacism*. That identifier should be read as a model label rather than as the best historical stage label.

Place in the cascade In the current inventory ordering, SC003 follows SC002 *PGmcGmSimplification* and precedes the Proto-West-Germanic developments that begin with SC004. In the implementation, however, it remains inside the same bundled Proto-Germanic consonant block that also contains final *z* deletion.

That placement still matters, but the early-rule harness now exposes SC003 to standalone first-break testing without changing the production bundle. The rule therefore has direct chronology support even though the live cascade remains bundled, and the historical prose can now say more clearly that the modeled stage name is earlier than the source-supported historical interpretation.

Order evidence Validated order evidence now exists through the temporary early-rule harness:

1. `Germanic/docs/sound_changes/order_tests/summaries/order_sensitivity_first_break_earl`
2. `Germanic/docs/sound_changes/order_tests/summaries/order_sensitivity_first_break_earl`
3. `Germanic/docs/sound_changes/order_tests/summaries/order_sensitivity_first_break_earl`

The earlier search moved SC003 safely across SC002 down to order *2* and then reached the left edge of the tested historical chain with no real break. That side is therefore boundary-only rather than a positive earlier chronology constraint.

The later search does find a real historical break at order *44* across `SC044` OE Breaking. If PGmc Rhotacism is delayed past that stage, PGmc líznōjaną yields lirnian rather than liornian, líznōθi yields lirnaþ rather than liornaþ, líznô yields lirna rather than liorna, and mízdai yields merde rather than meorde.

That later boundary is historically interpretable, but it is broad/far rather than a tight local adjacency claim.

Interpretation SC003 is now substantially stronger than SC002 on both the chronology and source sides. It has a validated one-sided historical boundary and exact wrong-output diagnostics, and the literature now supports a cautious historical label: post-PWGmc West

Germanic rhotacism. That is enough for a cautious singleton report, provided the prose keeps the CAPR rule name and the historical label clearly distinct.

Remaining cautions Two cautions still matter most. First, backend prose should keep rhotacism distinct from final *z* deletion, because the literature treats them as separate events with their own chronology. Second, the later boundary with `SC044` is broad/far rather than a tight local adjacency claim. The prose should therefore continue to describe the historical stage as post-PWGmc West Germanic while reserving *EAFRhotacism* for the CAPR rule label.

PWGmc ai-monophthongization note

Sound-change report

> **Corrected PROTOFORM pass.** SC004 is the stressed/root change ái > ā only. The unstressed development *ai* > ē (final and nonfinal) is the separate, earlier SC014 *PNWGmcUnstressedAiMonophthongization*. Application analysis uses the Old-English-row *PROTOFORM* (the production input): under it loam (láimą) is stressed and whine (xwī́naną) carries no *ai*. Implemented on branch historical-cascade-order (split commit f59b758d, PROTOFORM correction 9c71aed3).

Historical formulation SC004 *EAFAiMonophthongization* is the monophthongization of stressed/root ái to ā in the English line (later fronted to ǣ in the relevant OE environments). In the trace it is visible across 24 stressed families such as soul, stone, bone, deal, ghost, and loam. It is best understood as a North Sea Germanic / Anglo-Frisian areal development, not as a single dated node.

Source tradition Ringe and Taylor treat the monophthongization of *ai* among the widespread post-PNWGmc vowel developments of the English line (Ringe and Taylor 2014, 40–41, §6.1.5); Fulk lists *ai*/*au* among the North/West-Germanic shared innovations against Gothic (Fulk 2018, sect. 5.2); Campbell describes the Anglo-Frisian *ai* > ā (later fronted) as an English-line development (Campbell 1959, sects 133–134, §417). Versloot 2017 (consulted directly) argues that stressed/root *ai* monophthongization spread in two areal waves through a North Sea Germanic dialect continuum (c. AD 400–900), a diffusion rather than a single inherited Proto-Anglo-Frisian node, with Old English among the broadest monophthongizers. That source layer supports precisely the stressed side modelled here; the unstressed side is SC014's evidence.

CAPR implementation CAPR models the change as a single EAF-stage rule in the EAF corridor:

```
define EAFAiMonophthongization [
    {*ái} -> {*ā}
];
```

The rule targets the stressed diphthong ái only (a distinct segment from unstressed *ai*), so it is independent of SC014. No nonfinal unstressed root *ai* remains in the corpus once PROTOFORM is used, so no {*ai*} -> {ā} branch is required; the earlier unrestricted branch was an artefact of the PROTO-based misclassification of loam and whine. The former bundled identifier *PWGmcAiMonophthongization* is retained as a documented compatibility alias but is not composed in any pipeline.

Place in the cascade SC004 executes at cascade position 25, immediately after SC028 *PNWGmcPreconsonantalXLoss* and before SC029 *OEAwjGlideFormation* (the EAF corridor). Its SC number stays SC004 even though its executable position no longer follows numerical order.

Order evidence First-break testing with the corrected stressed-only rule reproduces the one historical boundary at order 33 across `SC036` OE Inter Stress Raising: delaying SC004 past that stage makes PGmc sáiwalō yield sāwel rather than expected OE sāwol (371/372 match at the break). The current placement (pos 25) sits safely before that boundary.

The earlier side has no corpus break toward the head. The formal crossing analysis (`sc004_sc014_interaction_report.md`) shows SC004's only genuine dependency is on SC036; its other non-commutations are feeding artefacts on non-corpus *EnglishProtoInput* forms.

Interpretation SC004 is the later, areally diffused North Sea Germanic monophthongization of stressed *ai*. The EAF corridor is an operational modelling home for a change whose real history is a dialect-continuum diffusion rather than a discrete node; the SC036 soul boundary is its one usable chronological anchor.

Remaining cautions The EAF stage is a modelling corridor, not a demonstrated discrete Proto-Anglo-Frisian node; the prose should say so. Old English ā is fronted to ǣ by later change in the relevant environments, so the surface reflex is not always ā. loam is a stressed witness; whine is not an ai-monophthongization case.

Unstressed a-raising before final m note

Sound-change report

Historical formulation SC005 *PNWGmcAToUBeforeM* isolates the raising of unstressed noninitial *a* to *u* before final *m* in a narrow set of inflectional environments. The inventory currently labels it Northwest Germanic and gives shoulder as the compact-trace witness, but the historical discussion recovered in this pass is broader and more morphologically oriented than that single lexical example suggests.

That mismatch is important. The change is historically plausible, but the source layer does not yet settle whether the best label is strictly Northwest Germanic, a broader North-and-West-Germanic shared development, or a still earlier PWGmc-stage adjustment inside inflectional endings. The existing needs_human_review=yes flag should therefore remain explicit.

Source tradition Campbell describes an early development in inflectional endings by which *a* > *o* > *u* before m, citing dat.pl. endings and related finite-verb morphology (Campbell 1959, sect. 331(6)). Sievers/Brunner likewise treats old unstressed *o* in derivational and inflectional syllables before *m* as yielding *u* in Old English (Brunner 1965, sect. 44). Fulk places the development of early Proto-Germanic unstressed *o* to *u* before *m* among the important similarities shared by North and West Germanic (Fulk 2018, sect. 5.2).

Those sources support the underlying phenomenon, but they do not align perfectly with CAPR's present label. They point most clearly to a morphologized pre-*m* raising in endings and to a wider North/West-Germanic distribution, not to a settled standalone Northwest-Germanic rule named from the single lexical trace item shoulder. The source base is therefore usable but not yet decisive on the staging question.

CAPR implementation CAPR models the change as a narrow pre-*m* raising rule:

```
define PNWGmcAToUBeforeM [
    {*a} -> {*u} || EnglishStarVocalic EnglishStarConsonant+ _ {*m} ({*i})? ({*z})? .#.
];
```

This rule is intentionally tight. It targets unstressed ending material before final *m*, with optional *i* and *z* tails, rather than a general stem-wide vowel shift. The implementation is therefore more precise than the broader morphophonological prose in the handbooks, but it also inherits their uncertainty about exact stage placement.

Place in the cascade In the inventory ordering, SC005 follows SC004 *PWGmcAiMonophthongization* and precedes SC006 *PWGmcEarlyIApocope*. In the live pipeline it remains part of bundled *EarlyEnglishLineChanges*, and the inventory itself flags the stage label for review because the rule is Northwest-Germanic in name but still embedded inside the Proto-West-Germanic pipeline bundle.

That bundle position no longer blocks chronology testing in principle. The current first-break runner can expose SC005 through the expanded-pwgmc order profile, but the historical naming issue remains unresolved even if the computation path is now available.

Order evidence Validated order evidence now exists through the expanded-PWGmc first-break output family:

1. `Germanic/docs/sound_changes/order_tests/summaries/order_sensitivity_first_break_expa`
2. `Germanic/docs/sound_changes/order_tests/summaries/order_sensitivity_first_break_expa`
3. `Germanic/docs/sound_changes/order_tests/summaries/order_sensitivity_first_break_expa`

The earlier search moved SC005 safely across SC004 down to order *4* and then reached the left edge of the tested expanded-PWGmc chain with no real break. That side is therefore boundary-only rather than a positive chronology constraint.

The later search does find a real historical break at order *17* across `SC017` NWGmc U Lowering. If NWGmc A To U Before M is delayed that far, PGmc skúldramiz yields sċoldrum rather than expected OE sċuldrum.

That later boundary is historically interpretable, but it is broad/far rather than a tight local adjacency claim.

Interpretation SC005 can now stand as a cautious singleton note. The source support points to a real unstressed-vowel development before final *m* in inflectional material, and the validated chronology card adds a real later boundary even though that boundary is broad/far rather than local. The section should not make the whole case rest on the compact trace witness shoulder.

Remaining cautions Three cautions remain explicit. First, the stage label is unresolved: the current literature support fits a wider North/West-Germanic inflectional development at least as well as a narrowly Northwest-Germanic singleton. Second, the compact trace currently gives only one lexical witness, shoulder, while the source prose is mostly morphological rather than lexical. Third, the chronology card is one-sided and broad/far on its later side. Those limits should remain visible in prose rather than being used as grounds for omission.

Early i-apocope note

Sound-change report

Historical formulation SC006 *PWGmcEarlyIApocope* isolates the early loss of final *i* after unstressed syllables in forms far enough from the main stress that later i-umlaut no longer sees the trigger. In the trace output it appears in forms such as thousand, bore (3sg), have, learn (3sg), and lick (3sg).

That is historically recognizable and more strongly sourced than SC005. The main historical value of the rule is not the individual lexemes alone, but the broader argument that early final *i* loss must precede later umlaut and related vocalic developments.

Source tradition Sievers/Brunner treats the early Common-Germanic loss of final -*i* after unstressed syllables as established by the fact that these endings no longer triggered i-umlaut in Old English (Brunner 1965, sects 145–146). Ringe and Taylor give the classic youth example directly: PWGmc jugunþi > juguþ > OE geoguþ ~ iuguþ (Ringe and Taylor 2014, 141). Campbell likewise cites dugup and geogup among the relevant outcomes of early final high-vowel loss (Campbell 1959, sect. 332).

That is good support for the historical phenomenon itself. The present source pass is less complete for the full spread of CAPR's inventory examples, since the retrieved prose is strongest on endings and on the youth family rather than on every trace witness such as have or lick (3sg). The source layer is therefore solid but not yet exhaustive.

CAPR implementation CAPR models the early apocope as a targeted deletion of final *i* in sufficiently remote unstressed syllables:

```
define PWGmcEarlyIApocope [
    {*i} -> 0 || PGmcStarStressedVowel PGmcStarConsonant+ PGmcStarVocalic PGmcStarConsor
    {*i} -> 0 || PGmcStarStressedVowel PGmcStarConsonant+ PGmcStarVocalic PGmcStarConsor
];
```

The implementation makes explicit what the historical discussion implies: the decisive issue is relative stress and syllable count, not just absolute word-final position. This is why the rule is modeled separately from later syncope and apocope.

Place in the cascade In the inventory ordering, SC006 follows SC005 *PNWGmcAToUBeforeM* and precedes SC007 *PWGmcFinalOrLowering*. Like SC004 and SC005, it is still live inside bundled *EarlyEnglishLineChanges*, but the current runner can now expose it directly through the expanded-pwgmc first-break profile.

That means the main backend blocker is no longer runner visibility. It is simply the absence of actual first-break TSV output for this specific change.

Order evidence Validated order evidence now exists through the expanded-PWGmc first-break output family:

1. `Germanic/docs/sound_changes/order_tests/summaries/order_sensitivity_first_break_expa`
2. `Germanic/docs/sound_changes/order_tests/summaries/order_sensitivity_first_break_expa`
3. `Germanic/docs/sound_changes/order_tests/summaries/order_sensitivity_first_break_expa`

The earlier search moved SC006 safely across SC005 and SC004 down to order *4* and then reached the left edge of the tested expanded-PWGmc chain with no real break. That side is therefore boundary-only rather than a positive chronology constraint.

The later search does find a real historical break at order *34* across `SC034` OE Aw Long Diphthong. If PWGmc Early I Apocope is delayed that far, PGmc skáwōθi yields sċēaweþ rather than expected OE sċēawaþ.

That later boundary is historically interpretable, but it is broad/far rather than a tight local adjacency claim.

Interpretation SC006 works well as a short singleton note. The historical phenomenon is well supported, and the chronology layer now supplies a usable one-sided later boundary. The report's center of gravity should remain the anti-umlaut and suffixal evidence rather than an attempt to distribute equal weight across every current trace witness.

Remaining cautions The major caution is not the existence of the rule but its scale. The retrieved source discussion is strongest for suffixal evidence and geoguþ rather than for every inventory witness, and the later `SC034` relation is broad/far rather than local. Those limits should remain visible even in a cautious singleton note.

Final *ō-lowering before* r note

Sound-change report

Historical formulation SC007 *PWGmcFinalOrLowering* isolates the lowering of final bimoric ō to *a* before word-final *r*. In the compact trace it is concentrated in a very small witness set, above all four and water.

That narrowness matters. This is not a general long-ō lowering rule, but a specific final/pre-final environment whose historical value lies in a small set of well-known forms.

Source tradition Ringe and Taylor state that surviving bimoric long ō became PWGmc *a* word-finally and before word-final *r*, and they illustrate the outcome with four and water (Ringe and Taylor 2014, 58–59). Fulk likewise notes that final *r* was preserved and that ō before it developed to *a* in West Germanic (Fulk 2018, sect. 5.3).

That is solid support for the underlying phenomenon. It is not broad support for a large lexical class, because the historical discussion is anchored chiefly in four- and water-type material. The report should therefore remain explicit about the rule's narrow environment.

CAPR implementation CAPR models the change as a single explicit environment:

```
define PWGmcFinalOrLowering [
    {*ō} -> {*a} || _ {*r} .#.
];
```

The implementation matches the narrow final environment described in the sources. It is also the place where an older duplicated shortening treatment was consolidated in the FST comments, so the current CAPR rule should be read as a compact formalization of a very specific historical setting.

Place in the cascade In the inventory ordering, SC007 follows SC006 *PWGmcEarlyIApocope* and precedes SC008 *PWGmcCoronalWAssimilation*. In the live cascade it remains inside bundled *EarlyEnglishLineChanges*, but the first-break runner can now expose it directly through the expanded-pwgmc order profile.

That means the rule already has a clean chronology-test path even though it remains bundled in production.

Order evidence Validated order evidence now exists through the expanded-PWGmc first-break output family:

1. `Germanic/docs/sound_changes/order_tests/summaries/order_sensitivity_first_break_expa`
2. `Germanic/docs/sound_changes/order_tests/summaries/order_sensitivity_first_break_expa`
3. `Germanic/docs/sound_changes/order_tests/summaries/order_sensitivity_first_break_expa`

The earlier search moved SC007 safely across SC006, SC005, and SC004 down to order *4* and then reached the left edge of the tested expanded-PWGmc chain with no real break. That side is therefore boundary-only rather than a positive chronology constraint.

The later search does find a real historical break at order *43* across `SC043` Anglo Frisian Brightening. If PWGmc Final Or Lowering is delayed that far, PGmc wátōr yields water rather than expected OE wæter.

That later boundary is historically interpretable, but it is broad/far rather than a tight local adjacency claim.

Interpretation SC007 works best as a narrow singleton note. The source support is real, the chronology layer now yields one usable later boundary, and the witness set is coherent even if small. That is enough for a cautious manifest-backed note, provided the report keeps the narrow environment and limited lexical base explicit.

Remaining cautions The chief caution is scope. The rule is narrowly conditioned and strongly tied to the four and water evidence. The earlier side of the chronology card is also only boundary-only, while the later `SC043` relation is broad/far rather than local. Any later prose should resist turning this into a broad long-vowel chapter or implying that the small witness set supports a much larger phenomenon than the sources actually state.

Coronal-w assimilation note

Sound-change report

Historical formulation SC008 *PWGmcCoronalWAssimilation* isolates the assimilation of coronal obstruents before *w*, yielding *ww*. In the compact trace the clearest lexical witness is four, while the historical discussion also depends on pronominal forms such as the plural dative and possessive `you / your`.

That makes the rule historically recognizable, but also narrow. The historical case rests on a very small number of input clusters and should be presented as such.

Source tradition Ringe and Taylor call the change a shared Proto-West-Germanic innovation: the intervocalic sequences *zw* and *dw* were assimilated to *ww* (Ringe and Taylor 2014, 56–57). Their supporting examples are PGmc fedwor four, PGmc izwiz `you (dat.pl.)`, and PGmc izweraz your (pl.), and they note that although there is essentially one clear lexical example of each cluster, the basic nature of the lexemes makes the change virtually certain (Ringe and Taylor 2014, 56–57).

That is enough for a backend report. It is also a warning against overstatement: the historical support is compelling but quite concentrated.

CAPR implementation CAPR models the assimilation with one formal rule covering both coronal inputs:

```
define PWGmcCoronalWAssimilation [
    {*d} -> {*w} || _ {*w},
    {*z} -> {*w} || _ {*w}
];
```

This implementation is broader than any one lexical example, but it is still directly tied to the two historically motivated cluster types recovered in the sources.

Place in the cascade In the inventory ordering, SC008 follows SC007 *PWGmcFinalOrLowering* and precedes SC009 *PWGmcIjContraction*. In the production cascade it remains part of bundled *EarlyEnglishLineChanges*, but the expanded-PWGmc first-break mode already exposes it directly for chronology testing.

That keeps the chronology path straightforward even though the live bundled cascade is unchanged.

Order evidence Validated order evidence now exists through the expanded-PWGmc first-break output family:

1. `Germanic/docs/sound_changes/order_tests/summaries/order_sensitivity_first_break_expa`
2. `Germanic/docs/sound_changes/order_tests/summaries/order_sensitivity_first_break_expa`
3. `Germanic/docs/sound_changes/order_tests/summaries/order_sensitivity_first_break_expa`

The earlier search moved SC008 safely across SC007, SC006, SC005, and SC004 down to order *4* and then reached the left edge of the tested expanded-PWGmc chain with no real break. That side is therefore boundary-only rather than a positive chronology constraint.

The later search does find a real historical break at order *31* across `SC031` OE WW Simplification. If PWGmc Coronal W Assimilation is delayed that far, PGmc fédwōr yields fēowwer rather than expected OE fēower.

That later boundary is historically interpretable, but it is broad/far rather than a tight local adjacency claim.

Interpretation SC008 works well as a narrow singleton note. The source support for the phenomenon is good, the chronology layer now yields a usable later boundary, and the pronominal support keeps the rule from collapsing into a single-example convenience. That is enough for a cautious manifest-backed note.

Remaining cautions The main caution is concentration of evidence. The historical case depends on a small cluster of forms, especially four and the plural-pronominal material. The earlier side of the chronology card is also only boundary-only, while the later `SC031` relation is broad/far rather than local. Any later prose should keep the rule tightly bounded and should not let the broad formal implementation read as if the sources had established a much wider lexical scope.

Ij-contraction in friend note

Sound-change report

Historical formulation SC009 *PWGmcIjContraction* isolates a contraction of ijo to *iu* in the friend family. In the current trace and source material, that family is essentially the whole historical argument.

That narrowness is not incidental. This is exactly the kind of rule whose existence may be historically real while still being too lexically restricted to support broad generalization.

Source tradition Ringe and Taylor describe a roughly similar change of ijo to *iu* in the word friend, giving the pathway PGmc frijond- > PWGmc friund > OE friond and related WGmc forms (Ringe and Taylor 2014, 62). They immediately warn, however, that the word is unique: the sequence ijo with stressed i* is so singular that it is inadvisable to attempt wider generalizations from this single history (Ringe and Taylor 2014, 62).

That is enough to justify backend documentation of the phenomenon. It is also strong reason to keep the report modest and explicit about its lexical narrowness.

CAPR implementation CAPR models the contraction with an explicit environment:

```
define PWGmcIjContraction [
    {*i} {*j} {*ō} -> {*iu} || _ EnglishStarConsonant,
    {*í} {*j} {*ō} -> {*íu} || _ EnglishStarConsonant
];
```

The implementation carries a stronger general shape than the historical source base does. It should therefore be read as a CAPR formalization of a very narrow lexical development, not as a broad rule independently supported across many families.

Place in the cascade In the inventory ordering, SC009 follows SC008 *PWGmcCoronalWAssimilation* and precedes SC010 *PWGmcJGemination*. In the production cascade it remains inside bundled *EarlyEnglishLineChanges*, but the expanded-PWGmc first-break mode already exposes it directly for chronology testing.

That means chronology testing is procedurally ready even though the historical source base remains narrow.

Order evidence Validated order evidence now exists through the expanded-PWGmc first-break output family:

1. `Germanic/docs/sound_changes/order_tests/summaries/order_sensitivity_first_break_expa`
2. `Germanic/docs/sound_changes/order_tests/summaries/order_sensitivity_first_break_expa`
3. `Germanic/docs/sound_changes/order_tests/summaries/order_sensitivity_first_break_expa`

The earlier search moved SC009 safely across SC008, SC007, SC006, SC005, and SC004 down to order *4* and then reached the left edge of the tested expanded-PWGmc chain with no real break. That side is therefore boundary-only rather than a positive chronology constraint.

The later search does find a real historical break at order *32* across `SC032` OE Diphthong Leveling. If PWGmc Ij Contraction is delayed that far, PGmc fríjōndz yields friund rather than expected OE frēond.

That later boundary is historically interpretable, but it is broad/far rather than a tight local adjacency claim.

Interpretation SC009 can now stand as a short lexical singleton note. The historical change is real enough to explain the friend family explicitly, and the validated chronology shows that the rule cannot simply be delayed indefinitely. The uniqueness of the friend family should remain part of the chapter rather than a reason to omit it.

Remaining cautions The key caution is lexical uniqueness. The later `SC032` boundary is real but broad/far, while the friend family remains effectively the whole historical case. Any later prose should keep that family at the center and should not let CAPR's formal rule read like a major chapter-scale West Germanic vowel change.

West Germanic j-gemination note

Sound-change report

Historical formulation SC010 *PWGmcJGemination* isolates West Germanic consonant gemination before *j* after a short vowel. In the compact trace it appears across several familiar families such as hedge, net, set, sit, and will, and the trace occurrence count is therefore appreciably larger than for some of the surrounding early PWGmc rules.

This is historically recognizable and not merely a one-lexeme convenience. Even so, the report should keep its scope narrow: the rule concerns gemination before *j* after a light syllable, not a general discussion of all consonant doubling in early West Germanic.

Source tradition Fulk treats West Germanic consonant gemination before *j* as regular after a short vowel and states that the change applies to any consonant other than *r* (including *r* < *z*) in that setting (Fulk 2018, sect. 6.15). He illustrates the rule with a wide set of examples including OE scieppan, settan, lecgan, fremman, wennan, and sellan (Fulk 2018, sect. 6.15).

That is strong support for the historical phenomenon itself. It is less explicit about the exact interaction with the later CAPR rules that follow it, because those interactions belong to the modeled cascade rather than to the handbooks. The current source layer is therefore good for the existence and scope of the rule, but not yet for every internal chronology edge.

CAPR implementation CAPR models the gemination as a large explicit set of consonants doubling before *j* after a short vowel:

```
define PWGmcJGemination [
    {*p} -> {*p} {*p} || EnglishStarShortVowel _ {*j},
    {*b} -> {*b} {*b} || EnglishStarShortVowel _ {*j},
    {*t} -> {*t} {*t} || EnglishStarShortVowel _ {*j},
    {*d} -> {*d} {*d} || EnglishStarShortVowel _ {*j},
    {*k} -> {*k} {*k} || EnglishStarShortVowel _ {*j},
    {*g} -> {*g} {*g} || EnglishStarShortVowel _ {*j},
    {*f} -> {*f} {*f} || EnglishStarShortVowel _ {*j},
    {*s} -> {*s} {*s} || EnglishStarShortVowel _ {*j},
    {*m} -> {*m} {*m} || EnglishStarShortVowel _ {*j},
    {*n} -> {*n} {*n} || EnglishStarShortVowel _ {*j},
    {*l} -> {*l} {*l} || EnglishStarShortVowel _ {*j},
    {*ŋ} -> {*ŋ} {*ŋ} || EnglishStarShortVowel _ {*j},
    {*x} -> {*x} {*x} || EnglishStarShortVowel _ {*j}
];
```

This is broader and more explicit than any single handbook example, but it follows the general historical statement closely. The exact enumerated set belongs to the CAPR implementation layer rather than to the historical source tradition.

Place in the cascade In the inventory ordering, SC010 follows SC009 *PWGmcIjContraction* and precedes SC011 *PWGmcSyllabicJ*. In the live production cascade it remains part of bundled *Early-EnglishLineChanges*, but the expanded-PWGmc first-break mode already exposes it directly for chronology testing.

The local implementation logic also makes SC010 important for the next rule. In CAPR, j-gemination must precede SC011 so that newly heavy stems block later syllabic-j vocalization. That interaction is part of the modeled cascade and should stay explicit in backend documentation.

Order evidence Validated expanded-PWGmc first-break TSV output now exists for SC010, and the chronology card is complete. The earlier search moved safely across SC009, SC008, SC007, SC006, SC005, and SC004 to order *4* and then reached the left edge of the tested expanded-PWGmc chain with no real break, so the earlier side is boundary-only.

The later search finds an immediate real break at SC011 *PWGmcSyllabicJ*: if SC010 is delayed to order *11*, PGmc nátją yields nete instead of expected OE nett. That is a tight local reciprocal boundary with SC011 rather than a broad/far rightward limit.

Interpretation SC010 can now stand as a cautious singleton note. The source support is good, the trace presence is substantial, and the validated order evidence recovers a tight local seam with SC011 rather than only a distant broad/far boundary.

Remaining cautions The main caution is scope. The rule is well supported historically, but the earlier side remains boundary-only at the left edge of the tested expanded-PWGmc chain. Any later prose should keep the relation to SC011 visible without turning the pair into a mandatory grouped chapter or enlarging SC010 into a general consonant-doubling chapter.

Syllabic j after final-vowel loss note

Sound-change report

Historical formulation SC011 *PWGmcSyllabicJ* isolates the development by which postconsonantal *j* becomes syllabic *i* after the loss of unstressed final *a* and *ą*. In the current inventory the trace occurrence count is *0*, which immediately marks this rule as backend-useful but observationally weak inside the present compact trace.

That does not make the rule unreal. It does mean the report should stay modest and should not pretend that the current lexical dataset exhibits a broad live witness set for the change.

Source tradition Ringe and Taylor state directly that upon the loss of unstressed *a* and *ą*, preceding postconsonantal *j* and *w* became syllabic *i* and *u* respectively (Ringe and Taylor 2014, 46). Their examples include PGmc harjaz > PWGmc hari, PGmc andijaz > PWGmc andi, and PGmc rikija > PWGmc riki (Ringe and Taylor 2014, 46).

That is good direct support for the historical phenomenon. The weakness lies not in the source tradition but in the trace layer: the current compact trace does not surface direct SC011 hits, so the report must keep the distinction between source-backed history and present trace visibility clear.

CAPR implementation CAPR models the syllabic-j development as:

```
define PWGmcSyllabicJ [
    {*j} {*a} -> {*i} || EnglishStarShortVowel EnglishStarConsonant _ .#.,
    {*j} {*ą} -> {*i} || EnglishStarShortVowel EnglishStarConsonant _ .#.
];
```

This is a deliberately tight formalization of the environment described by Ringe and Taylor. It keeps the focus on postconsonantal *j* after final-vowel loss, rather than widening the rule into a generic fronting or vocalization note.

Place in the cascade In the inventory ordering, SC011 follows SC010 *PWGmcJGemination* and precedes SC012 *EAFLThVoicing*. In the production cascade it remains inside bundled *EarlyEnglishLineChanges*, but the expanded-PWGmc first-break mode already exposes it directly for chronology testing.

Its most important local relation is to SC010. In CAPR, j-gemination must precede syllabic-j vocalization, because gemination creates heavy stems that no longer match the light-syllable environment for SC011. That is a modeled-cascade claim rather than a validated chronology card result at present.

Order evidence Validated expanded-PWGmc first-break TSV output now exists for SC011, and the chronology card is complete. The earlier search finds an immediate real break at SC010 *PWGmcJGemination*: if SC011 is moved earlier to order *10*, PGmc nátją yields nete instead of expected OE nett.

The later search then reaches order *86* with no real break before the current SC087 boundary, so the later side is boundary-only rather than a positive historical limit.

Interpretation SC011 can now stand as a cautious singleton note. The handbook support is good and the validated order evidence recovers a real local seam with SC010, even though the current compact trace still gives the rule a zero direct occurrence count of its own.

Remaining cautions The main caution is the mismatch between the validated boundary and the present trace layer. The historical phenomenon is real and the SC010/SC011 seam is now explicit, but the compact trace still yields no direct SC011 hits. Any later prose should keep that limitation visible and should not inflate the rule into a broader high-vowel chapter.

Lþ-voicing note

Sound-change report

Historical formulation SC012 *EAFLThVoicing* isolates the development of word-internal lþ to *ld*, reflected in trace examples such as field, fold, gold, and wold. The historical phenomenon is clear enough to document, but the stage label is already more delicate than the current inventory makes it sound.

The source support recovered here points most clearly to a **northern West Germanic** development rather than to an unquestioned pan-PWGmc rule. That caution should remain visible from the beginning.

Source tradition Ringe and Taylor state that word-internal lþ became *ld* by regular sound change in northern WGmc and give examples such as fealdan, beald, wuldor, and gylden (Ringe and Taylor 2014, 170–71). Campbell likewise states that medial lþ became *ld* in West Germanic and illustrates the outcome with forms such as fealdan, wuldor, beald, gold, and feld (Campbell 1959, sect. 414).

That is enough to support the historical development itself. It is less enough to license a fully settled PWGmc label, because the strongest Ringe and Taylor wording is explicitly narrower than the inventory's present stage label.

CAPR implementation CAPR models the change as:

```
define EAFLThVoicing [
    {*θ} -> {*d} || {*l} _
];
```

This is a compact formalization of the lþ > *ld* development. As the FST comments already note, some lexical families may also intersect with Verner's-law alternation, but the implementation keeps the historical center of gravity on the consonant cluster itself.

Place in the cascade In the inventory ordering, SC012 follows SC011 *PWGmcSyllabicJ* and precedes SC013 *PWGmcDentalHardening*. In the live cascade it remains inside bundled *EarlyEnglishLineChanges*, but the expanded-PWGmc first-break mode already exposes it directly for chronology testing.

That means the chronology path is procedurally available even though the source-layer stage label still needs caution.

Order evidence Validated expanded-PWGmc first-break TSV output now exists for SC012, and the chronology card is complete. The earlier search moved safely across SC011, SC010, SC009, SC008, SC007, SC006, SC005, and SC004 to order *4* and then reached the left edge of the tested expanded-PWGmc chain with no real break.

The later search likewise reached order *86* with no real break before the current SC087 boundary. Both sides are therefore boundary-only / chronology-negative in current testing.

Interpretation SC012 can now stand as a cautious singleton note. The underlying sound change is credible and source-supported, and the sequence should account for it explicitly even though the validated chronology card supplies no positive local boundary on either side.

Remaining cautions Two cautions matter most. First, the historical stage may be narrower than the inventory's plain PWGmc framing, since Ringe and Taylor describe the change as northern WGmc. Second, the validated chronology card is negative on both sides and therefore supplies no positive local ordering claim. The chapter should keep both cautions visible and should not present the rule as a tightly anchored local seam.

Dental hardening note

Sound-change report

Historical formulation SC013 *PWGmcDentalHardening* isolates the hardening of voiced dental fricative ð to stop *d*. In the current trace layer it appears through forms such as lade, needle, find, and cud, but the historical point is broader than any one lexical family: this is a systemic statement about the status of PWGmc *d*.

That gives the rule a stronger structural footing than some neighboring early PWGmc items, even though the present lexical witness set is still modest.

Source tradition Ringe and Taylor state directly that in PWGmc the non-coronal voiced obstruents continued to show allophony, but *d* became a stop in all positions (Ringe and Taylor 2014, 43). That is unusually strong and explicit support for the historical phenomenon itself.

What the source layer does not yet provide in this pass is a fully expanded lexical dossier for every current trace witness. The systemic historical claim is well supported; the individual witness mapping is still lighter than it could be.

CAPR implementation CAPR models the change as:

```
define PWGmcDentalHardening [
    {*ð} -> {*d}
];
```

This is a direct formalization of the historical statement recovered from Ringe and Taylor. It is one of the cleaner cases in the early PWGmc bundle because the source tradition and the modeled rule align closely.

Place in the cascade In the inventory ordering, SC013 follows SC012 *EAFLThVoicing* and stands as the last internal PWGmc component before the already established SC014 onward sequence. In the production cascade it remains inside bundled *EarlyEnglishLineChanges*, but the expanded-PWGmc first-break mode already exposes it directly for chronology testing.

That makes SC013 a natural right-edge singleton in the present backend-preparation sequence.

Order evidence Validated expanded-PWGmc first-break TSV output now exists for SC013, and the chronology card is complete. The earlier search moved safely across SC012, SC011, SC010, SC009, SC008, SC007, SC006, SC005, and SC004 to order *4* and then reached the left edge of the tested expanded-PWGmc chain with no real break.

The later search likewise reached order *86* with no real break before the current SC087 boundary. Both sides are therefore boundary-only / chronology-negative in current testing.

Interpretation SC013 can still stand as a short finished singleton note. The source support for the basic change is direct, the implementation is straightforward, and the negative chronology card is acceptable so long as the note stays precise and does not claim a positive local ordering that current testing does not show.

Remaining cautions The chief caution is methodological. The systemic historical statement is strong, but both sides of the validated chronology card are boundary-only. Any later prose should keep the rule precise and should not turn the tested-chain left edge or the SC087 search boundary into historical anchors.

Opening vowel prelude

Sound-change report

> **Corrected PROTOFORM pass (supersedes the split note below).** SC014 is the unstressed monophthongization *ai* > ē (final AND nonfinal), live Foma rule {*ai*} -> {ē}. Under the production *PROTOFORM* it is corpus-active, with two dat.sg witnesses span (> spanne) and meed (> meorde) and a later first-break boundary at SC072 (order 69). **Read every "corpus-inert" / "chronology-negative" / "word-final only" below as superseded**: SC014 is corpus-active and covers final and nonfinal unstressed *ai*. It executes at the head of the cascade (position 1), so it is no longer adjacent to SC015 *PNWGmcILowering* (position 11). The separate stressed change is SC004 *EAFAiMonophthongization* ({ái} -> {ā}; see `004-pwgmc-ai-monophthongization.md`). SC015's substantive text is unchanged. Implemented on branch historical-cascade-order (split f59b758d, PROTOFORM correction 9c71aed3).

Historical formulation `SC014-SC015` appears here as a short, cautious opening bridge report, not as a claim that the traditional grammars present one robust textbook chapter called "opening vowel prelude." The grouped report is useful for narrative shape because these are the earliest ordinary FST changes in the volume, and the sound-change narrative reads more clearly if it begins with an explicit opening prelude rather than jumping straight into the more developed `SC016-SC020` corridor.

The internal hierarchy of the pair should stay explicit. SC014 NWGmc Unstressed Ai Monophthongization is the early opening member: it is the word-final unstressed -ai > -ē change, historically plausible and source-backed but corpus-inert (zero corpus applications), so it imposes no chronology constraint. SC015 NWGmc I Lowering is the stronger member and carries most of the report. It belongs to the broader history of early unstressed front-vowel leveling and has one real, though broad/far, later chronology relation through world.

Source tradition The source tradition supports an early Northwest Germanic unstressed-vowel prelude, but not a sharp two-step local corridor. Ringe and Taylor give the clearest comparative statement for SC014: unstressed *ai* was usually monophthongized and merged with unstressed *e* across much of the Northwest Germanic area (Ringe and Taylor 2014, 37–41). Campbell independently treats unaccented medial *ai* in West Germanic as monophthongized, though with sparse direct evidence, which is exactly the kind of source profile that justifies brief inclusion without overstating local chronology (Campbell 1959, sect. 331.7). Hogg adds the structural observation that diphthongs do not survive in Old English unstressed syllables, reinforcing the broad historical type behind SC014 without turning it into a chapter center (Hogg 1992, 112).

SC015 is better anchored. Campbell states that unaccented front vowels fell together in Old English (Campbell 1959, sect. 369), and Hogg likewise argues that by about 700 unstressed front vowels had broadly converged on `/e/` (Hogg 1992, 117). The world material then makes the rule more concrete. Campbell records weorold / weoruld variation (Campbell 1959, sects 338–339), while Ringe and Taylor derive the word through `weraldu > weruld > weorold ~ worold` (Ringe and Taylor 2014, sect. 6.3.3). That does not create a tight local chapter, but it does make SC015 a historically intelligible opening hinge rather than a purely model-internal adjustment.

CAPR implementation CAPR turns that broad unstressed-vowel background into two explicit opening rules at the left edge of the ordinary cascade.

SC014 *PNWGmcUnstressedAiMonophthongization* removes the unstressed diphthongal quality of *ai*, formalizing a comparative development that the grammars describe more diffusely. SC015 *PNWGmcILowering* then lowers or levels early unstressed front-vowel quality farther. That sequence is more explicit than the handbook prose, but it is still historically defensible so long as the report states clearly that the two rules do not form a strong local reciprocal corridor.

Place in the cascade This report sits at the very opening of the volume and hands off directly to the pilot `SC016-SC020` **Early vocalic/final corridor**.

```
.o. PNWGmcUnstressedAiMonophthongization
.o. PNWGmcILowering
.o. OEWsPalatalGlide
.o. PNWGmcULowering
.o. PNWGmcStressedMonosyllableORaising
.o. PNWGmcFinalLongORaising
.o. EAFFinalZDeletion
```

That immediate handoff is the main structural reason to keep the row here. The report gives the volume an explicit opening without absorbing the `SC016-SC020` corridor to the right. Its outward links should remain restrained. Bundled *EarlyEnglishLineChanges* bounds the earlier search space for both rules, but that is a runner limitation rather than a positive left boundary. SC015 points forward to SC036 through world, but that relation should remain a cross-reference only, not a larger chapter claim.

Order evidence The order evidence justifies the grouping only if the asymmetry remains explicit.

SC014 is source-backed but chronology-negative in current testing. The earlier search runs safely down to bundled *EarlyEnglishLineChanges* with no real break, and the later search finds no real break before the current search boundary at SC087. Those results should not be rewritten into positive chronology claims. The card supports SC014 as a brief opening note, not as a strongly bounded local law.

SC015 is the stronger member. Its earlier search is likewise runner-bounded at bundled *Early-EnglishLineChanges*, so that side is not a positive earlier boundary either. The later side is real, but broad/far: SC015 must precede SC036 OE Inter Stress Raising. If SC015 is delayed later than SC036, the world derivation yields wuruld rather than expected weorold. That is a genuine historical ordering statement, but it is still a forward cross-reference rather than a reason to build a non-contiguous `SC015-SC036` chapter.

Taken together, the cards support a short opening bridge with one weak boundary-limited member and one stronger forward-looking member. They do not support a local reciprocal corridor.

Interpretation The value of this report is architectural as much as historical. It lets the volume begin with explicit prose for the earliest ordinary Northwest Germanic vowel adjustments instead of leaving the narrative to start abruptly at the `SC016-SC020` corridor. That is enough to justify prose even though the two members do not contribute equal weight.

SC014 should therefore stay brief: historically plausible, source-backed, and useful as an opening note, but not a major chronology anchor. SC015 carries the report because it belongs to the broader history of unstressed front-vowel leveling and because its world relation gives the pair one real positive ordering result. The grouped unit works best when it is described as a short SC015-led prelude rather than as a symmetrical two-rule chapter.

Remaining cautions The cautions here are straightforward but important. Do not treat bundled *EarlyEnglishLineChanges* as a historical left boundary for either rule. Do not treat SC014's no-break-before-boundary result as proof that it must precede the whole rest of the half. Do not let the forward `SC015 < SC036` relation pull this opening report into a non-contiguous chapter with the later `SC035-SC037` bridge. And do not let the prelude duplicate the neighboring `SC016-SC020` corridor. The report should remain exactly what it is: a short cautious opening bridge with SC014 brief and SC015 carrying most of the prose.

Early vocalic/final corridor

Sound-change report

Historical formulation This entry treats **SC016, SC017, SC019, and SC020** as one corridor because the four changes interact in actual derivations, not because the handbooks present them as a ready-made chapter. The corridor begins with the West Saxon `geoc / geong / geoguþ` phenomenon, passes through the standard lowering of Germanic *u* before a following non-high vowel, turns into final-vowel structure through unstressed -ō > *-u*, and closes with West Germanic loss of final *z* before rhotacism (Kaluza 1906, sect. 90; Bülbring 1902, sects 298–299; Campbell 1959, sect. 44; 1959, sect. 115; Ringe and Taylor 2014, 267; Crist 2002).

The sequence is therefore historical but uneven. SC017 is the clearest standard handbook sound change. SC016 is real, but the literature usually presents it as a West Saxon palatal-glide or rising-diphthong phenomenon rather than as a widely named law (Campbell 1959, sect. 44). SC019 is historical, but the literature usually frames it through ō-stem and final-vowel behavior rather than under the CAPR label NWGmc Final Long O Raising (Ringe and Taylor 2014, 267; Fulk 2018). SC020 is likewise historical, but the clearest formulation is the loss of word-final *z* in unstressed syllables with major morphological consequences, not the exact one-line FOMA rule used by CAPR (Crist 2001, 2002; Kilday 2024).

Source tradition The source tradition begins at the left edge with the older Old English grammarians. Kaluza and Bülbring treat geoc, geong, and geoguþ as part of the West Saxon reflexes of initial Germanic *j*, while Campbell gives the most usable short formulation for book prose: the spellings in forms such as geoc reflect rising diphthongs formed when palatal glides developed before back vowels (Kaluza 1906, sect. 90; Bülbring 1902, sects 298–299; Campbell 1959, sect. 44). That is enough to justify SC016 as a real phenomenon, but it also shows why the subsection should be phrased cautiously: the sources describe a real West Saxon development more often than they name a discrete sound law.

For SC017 the handbook tradition is firmer. Kaluza states the rule early and compactly, Campbell gives the standard textbook wording "`u > o before mid and low vowels`," and Fulk provides the clearest modern formulation with the relevant blocking conditions (Kaluza 1906, sect. 66; Campbell 1959, sect. 115; Fulk 2018, sect. 4.3). Luick is especially useful here as a cautionary witness, because he shows that the outputs could later fluctuate analogically, as in the `scufel / scofel` type (Luick 1914--1940). The corridor should therefore treat SC017 as the strongest standard change in the set, while still recognizing that the distribution of actual forms is not mechanically uniform.

The tradition for SC019 is more ending-centered. Luick treats final ō > *u* through final-vowel shortening, Ringe and Taylor give the clearest concise statement that Proto-Germanic word-final non-nasalized long -ō became unstressed -*u* in Proto-Northwest-Germanic, and Fulk translates that comparative claim into the ō-stem and light-stem endings familiar from Old English (Luick 1914--1940; Ringe and Taylor 2014, 267; Fulk 2018). That makes SC019 a historical change with strong comparative support, but not one whose CAPR label should necessarily become the prose heading unchanged.

SC020 continues the same move from root vocalism into endings and weak tails. Campbell and Hogg preserve the textbook frame that Germanic *z* is later lost or rhotacized, while Crist's dissertation and 2002 handout formulate the key historical claim more sharply as West Germanic loss of word-final *z* in unstressed syllables before rhotacism; Kilday's recent summary treats that core development as already established background (Campbell 1959; Hogg 1992; Crist 2001, 2002; Kilday 2024). This means the right edge of the corridor is also historically real, but its literature is more morphology-heavy than the bare CAPR rule suggests.

CAPR implementation CAPR turns those historical phenomena into four explicit ordered stages. SC016 formalizes the West Saxon palatal-glide / rising-diphthong behavior by rewriting initial palatal + `u/ú` sequences to palatal + `eu/éu`, extended across the initial palatal inventory used in the model. SC017 is the most ordinary sound-law implementation of the four: it lowers stressed `u/ú` to `o/ó` before a following non-high vowel while explicitly blocking the relevant nasal and *j* environments. SC019 formalizes final unstressed -ō > -*u* as a separate stage, while the adjacent monosyllabic ō rule is split off into SC018. SC020 then deletes final *z* outright by applying a simple word-final deletion step, even though the literature states the historical core more narrowly as loss of word-final *z* in unstressed syllables (Campbell 1959; Fulk 2018; Ringe and Taylor 2014; Crist 2002).

That explicitness is useful, but it is not neutral. SC016 is more sharply bounded in CAPR than in the grammars. SC017 is closest to a direct handbook implementation. SC019 isolates a literature-backed ending development as a single stage with exact structural guards. SC020 is the strongest case where the formal model is broader than the cleanest handbook formulation.

Place in the cascade The relevant cascade slice is:

```
.o. PNWGmcUnstressedAiMonophthongization
.o. PNWGmcILowering
.o. OEWsPalatalGlide
.o. PNWGmcULowering
.o. PNWGmcStressedMonosyllableORaising
.o. PNWGmcFinalLongORaising
.o. EAFFinalZDeletion
```

This is the local neighborhood that makes the corridor visible. SC016, SC017, SC019, and SC020 form the meaningful derivational spine. SC018 remains adjacent in the implementation, but its chronology card is negative / boundary-limited, so it is not chapter-forming in the same way. SC020 also points forward: once final *z* is gone, the model is already moving into the later final-syllable and weak-tail territory that becomes more prominent in subsequent chapters.

Order evidence The chronology cards recover three local order-sensitive links. First, `SC016 < SC017` is visible in yoke: PGmc júką yields expected OE ġeoc, but the wrong ordering produces ġoc. The literature independently supports both the geoc phenomenon and the *u* > *o* rule, but not this exact local ordering statement; that is the contribution of CAPR's order testing.

Second, `SC017 < SC019` is visible in nose, shovel, and sorrow. If *u*-lowering is delayed too long, the lowered root vowel never appears and the derivations come out as nusu, sċufl, and surg instead of nosu, sċofl, and sorg. This boundary is literature-compatible, especially because Fulk places *u*-lowering before the final -ō > -*u* development (Fulk 2018, sect. 4.3), but the exact local three-word test remains a CAPR result. *Nose* should also be treated cautiously in polished prose because Kroonen gives the etymon as nasō- ~ nusō- rather than as one uncontested preform (Kroonen 2013).

Third, `SC019 < SC020` is visible in rest. If final *z* is deleted too early, final -ō never receives the treatment it needs and PGmc rástōz comes out as rast instead of expected ræste. The literature strongly supports both ingredients of this interaction, but it does not itself formulate the rest boundary as a local chronology theorem. Here again CAPR supplies the explicit derivational hinge.

SC020 also has later broad/far links, especially toward SC040, but those should be treated only as forward-looking context. They show that the corridor leads into the later weak-tail system; they do not justify expanding this entry into an undifferentiated graph chapter.

Interpretation The structural value of this corridor is methodological as well as historical. The literature gives four real phenomena, but it gives them unevenly: one is a strong standard sound law, one is a real but weakly named West Saxon sub-rule, and two belong most naturally to ending structure and final-syllable history. The CAPR cascade then forces those phenomena into explicit ordered rule form. The chronology cards, finally, show where specific derivations break if that order is disturbed.

That combination is what makes the corridor worth its own report. It shows how the sound-change half of the book can move from scholarship to model to controlled inference without confusing the three. The handbooks do not give a ready-made four-part chapter, but they do provide the historical material from which the corridor can be written responsibly.

Remaining cautions SC016 should probably be described in final prose as a **modeled West Saxon palatal-glide / rising-diphthong sub-rule** rather than as a universally named sound law. SC019's label may likewise need softening toward **final unstressed -ō > -*u*** language. SC020's literature-backed core is narrower than CAPR's unconditioned final-*z* deletion rule, so the prose should keep the handbook formulation and the model formulation distinct (Crist 2002; Kilday 2024). *Nose* should continue to carry an ablaut caution because of Kroonen's nasō- ~ nusō- reconstruction (Kroonen 2013). Finally, the later SC020 links to SC041, SC042, and SC054 should remain background orientation only; they should not expand this entry into a graph-driven chapter.

Stressed monosyllable o-raising note

Sound-change report

Historical formulation SC018 *PNWGmcStressedMonosyllableORaising* appears here as a **short finished singleton note**. It is historically legible enough to remain explicit in the book, but the current chronology card is boundary-limited on both sides. The right reading is therefore modest: this is a short standalone note in strict chronological position, not a reason to expand the `SC016-SC020` corridor or to merge the rule into the following SC021 singleton.

Source tradition The source tradition is sufficient for a short finished note. Campbell treats the Northwest Germanic and Old English long-vowel region behind stressed monosyllabic *o*-raising as ordinary handbook material, and the same broader background also feeds the earlier `SC016-SC020` corridor (Campbell 1959, sect. 138; Ringe and Taylor 2014; Fulk 2018).

That is enough to keep SC018 visible, but not enough to inflate it into a major chapter. The historical claim stays narrow: the rule belongs in the book, even though current order testing does not recover a positive local chronology for it.

CAPR implementation CAPR isolates this background as one explicit step: SC018 *PNWGmcStressedMonosyllableORaising*. That formalization is sharper than most handbook summaries, but it is useful because it lets the book keep the rule explicit without pretending that it anchors a larger report of its own.

Place in the cascade This note belongs immediately after the pilot `SC016-SC020` **Early vocalic/final corridor** and immediately before the `SC021` **Unstressed o-raising note**. It should remain a short standalone note in that position rather than being folded backward into the `SC016-SC020` corridor or forward into the later singleton sequence.

Order evidence Current testing does not identify a positive historical first-break boundary for SC018 in either direction. On the earlier side, the search reaches bundled *EarlyEnglishLineChanges*; that is a methodological runner limit, not a detected historical boundary. On the later side, the search reaches the current SC087 boundary with no real break; that is likewise a no-break-before-boundary result bounded by the present search space, not a claim that SC018 must precede SC087.

This means the card is useful as a negative computational result. It shows that SC018 remains historically visible but does not currently supply a positive local chronology claim of its own.

Interpretation SC018 belongs here because the final volume should discuss every ordinary sound change explicitly, and SC018 is historically legible enough to deserve final prose. The honest final prose form is short: the rule stays visible, but its current chronology remains boundary-limited rather than positively ordered.

Remaining cautions The caution is methodological. Bundled *EarlyEnglishLineChanges* and the current SC087 search boundary are limits of the searchable corridor, not historical anchors for SC018. This note should therefore stay brief and should not be merged into either the `SC016-SC020` corridor on the left or the SC021 singleton on the right.

Unstressed o-raising note

Sound-change report

Historical formulation SC021 *PNWGmcUnstressedORaising* appears here as a **short singleton unstressed-vowel note**. It is not a claim that the handbooks isolate one large independent chapter under exactly this CAPR label, and it is not a reason to build a non-contiguous chapter with the later `SC039-SC040` medial-vowel core. The historical point is narrower: the heaven / heofon material keeps this change legible, and the split early Northwest Germanic zone is clearer if its strongest one-sided member is made explicit in strict chronological position.

Source tradition The source tradition is modest but real. Campbell explicitly derives heofon from an older hefun, with the later visible -o- belonging to the broader history of unstressed vowels (Campbell 1959, sect. 373). Hogg likewise treats the heofon / heofun region inside later unstressed-vowel development, and Ringe and Taylor also keep heaven historically legible inside the wider West Germanic and Old English vocalic record (Hogg 1992; Ringe and Taylor 2014).

That is enough support for a short note. The prose should keep SC021 focused on the narrow unstressed-vowel hinge behind heofon, not inflate it into a corridor report or a hidden left member of the later `SC039-SC040` unit.

CAPR implementation CAPR sharpens that broad unstressed-vowel background into one explicit step: SC021 *PNWGmcUnstressedORaising* isolates the unstressed-vowel adjustment that lets the heaven derivation reach heofon rather than heofun. This is more specific than the handbook phrasing, but that specificity is exactly why the note is useful. It lets the book mark the first singleton in the split early Northwest Germanic zone without pretending that the whole zone forms one adjacent chapter.

Place in the cascade This note belongs immediately after the `SC018` **Stressed monosyllable o-raising note** and immediately before the `SC022` **Mn dissimilation note**. Its meaningful positive chronology link points outward on the right to the earlier `SC039-SC040` **Medial unstressed vowel core**, but that relation should remain a cross-reference rather than a reason to pull SC021 out of strict chronological order.

That placement matters because SC021 is the first full singleton to emerge from the split early Northwest Germanic zone, and it establishes the model for possible later singleton treatments of SC023 and SC024 without flattening the whole region into one non-local chapter.

Order evidence The chronology card gives SC021 a real but one-sided window: its earlier side is runner-bounded at bundled *EarlyEnglishLineChanges*, while its later side yields the broad/far boundary `SC021 < SC040`.

On the earlier side, the current runner can move SC021 back only to the bundled *EarlyEnglishLineChanges* boundary. That is a methodological runner limitation, not a positive earlier boundary, so the note must not rewrite it as one.

On the later side, SC021 must precede SC040. If NWGmc unstressed *o*-raising is delayed across the medial-vowel core, PGmc xémonų yields heofun rather than expected OE heofon. That later edge is historically real, but it is broad/far and belongs as a rightward cross-reference to `SC039-SC040` rather than as a larger chapter claim inside one shared report.

Interpretation SC021 belongs here because the modest reading is now the best one. The heaven evidence is narrow, but it is genuine, and the volume is clearer when the earliest singleton in this split zone is stated explicitly. The result is a cleaner sequence: SC018, SC021, SC022, then the remaining early Northwest Germanic singleton notes before the `SC026-SC027` corridor.

Remaining cautions The cautions are mostly structural. SC021 should not be merged with the `SC039-SC040` report just because its one positive edge points there. Its earlier boundary against bundled *EarlyEnglishLineChanges* must remain a runner limitation rather than a historical claim. And the note should stay modest: it is a short architectural singleton, not a new corridor report and not a reason to treat SC023 or SC024 the same way.

Mn dissimilation note

Sound-change report

Historical formulation SC022 *PNWGmcMnDissimilation* appears here as a **short finished singleton note**. The source background is thinner and more descriptive than for the stronger nearby singletons, but the rule is still historically legible enough to remain explicit in the book. The right final form is therefore a short note, not a broader early Northwest Germanic chapter.

Source tradition The source tradition is modest but real. Campbell treats the preservation and simplification of *mn* material descriptively, including the special status of forms such as month inside the broader analogical and phonological history (Campbell 1959, sects 470, 484).

That support is enough to justify final prose, but only in a restrained form. SC022 belongs in the book because it is historically legible, not because the current record turns it into a strong chronology hinge.

CAPR implementation CAPR isolates this material as one explicit step: SC022 *PNWGmcMnDissimilation*. That is analytically useful because it keeps the rule visible in strict order even though the handbooks usually discuss the underlying pattern more descriptively than as a separate major chapter.

Place in the cascade This note belongs immediately after the `SC021` **Unstressed o-raising note** and immediately before the `SC023` **N-stem n-loss note**. It should remain a short standalone note in that position rather than being merged with either neighboring singleton or with any broader early NWGmc holding chapter.

Order evidence Current testing does not identify a positive historical first-break boundary for SC022 in either direction. On the earlier side, the search reaches bundled *EarlyEnglishLineChanges*; that is a methodological runner limit, not a detected historical boundary. On the later side, the search reaches the current SC087 boundary with no real break; that is a no-break-before-boundary result bounded by the present search space, not a claim that SC022 must precede SC087.

The note therefore records historically legible but chronology-light material. Its evidence is sufficient for explicit final prose, but not for a stronger order claim.

Interpretation SC022 belongs here because the volume should not leave ordinary sound changes in placeholder form when a short honest note will do. The correct final treatment is modest: keep the rule visible, keep the chronology language negative and methodological, and avoid inflating a thin descriptive background into a larger report.

Remaining cautions The methodological limits matter here. Bundled *EarlyEnglishLineChanges* and the current SC087 search boundary are search limits, not historical boundaries for SC022. This note should therefore stay short and should not be merged into SC021, SC023, or any broader early Northwest Germanic chapter.

N-stem n-loss note

Sound-change report

Historical formulation SC023 *PNWGmcNStemNLoss* appears here as a **short singleton n-stem-loss note**. It is not a claim that the handbooks isolate one broad morphology chapter under exactly this CAPR label, and it is not a reason to build a non-contiguous chapter with the later `SC046-SC048` restoration and nasal-tail corridor. The historical point is narrower: the *do* / dōn material keeps this reduction legible, and the split early Northwest Germanic zone is clearer if its next strongest one-sided member is made explicit in strict chronological position after SC021.

Source tradition The source tradition is modest but real. Ringe and Taylor describe heavy syncretism and reduction in the Germanic *n*-stems and separately discuss the development of the verb *do*, which is exactly the kind of morphologized material CAPR is isolating here (Ringe and Taylor 2014).

That is enough support for a short note. The prose should keep SC023 focused on the narrow n-stem-loss hinge behind *do* / dōn, not inflate it into a full n-stem morphology chapter or a hidden left member of the later `SC046-SC048` unit.

CAPR implementation CAPR sharpens that broad background into one explicit step: SC023 *PNWGmcNStemNLoss* removes the relevant n-stem *n* in the early Northwest Germanic zone so that the *do* derivation reaches expected OE dōn rather than collapsing later in the nasal-tail region. This is more specific than the handbook phrasing, but that specificity is exactly why the note is useful. It lets the book mark the second singleton in the split early Northwest Germanic zone without pretending that the whole zone forms one adjacent chapter.

Place in the cascade This note belongs immediately after the `SC022` **Mn dissimilation note** and immediately before the `SC024` **Long e-lowering note**. Its meaningful positive chronology link points outward on the right to the earlier `SC046-SC048` **Restoration and nasal-tail corridor**, but that relation should remain a cross-reference rather than a reason to pull SC023 out of strict chronological order.

That placement matters because SC023 is the second full singleton to emerge from the split early Northwest Germanic zone, and it extends the model for possible later singleton treatment of SC024 while leaving SC018, SC022, and SC025 as boundary/context notes.

Order evidence The chronology card gives SC023 a real but one-sided window: its earlier side is runner-bounded at bundled *EarlyEnglishLineChanges*, while its later side yields the broad/far boundary `SC023 < SC047`.

On the earlier side, the current runner can move SC023 back only to the bundled *EarlyEnglishLineChanges* boundary. That is a methodological runner limitation, not a positive earlier boundary, so the note must not rewrite it as one.

On the later side, SC023 must precede SC047. If NWGmc n-stem *n*-loss is delayed across the restoration and nasal-tail corridor, PGmc dōną no longer yields expected OE dōn; the later-shifted derivation records no output at all. That later edge is historically real, but it is broad/far and belongs as a rightward cross-reference to `SC046-SC048` rather than as a larger chapter claim inside one shared report.

Interpretation SC023 belongs here because the modest reading is now the best one. The *do* evidence is narrow, but it is genuine, and the volume is clearer when the next singleton in this split zone is stated explicitly. The result is a cleaner sequence: SC022, SC023, SC024, then the remaining early Northwest Germanic singleton notes before the `SC026-SC027` corridor.

Remaining cautions The cautions are mostly structural. SC023 should not be merged with the `SC046-SC048` report just because its one positive edge points there. Its earlier boundary against bundled *EarlyEnglishLineChanges* must remain a runner limitation rather than a historical claim. And the failed late derivation should be described as no output or derivational collapse, not as a competing Old English surface form. The note should stay modest: it is a short architectural singleton, not a new corridor report and not a reason to treat SC024 the same way.

Long e-lowering note

Sound-change report

Historical formulation SC024 *PNWGmcLongELowering* appears here as a **short singleton long-vowel note**. It is not a claim that the handbooks isolate one broad long-vowel chapter under exactly this CAPR label, and it is not a reason to build a non-contiguous chapter with the later `SC055-SC056` umlaut core. The historical point is narrower: the sheep / year material keeps this lowering legible, and the split early Northwest Germanic zone is clearer if its final one-sided candidate is made explicit in strict chronological position after SC023.

Source tradition The source tradition is clear enough for a short note. Campbell treats West Saxon sċēap and ġēar as outcomes of earlier long-vowel lowering/fronting in the relevant region (Campbell 1959, sect. 156). Ringe and Taylor likewise discuss the skēpa- / jēra- pathway behind these forms (Ringe and Taylor 2014).

That is enough support for a short note. The prose should keep SC024 focused on the narrow lowering/fronting hinge behind sċēap and ġēar, not inflate it into a broad long-vowel chapter or a hidden left member of the later `SC055-SC056` unit.

CAPR implementation CAPR sharpens that broad background into one explicit step: SC024 *PNWGmcLongELowering* lowers the relevant long ē in the early Northwest Germanic zone so that the sheep and year derivations reach expected OE sċēap and ġēar rather than later high-diphthong outcomes. This is more specific than the handbook phrasing, but that specificity is exactly why the note is useful. It lets the book mark the third singleton in the split early Northwest Germanic zone without pretending that the whole zone forms one adjacent chapter.

Place in the cascade This note belongs immediately after the `SC023` **N-stem n-loss note** and immediately before the `SC025` **Long e nasal-rounding note**. Its meaningful positive chronology link points outward on the right to the already `SC055-SC056` **Umlaut core and palatal-diphthongization follower** report, but that relation should remain a cross-reference rather than a reason to pull SC024 out of strict chronological order.

That placement matters because SC024 is the third full singleton to emerge from the split early Northwest Germanic zone, and it closes the current run of one-sided early-zone notes while leaving SC018, SC022, and SC025 as boundary/context notes.

Order evidence The chronology card gives SC024 a real but one-sided window: its earlier side is runner-bounded at bundled *EarlyEnglishLineChanges*, while its later side yields the broad/far boundary `SC024 < SC056`.

On the earlier side, the current runner can move SC024 back only to the bundled *EarlyEnglishLineChanges* boundary. That is a methodological runner limitation, not a positive earlier boundary, so the note must not rewrite it as one.

On the later side, SC024 must precede SC056. If NWGmc long-ē lowering is delayed across the umlaut core, PGmc skḗpą yields sċīep instead of expected OE sċēap, and PGmc jḗrą yields ġīer instead of expected ġēar. That later edge is historically real, but it is broad/far and belongs as a rightward cross-reference to `SC055-SC056` rather than as a larger chapter claim inside one shared report.

Interpretation SC024 belongs here because the modest reading is now the best one. The sheep and year evidence is narrow, but it is genuine, and the volume is clearer when the last singleton in this early-zone run is stated explicitly. The result is a cleaner sequence: SC023, SC024, SC025, then the `SC026-SC027` corridor.

Remaining cautions The cautions are mostly structural. SC024 should not be merged with the `SC055-SC056` report just because its one positive edge points there. Its earlier boundary against bundled *EarlyEnglishLineChanges* must remain a runner limitation rather than a historical claim. And the note should stay modest: it is a short architectural singleton, not a new corridor report and not a reason to reopen SC018, SC022, or SC025 the same way.

Long e nasal-rounding note

Sound-change report

Historical formulation SC025 *PNWGmcLongENasalRounding* appears here as a **short finished singleton note**. The long-ē-before-nasal background is historically legible enough to remain explicit, but the chronology card is boundary-limited on both sides. The right final form is therefore a short standalone note, not a hidden extension of SC024 or `SC026-SC027`.

Source tradition The source tradition is modest but clear enough for final prose. Campbell explicitly cites forms such as moon and month from earlier mēnō / mēnōþ, and Ringe and Taylor treat the same long-vowel-before-nasal region inside the broader West Germanic development (Campbell 1959, sect. 129; Ringe and Taylor 2014).

That background is enough to keep SC025 explicit in the book. It is not enough to turn the rule into a larger report or to claim that current testing has recovered a positive local chronology.

CAPR implementation CAPR isolates this background as one explicit step: SC025 *PNWGmcLongENasalRounding*. That sharper implementation is useful because it lets the book keep the long-ē-before-nasal adjustment visible in strict order without overstating its chronology.

Place in the cascade This note belongs immediately after the `SC024` **Long e-lowering note** and immediately before the `SC026-SC027` **Nasal spirant corridor**. It should remain a short standalone note in that position rather than being merged backward into SC024 or forward into the nasal spirant corridor.

Order evidence Current testing does not identify a positive historical first-break boundary for SC025 in either direction. On the earlier side, the search reaches bundled *EarlyEnglishLineChanges*; that is a methodological runner limit, not a detected historical boundary. On the later side, the search reaches the current SC087 boundary with no real break; that is a no-break-before-boundary result bounded by the present search space, not a claim that SC025 must precede SC087.

The note therefore records a historically intelligible rule with boundary-limited chronology. The testing keeps the rule explicit but does not license a stronger order claim.

Interpretation SC025 belongs here because residual completion requires final prose even for chronology-light rules. The honest final treatment is a short singleton note: keep the long-ē-before-nasal background visible, keep the chronology phrasing strictly methodological, and avoid inflating the rule into a corridor chapter.

Remaining cautions The main caution is not to turn methodological search limits into history. Bundled *EarlyEnglishLineChanges* and the current SC087 boundary are not positive chronology anchors for SC025. This note should therefore stay brief and should not be merged into the stronger neighboring SC024 or `SC026-SC027` reports.

Nasal spirant corridor

Sound-change report

Historical formulation SC026 and SC027 together represent the bundled North Sea Germanic / Ingvaeonic development in which nasals are lost before voiceless fricatives, with compensatory lengthening and often an intermediate nasalized stage in the preceding vowel. That bundled process is the historical core supported by the handbook tradition. CAPR's labels NWGmc Nasal Spirant Lengthening and NWGmc Nasal Spirant Loss should therefore be read as an analytic split of one broader development, not as two independently standard named laws (Campbell 1959, sect. 121; Fulk 2018, sect. 4.11; Luick 1914--1940, sect. 301.1; Brunner 1965, sect. 186.1).

Source tradition Campbell, Fulk, and Sievers-Brunner give the clearest traditional formulation of the bundled process: before voiceless fricatives, the nasal disappears and the preceding vowel is lengthened, often with nasalization in the historical description (Campbell 1959, sect. 121; Fulk 2018, sect. 4.11; Brunner 1965, sect. 186.1). Hogg is especially useful for the compact ansuz > ōs type, where rounding, nasal loss, and compensatory lengthening are kept together as one connected development (Hogg 1992). Ringe and Taylor help most with stage assignment: they treat forms such as goose and youth as outcomes of a broader northern West Germanic development rather than as a late isolated Old English innovation (Ringe and Taylor 2014, 140–41). Luick adds the most useful internal phonological refinement by describing a long nasalized-vowel stage that later loses its nasal quality and by distinguishing the special *a*-branch without turning the process into separately named laws (Luick 1914--1940, sects 299, 301.1).

CAPR implementation CAPR makes the historical bundle explicit as two adjacent rules. `SC026` adjusts vowel quality and quantity while the nasal + voiceless-fricative conditioning environment is still present; `SC027` then deletes the nasal before the spirant. That split is stricter than ordinary handbook prose, but it is defensible as a model articulation of the same historical development: the transducer needs the conditioning string to remain visible long enough for the vowel effects to apply before the nasal is removed. The report should therefore present the split as a formal clarification inside CAPR, not as direct proof that the traditional literature recognized two separately named sound laws (Fulk 2018, sect. 4.11; Luick 1914--1940, sect. 301.1).

Place in the cascade `SC026-SC027` sits after the early unstressed and boundary-limited Northwest Germanic zone and before the early *x*-loss and glide/fronting entry zone. The pair follows nearby left-side context such as `SC025`, but its strongest local identity is internal: `SC026` and `SC027` form a tight reciprocal center. On the right, `SC028` provides useful context, especially for forms such as fist, because later *x*-loss and fronting material still help shape the final Old English outcome. But that neighboring context should remain contextual rather than being folded into this unit.

Order evidence The chronology cards give the pair an unusually clear local result. `SC026` cannot move later across `SC027`, and `SC027` cannot move earlier across `SC026`. The shared failure set is fist, goose, and youth: if the nasal is deleted too early or the vowel adjustment delayed too long, the model loses the conditioning environment and restores the wrong vocalism. The positive local claim is therefore straightforward: in the present system the corridor requires `SC026 < SC027`.

Interpretation For book purposes, this is the first strong paired report after the singleton treatments because it shows how CAPR can sharpen a historically bundled process into a formally ordered corridor without pretending that the handbooks already drew the same line. The historical claim remains the bundled North Sea Germanic / Ingvaeonic nasal-loss development. CAPR then makes the internal sequencing explicit, and the chronology evidence shows that once this modeling choice is made only one local order works. The chapter should therefore present the pair as a disciplined bridge between handbook tradition and formal implementation.

Remaining cautions The cautions are as important as the positive result. The earlier side of `SC026` remains runner-limited at bundled *EarlyEnglishLineChanges*, so the current evidence does not identify an earlier historical boundary for the pair. The later side of `SC027` is a no-break-before-boundary result through order *86*, so the current evidence does not justify a claim that the corridor must precede `SC087`. More broadly, CAPR's two-rule split is best treated as a defensible model articulation rather than as proof of two separately named historical laws. The stage label also needs care: CAPR files the pair as NWGmc, whereas much of the literature frames it more broadly as North Sea Germanic or Ingvaeonic (Ringe and Taylor 2014, 140–41; Fulk 2018, sect. 4.11). Finally, fist is a valuable chronology example, but it is a multi-stage derivation and should not be used as if this corridor alone explained the whole Old English output.

Preconsonantal x-loss note

Sound-change report

Historical formulation SC028 *PNWGmcPreconsonantalXLoss* appears here as a **short finished singleton note**. It is historically legible as preconsonantal *x*-loss and deserves explicit final prose, but its chronology card remains boundary-limited on both sides. The right final form is therefore a short standalone note, not a reason to reopen either neighboring bridge/core report.

Source tradition The source tradition is sufficient for a short note. Campbell explicitly cites forms such as fléam and héla as examples of loss of *x*, and Ringe and Taylor likewise preserve historically legible `h/x`-bearing derivations in the same broader region (Campbell 1959, sect. 461; Ringe and Taylor 2014, sect. 6.2.1).

That background is enough to keep SC028 visible in the book. It does not turn the rule into a strong chronology anchor, which is why the final prose should stay short and explicit rather than becoming a larger entry chapter.

CAPR implementation CAPR isolates this background as one explicit step: SC028 *PNWGmcPreconsonantalXLoss*. That sharper implementation is useful because it lets the book keep the left edge of the glide/fronting entry zone visible without pretending that the rule carries the same local chronology weight as the earlier `SC029-SC030` core.

Place in the cascade This note belongs immediately after the `SC026-SC027` **Nasal spirant corridor** and immediately before the `SC029-SC030` **Awj glide and au-fronting core**. It should remain a short standalone note in that position rather than being merged backward into the nasal spirant corridor or forward into the glide/fronting core.

Order evidence Current testing does not identify a positive historical first-break boundary for SC028 in either direction. On the earlier side, the search reaches bundled *EarlyEnglishLineChanges*; that is a methodological runner limit, not a detected historical boundary. On the later side, the search reaches the current SC087 boundary with no real break; that is a no-break-before-boundary result bounded by the present search space, not a claim that SC028 must precede SC087.

This means the note records a historically intelligible rule whose current order evidence is negative or boundary-limited rather than positively local.

Interpretation SC028 belongs here because residual completion requires final prose even where the chronology is light. The rule remains explicit, the note stays short, and the stronger local story remains with the earlier `SC029-SC030` core to its right.

Remaining cautions The methodological boundaries matter here. Bundled *EarlyEnglishLineChanges* and the current SC087 search boundary are search limits, not historical anchors for SC028. This note should therefore stay brief and should not be merged into the `SC026-SC027` corridor or the `SC029-SC030` core.

Awj glide and au-fronting core

Sound-change report

Historical formulation `SC029-SC030` appears here as a **compact adjacent glide/fronting core report**, not as a claim that the handbooks present one named two-change chapter with exactly these CAPR labels. The grouping is justified because the local source support and the ordinary chronology align most strongly on this pair. SC029 OE Awj Glide Formation is the left member: it belongs to the early awj / auj developments behind hay and strew, but its earlier side remains runner-bounded. SC030 OE Au Fronting is the local center and right member: it reciprocates the hay / strew evidence on its left and then hands the derivations forward into the later diphthongal region on its right.

Source tradition The source tradition is unusually cooperative for this narrow pair. Campbell treats the auj material directly and gives forms such as `hég / hig`, ieg, and strégan, which are exactly the historical region that the CAPR pair is trying to keep explicit (Campbell 1959, sect. 120). Ringe and Taylor sharpen the same region comparatively: they argue that the Old English outcomes reflect restored auj sequences rather than simple persistence of older aw'w, and their examples include hawja > hieg and strawjaną > *Angl.* OE strégan (Ringe and Taylor 2014, sect. 3.1.3; 2014, sect. 6.1.2). Luick reinforces that this material belongs to the normal Old English diphthongal history rather than to a narrow lexical accident (Luick 1914--1940, sect. 116).

That combination is enough to justify final prose for `SC029-SC030` without pulling SC028 in as a coequal member. The handbooks support the historical region well; CAPR then isolates the tight local seam inside it.

CAPR implementation CAPR makes the local sequence explicit by keeping two adjacent rules.

SC029 *OEAwjGlideFormation* handles the intermediate glide structure that turns the inherited awj / auj material into the form that later fronting reads. On its own it is still somewhat one-sided because the earlier search remains runner-bounded at bundled *EarlyEnglishLineChanges*, but it is the indispensable left-hand feeder of the pair.

SC030 *OEAuFronting* then fronts the relevant *au* material and delivers the hay / strew derivations into the broader West Saxon diphthongal region. That is why the pair is more than a convenient adjacency: SC029 feeds SC030, and SC030 is the rule that makes the local fronted outputs historically recognizable in book prose.

Place in the cascade This report belongs immediately after the `SC028` **Preconsonantal x-loss note** and immediately before the `SC031-SC034` **West Saxon diphthong chain**. On the left, the `SC026-SC027` nasal spirant corridor remains a separate earlier chapter. On the right, SC030's later relation to `SC032` provides a genuine handoff into the earlier diphthongal region, but that relation should remain a cross-reference rather than a reason to merge the two reports.

That position matters for narrative shape. The pair gives the volume a real local seam between the nasal spirant corridor and the broader diphthong chain while leaving SC028 visible as a separate left-edge context note.

Order evidence The order evidence is the clearest reason to keep this pair together.

SC029 has one live positive historical boundary, and it is rightward. If OE Awj Glide Formation is delayed past SC030, PGmc xáwwją yields hauġ instead of expected OE hīeġ, and stráwjaną yields strauian instead of expected strīeġan. That makes `SC029 < SC030` a real local chronology claim. The earlier side is not comparable: the current runner can move SC029 safely back to order *13*, but the search then stops at bundled *EarlyEnglishLineChanges*, so the present result does not identify any positive earlier boundary for SC029.

SC030 reciprocates that local evidence on its left and adds one broader rightward boundary of its own. Moving SC030 earlier than SC029 again produces hauġ and strauian instead of hīeġ and strīeġan, so the pair forms a genuine reciprocal local core. Moving SC030 later than SC032 causes a broader set of derivations to fail outright rather than to produce competing surface forms: believe, bow, bread, dream, and flea all collapse to no output when OE Au Fronting is delayed too far. That later relation is real, but it is best read as a rightward handoff into the next region rather than as a second local reciprocal pair.

Taken together, the cards support one sharp adjacent claim and one broader forward orientation: `SC029 < SC030 < SC032`, with only the `SC029 < SC030` side functioning as a tight local reciprocal corridor.

Interpretation This report works because it is narrower and more honest than the older three-change version. SC028 remains visible as the historically legible but chronology-negative *x*-loss preface. SC029 and SC030 carry the actual report because the same hay / strew material that the handbook tradition discusses also defines the local reciprocal boundary. And SC030's later SC032 relation keeps the handoff into the West Saxon diphthong chain explicit without making the two reports collapse into one chapter.

The result is a short full report that fits the book's narrative shape better than the old grouped entry bridge. It keeps the strongest local seam explicit, preserves strict chronology, and leaves both the short left note and the larger right-hand diphthong chain in their own chapters.

Remaining cautions The cautions are structural as much as evidential. SC029's runner-bounded earlier side must not be turned into a hidden positive boundary against bundled *EarlyEnglishLineChanges*. SC028 should remain visible to the left as a separate context note rather than being treated as a coequal member of this report. SC030's later relation to SC032 is real, but its failure set must be described as no-output / derivational collapse rather than as a cluster of competing Old English surface forms. And this report should not be expanded backward into `SC026-SC027` or forward into `SC031-SC034`.

West Saxon diphthong chain

Sound-change report

Historical formulation `SC031-SC034` appears here as a **cautious West Saxon diphthong-chain report**: a real four-change corridor in the model, not a claim that the handbooks themselves present exactly four separate laws matching CAPR's present segmentation (Campbell 1959, sect. 120; 1959, sects 170–176, 185, 222–227; Hogg 1992, 106–7, 111–12; Luick 1914--1940, sects 115–119, 168–176, 182–183, 195; Ringe and Taylor 2014, sects 6.1.2, 6.5.1, 6.10.1; Fulk 2018, sects 4.7, 4.13). `SC031` OE WW Simplification and `SC034` OE Aw Long Diphthong are the local center of the chapter, while `SC032` OE Diphthong Leveling and `SC033` OE Ew Long Diphthong are retained as real but less central flank material.

Source tradition The source tradition supports a real West Saxon diphthongal region, but not one neat four-step textbook chain. Campbell and Luick are the clearest evidence for that mixed picture: both treat *ww*-based and inherited diphthong developments, later palatal diphthongization, and later smoothing or leveling material as related but distinct layers rather than as four coequal chapter headings (Campbell 1959, sect. 120; 1959, sects 170–176, 185, 222–227; Luick 1914--1940, sects 115–119, 168–176, 182–183, 195). Hogg, Ringe and Taylor, and Fulk sharpen the same caution from different angles: the palatal and later diphthongal material is real, but it sits inside a wider chronology of brightening, breaking, smoothing, and front mutation rather than inside one simple handbook chain (Hogg 1992, 106–7, 111–12; Ringe and Taylor 2014, sects 6.1.2, 6.5.1, 6.10.1; Fulk 2018, sects 4.7, 4.13).

That makes `SC031` and `SC034` the most natural production core. Their witness sets belong to the same dēaw / hēawan side of the region, while `SC032` and `SC033` are better read as subordinate members of the same corridor than as equal headline laws. `SC032` maps less neatly onto a single handbook rule, and `SC033` points more broadly toward the later breaking context on its right.

CAPR implementation CAPR makes the corridor more explicit than the handbooks do by keeping four adjacent rules. *OEWWSimplification* handles the reduction of *ww* sequences that feed the dēaw / hēawan set. *OEAwLongDiphthong* then handles the long ēaw side of that same local neighborhood. *OEEwLongDiphthong* captures the parallel long-ēow material, while *OEDiphthongLeveling* handles a later regularizing zone that is useful inside the model even if it does not map cleanly onto one classical handbook label.

This sharper segmentation is a feature rather than a defect, so long as the prose states clearly what it is doing. CAPR is not pretending that the traditional grammars literally enumerate these four rules in this exact form. It is using four rules to model a historically real diphthongal corridor whose source tradition is more overlapped and asymmetrical than the transducer layer.

Place in the cascade This report belongs immediately after `SC030` OE Au Fronting and before the `SC035-SC037` **Prefix and compound adjustments** bridge. On its right, it reaches toward three later contexts without collapsing into them: `SC032` points forward to `SC040` OE Med Unstressed U Lowering, `SC034` points forward to `SC043` **Anglo-Frisian Brightening**, and `SC033` points forward to `SC044` OE Breaking.

That position matters for narrative shape. The chapter can now cover the local West Saxon diphthong corridor as a coherent middle zone, while the already `SC043` report remains the dedicated right-hand chapter for the brightening pivot rather than being duplicated here.

Order evidence The chronology cards justify the grouping, but only if the internal asymmetry stays visible. The strongest local relation is the reciprocal `SC031 < SC034` / `SC034` after `SC031` pair. On both sides the witnesses are dew and hew: moving `SC031` later than `SC034`, or `SC034` earlier than `SC031`, restores the unsimplified *aw* sequence and breaks the expected dēaw / hēawan outcomes.

`SC032` has a real two-sided profile, but it is the least textbook-like member of the corridor. It must follow `SC030`, with believe, bow, bread, dream, and flea failing when the rule is pulled too far left; but those earlier failures are no-output derivational collapses rather than ordinary alternative surface reflexes. On its right, `SC032` must precede `SC040`, with head showing the wrong unstressed vowel if the rule moves too late.

`SC033` also has a real right-hand boundary, but it is broad rather than tightly local. It must precede `SC044`, with chew, four, and knee losing their long-diphthong outputs if the rule is delayed too far. Its earlier-side default card does not yield an ordinary historical boundary before the bundled *EarlyEnglishLineChanges* runner limit, and the expanded-PWGmc note across `SC008` is supplementary only rather than a replacement for the default chronology claim.

`SC034` likewise has a real forward relation into the brightening region. It must precede `SC043`, with the show witnesses and the straw side of the *aw* set showing why the corridor points naturally toward the later brightening chapter without becoming identical with it. `SC031`'s own expanded-PWGmc earlier-side note across `SC011` is likewise supplementary only and must not be rewritten as the default left boundary of the report.

Interpretation This chapter is valuable precisely because it captures a real region without flattening its internal structure. `SC031` and `SC034` are the local reciprocal core. `SC032` and `SC033` remain inside the report because they belong to the same West Saxon diphthongal corridor, but they are presented as subordinate or flank material rather than as equal historical centers.

That makes the report a useful model for later grouped reports. CAPR can sustain a multi-change corridor even when the handbooks support the region more strongly than they support the model's exact segmentation. The result is more than a placeholder, but still careful about the difference between source tradition, rule implementation, and local chronology evidence.

Remaining cautions `SC032` should not be overstated: its no-output-heavy earlier failures show a real ordering constraint, but not a tidy set of alternate surface reflexes. `SC031` and `SC033` should not inherit ordinary historical left boundaries from their expanded-PWGmc supplementary notes. `SC033` should also remain somewhat open on the right, since its broad relation to `SC044` does not make it a tight local pair in the way that `SC031` and `SC034` are. Finally, this report should not duplicate the separate `SC043` brightening chapter. If later narrative shape ever demands a narrower chapter centered only on the `SC031` / `SC034` pair, that would be a revision of chapter shape, not a discovery that the present corridor was unreal.

Prefix and compound adjustments

Sound-change report

Historical formulation `SC035-SC037` appears here as a **cautious adjacent derivational bridge report**, not as a claim that the traditional grammars present one classic three-change chapter called "prefix and compound adjustments." The grouping is useful because all three rules are adjacent ordinary FST changes that need explicit prose somewhere in the volume, but the internal hierarchy is uneven and should stay explicit. SC035 OE Prefix A Reduction is a narrow prefix-vowel flank. SC036 OE Inter Stress Raising is the clear center of gravity, with the strongest source support and the strongest ordinary chronology in the row. SC037 OE Compound Linking Syncope is the technical/compound- juncture flank: historically intelligible, computationally important, but much less chapter-centered than SC036.

Source tradition The source tradition supports a real stress-sensitive and compounding region, but not one tidy handbook unit. Campbell is the clearest single anchor for the whole bridge because he touches all three relevant domains: reduction in pretonic prefixes, unstable low-stress medial vowels such as weorold / weoruld, and reduced-force compound members with variable linking vowels (Campbell 1959, sect. 354; 1959, sects 338–339; 1959, sects 356–357, 367, 386–387). Hogg sharpens the same general picture by stressing the reduction of unstressed-vowel contrasts and the permanently unstressed status of ge-, which makes both prefix reduction and medial unstressed adjustment historically legible rather than purely model-driven (Hogg 1992, sect. 3.3.1.3; 1992, 99).

The weight of that evidence is not equal across the three members. SC035 is source-legible through reduced pretonic prefixes and stress-weak prefix vocalism, and Ringe and Taylor's derivation of galaubijaną > geliefan gives the believe path a useful comparative anchor (Ringe and Taylor 2014, 245, 267). SC036 is stronger still: Campbell's weorold / weoruld material and Ringe and Taylor's weraldu > weruld > weorold ~ worold derivation place it directly in the historical space of low-stress or interstress vocalism (Campbell 1959, sects 338–339; Ringe and Taylor 2014, sect. 6.3.3). SC037 is different in kind. Campbell and Sievers-Brunner both support compound-juncture vowel behavior, reduced-force second elements, and linking-vowel instability, but they support SC037 best as a compound/technical note rather than as a major ordinary chapter (Campbell 1959, sects 356–357, 367, 386–387; Brunner 1965, sects 167–168).

CAPR implementation CAPR therefore sharpens one historically real but uneven corridor into three adjacent rules with different rhetorical weight.

SC035 *OEPrefixAReduction* reduces an early prefix vowel in weakly stressed prefixed forms. That gives the row a legitimate left flank, but only a narrow one. The rule is real enough to deserve prose, yet it does not carry a large independent chapter by itself.

SC036 *OEInterStressRaising* is the center of the report. It adjusts medial unstressed vocalism between stronger stress peaks, which is why it lands most naturally in the historical space described by Campbell, Hogg, and Ringe and Taylor. If this grouped report works as final prose, it works primarily because SC036 gives it a real historical middle.

SC037 *OECompoundLinkingSyncope* then handles the right-hand compound and linking-vowel cleanup. Here CAPR is more explicit than the handbooks: it keeps a distinct rule where the historical literature more often speaks in broader terms about compounding, weakened second elements, or unstable composition junctures. That is acceptable so long as the prose states clearly that SC037 is book-relevant without pretending it is a canonical textbook law on the same level as SC036.

Place in the cascade This bridge sits immediately after the `SC031-SC034` **West Saxon diphthong chain** and immediately before the `SC039-SC040` **Medial unstressed vowel core**, followed by the separate `SC041` final-loss note. That position is the main reason to keep the grouped row unchanged for now: it keeps a real seam in the middle of the half explicit without forcing three tiny chapters or a non-contiguous a larger chapter claim.

The outward chronology links should remain cross-references only. SC035 points rightward to SC043 through believe, but that does not justify pulling this report out of place toward the brightening chapter. SC036 points forward to SC040, but that later relation should remain a forward cross- reference rather than a reason to absorb this report into the `SC039-SC040` core. SC037 points rightward again through the rainbow evidence that later appears at the left edge of `SC049-SC050`, but that relation should likewise remain cross-reference only.

Order evidence The order evidence justifies the grouping only if the asymmetry stays visible.

SC035 has one real positive historical boundary, and it is rightward. The card shows that SC035 must precede SC043: if OE Prefix A Reduction is delayed beyond Anglo-Frisian Brightening, PGmc galáubijaną yields ġealīefan instead of expected OE ġelīefan. That is a real chronology claim. The earlier side is not. The runner can move SC035 safely back to order 13, but it then stops at bundled *EarlyEnglishLineChanges*, so the current result does not identify any earlier historical boundary and must not be rewritten as one.

SC036 is the strongest member of the report. It must follow SC019: moving it earlier causes soul to surface as sāwel instead of expected sāwol. It must also precede SC040: delaying it yields sāwul and weoruld instead of sāwol and weorold in soul and world. Those are fully historical card boundaries, and they make SC036 the real center of gravity here. At the same time, the earlier side reaches back to SC019 rather than to a tight local neighbor, so the prose should not overstate it as a crisp local adjacency inside this immediate corridor.

SC037 is weaker in ordinary chronology but still important. Its card currently finds no ordinary historical positive boundary in either direction. The earlier search stops at bundled *EarlyEnglish-LineChanges* with no real break. The later search does find a real computational break across SC038 OE Strip Secondary Stress: moving SC037 later produces reġnefoga instead of expected reġnboga in rainbow. But SC038 is a technical marker, so that result must remain exactly what the card says it is: non-historical / technical-marker evidence, not a normal chronology claim. The later rainbow relation to SC049 is real and book-useful, but it should remain a rightward cross-reference only.

Taken together, the cards support a grouped chapter with one strong center and two modest flanks: `SC035 < SC043`, `SC019 < SC036 < SC040`, and a technical-marker SC037/SC038 computational seam that should not be treated into ordinary chronology.

Interpretation This report is valuable because it keeps three adjacent ordinary changes visible without pretending that they have equal source weight or equal chronology strength. SC035 is included because the book still needs explicit prose for the prefix-reduction flank, not because it anchors a large historical chapter. SC036 carries the report: it has the clearest handbook support and the clearest ordinary chronology in the row. SC037 remains visible because the compound-linking cleanup is historically intelligible and computationally important, even though its own card evidence is technical rather than ordinary-historical.

That makes the unit a derivational bridge in the strongest useful sense. It is adjacent, chronological, and coherent enough to carry final prose, but the final prose should be explicit about hierarchy rather than flattening all three members into coequal textbook laws. If later architecture ever narrows the region around SC036, that would be a chapter-shape refinement rather than proof that the present bridge was mistaken.

Remaining cautions The cautions are structural as much as evidential. SC035's runner-bounded earlier side must not be converted into a hidden positive boundary. SC036's earlier relation to SC019 is historically real, but broad enough that it should not be narrated as a tight local adjacency claim. SC037's break across SC038 must not be rewritten as an ordinary historical ordering statement, because SC038 is only a technical marker. And none of the outward links to SC043, SC040, SC049, or SC038 should be allowed to expand this report into a non-contiguous chapter. The report should remain exactly what it is: a cautious adjacent derivational bridge with SC036 as its clear center of gravity.

Medial unstressed vowel core

Sound-change report

Historical formulation `SC039-SC040` appears here as a **narrow adjacent medial-vowel core report**, not as a claim that the handbooks present one named two-change chapter with exactly these CAPR labels. The grouping is justified because the source tradition and the ordinary chronology line up most strongly on this pair. SC039 OE WI Combinative U Umlaut is the narrow widow-based left flank: a historically legible *w*-conditioned low-stress *u* adjustment whose earlier side is still runner-bounded. SC040 OE Med Unstressed U Lowering is the stronger member and the clear center of gravity: it belongs to the historical space of low-stress medial vocalism and carries both the reciprocal widow relation on its left and the broader youth cross-reference on its right.

Source tradition The source tradition is unusually cooperative for this adjacent pair. Campbell discusses the *w*-conditioned or combinative *u* material directly, with wuduwe / widuwe-type forms that make SC039 historically intelligible rather than merely model-internal (Campbell 1959, sect. 218). He then treats low-stress vocalism in forms such as weorold and weoruld, which is the clearest single handbook anchor for SC040 (Campbell 1959, sects 338–339). Campbell's geogup material also keeps SC040's later youth relation inside the same broader unstressed-vowel zone, even though that right-hand boundary is far rather than local (Campbell 1959, sect. 332).

Ringe and Taylor sharpen the same picture comparatively. Their derivation widuwon- > wuduwe ~ widuwe supports the widow evidence behind both SC039 and SC040, while weraldu > weruld > weorold ~ worold places SC040 squarely in the historical space of low-stress medial-vowel adjustment (Ringe and Taylor 2014, 322; 2014, sect. 6.3.3). Their treatment of jugunpi > geogup ~ iugup likewise supports the later youth witness without turning SC072 into part of this chapter's architecture (Ringe and Taylor 2014, 267).

That combination is enough to justify final prose for the pair. The handbooks do not force one classical "`SC039-SC040` chapter," but they do support exactly the kind of narrow adjacent medial-vowel core that the chronology cards now isolate.

CAPR implementation CAPR makes the local sequence explicit by keeping two adjacent rules.

SC039 *OEWICombinativeUUmlaut* handles a narrow *w*-conditioned low-stress *u* adjustment in widow-type material. On its own it is a modest flank, not a large independent chapter. But it matters because it sets up the local vocalic shape that SC040 then reads.

SC040 *OEMedUnstressedULowering* is the broader and more chapter-carrying member. It regularizes medial unstressed *u* behavior across the low-stress system and is the rule that makes the world / weorold material historically recognizable in book prose.

That is why the split works. The pair is not just adjacent in the cascade; it is a real local core in both source support and order evidence. SC041 remains visible to the right as a separate final-loss note rather than a coequal member of this unit.

Place in the cascade This report belongs immediately after the `SC035-SC037` **Prefix and compound adjustments** bridge and immediately before the `SC041` **Final bare-a loss note**. The immediate rightward neighborhood then continues through `SC042` into the SC043 brightening pivot.

The outward chronology links should remain cross-references only. SC036 points forward to SC040, but that later relation should not pull this report leftward into the earlier derivational bridge. SC040 points rightward to SC072 through youth, but that relation is broad/far and should remain orientation rather than a larger chapter claim. SC041 now remains explicitly separate on the right so that the local medial-vowel core can be without flattening the broader final-loss hinge into the same chapter.

Order evidence The order evidence is the clearest reason to keep this pair together.

SC039 has one live positive historical boundary, and it is rightward. If OE WI Combinative U Umlaut is delayed past SC040, PGmc wíduwōn yields wudowe instead of expected OE wuduwe. That makes `SC039 < SC040` a real local chronology claim. The earlier side is not comparable: the current runner can move SC039 safely back to order *13*, but the search then stops at bundled *EarlyEnglishLineChanges*, so the present result does not identify any positive earlier historical boundary for SC039.

SC040 reciprocates that local evidence on its left and adds one broader rightward boundary of its own. Moving SC040 earlier than SC039 again produces wudowe instead of wuduwe, so the pair forms a genuine reciprocal local core. Moving SC040 later than SC072 yields ġeogoþ instead of expected ġeoguþ in youth, which gives SC040 a real later boundary as well. But that later relation is broad/far rather than locally chapter-defining.

Taken together, the cards support one sharp adjacent claim and one broader rightward orientation: `SC039 < SC040 < SC072`, with only the `SC039 < SC040` side functioning as a tight local reciprocal corridor.

Interpretation This report works because it is narrower and more honest than the old grouped bridge. SC039 remains visible as the one-sided widow flank. SC040 carries the report as the stronger source-backed historical center, with world and youth keeping the broader low-stress context in view. And SC041 is no longer forced into the same production unit when its own chronology points outward to SC020 and SC046 rather than forming an equally tight local pair here.

The result is a short full report that fits the book's architecture better than the previous older three-change version. It keeps the strongest local seam explicit, preserves strict chronology, and leaves the broader final-loss material visible without pretending it is part of the same immediate core.

Remaining cautions The cautions are structural as much as evidential. SC039's runner-bounded earlier side must not be turned into a hidden positive boundary against bundled *EarlyEnglishLineChanges*. SC040's later relation to SC072 is real, but it should remain a rightward cross-reference rather than a reason to expand this report into a non-local late-vowel chapter. And the separate SC041 note should stay separate: the split solves a real asymmetry in source weight and chronology, so the book should not casually recombine the pair with SC041 or absorb this report into the earlier `SC035-SC037` bridge.

Final bare-a loss note

Sound-change report

Historical formulation SC041 *PWGmcFinalBareALoss* appears here as a **short singleton final-loss note**, not as a claim that the handbooks present one isolated textbook chapter with exactly this CAPR label. The historical type is nevertheless real. The broader tradition supports early loss or erosion of final short low vowels, and that support is enough to justify explicit prose for the rule even though its chronology is broader than the tight local widow pair that now carries `SC039-SC040` to the left (Campbell 1959, sect. 341; Ringe and Taylor 2014, 60–61).

Source tradition The source tradition is broader than the CAPR label but still clear enough for a short note. Campbell treats the loss of the relevant unaccented final vowel as part of early Germanic and prehistoric Old English final-syllable erosion (Campbell 1959, sect. 341). Ringe and Taylor sharpen the same background comparatively by arguing that the loss of word-final short low vowels was already a Proto-West-Germanic change (Ringe and Taylor 2014, 60–61).

That source support does not make SC041 part of the `SC039-SC040` medial-vowel core. It instead makes SC041 historically legible as the first explicit final-loss hinge after that core: close enough to belong here in strict chronological order, but broad enough that it works better as a short singleton report than as a coequal member of the medial-vowel chapter on its left or the restoration corridor on its right.

CAPR implementation CAPR makes this broad historical background explicit through one narrow rule: SC041 *PWGmcFinalBareALoss* deletes a surviving bare final low vowel in word- final position. That implementation is intentionally sharper than the handbook phrasing. The grammars describe a broader region of final-vowel erosion and leveling; CAPR isolates one specific stage so that later interactions with brightening, restoration, and weak-tail developments can be tested directly.

That explicitness is the main reason the rule deserves its own production note. It lets the book state plainly where the model turns from medial unstressed vowels toward final-syllable loss without pretending that every later final-vowel effect belongs to this one rule.

Place in the cascade This note belongs immediately after the `SC039-SC040` **Medial unstressed vowel core** and immediately before the `SC042` **Surviving bimoric O unrounding context**. Its broader cross-references point outward rather than architecturally inward: leftward to the pilot `SC016-SC020` **Early vocalic/final corridor** through SC020, and rightward to the `SC046-SC048` **Restoration and nasal-tail corridor** through SC046.

That position is what makes SC041 useful as a singleton note. It closes the medial-lowering/final-loss seam without reopening the earlier `SC016-SC020` corridor or collapsing into the later restoration chapter.

Order evidence The chronology card places SC041 in a real but broad window: `SC020 < SC041 < SC046`.

On the earlier side, SC041 must follow SC020. If final bare-*a* loss is moved too early across PGmc Final Z Deletion, a broad class of spurious final -*a* outputs survives. Representative failures include beard, bosom, bottom, calf, and coat; concretely, PGmc bárdaz yields bearda rather than expected OE beard, and kámbaz yields camba rather than camb. That earlier boundary is historically real, but it is broad/far and should be read as a leftward cross-reference to the `SC016-SC020` corridor rather than as local chapter claim.

On the later side, SC041 must precede SC046. If final bare-*a* loss is delayed too long, the later restoration environment changes and fronted outcomes are lost. Representative failures include craft, dale, day, hazel, and mast; concretely, PGmc kráftaz yields craft rather than expected OE cræft, and dágaz yields dag rather than expected dæġ. This later boundary is also historically real, but it should remain a rightward cross-reference to the restoration corridor rather than a reason to merge the two units.

Taken together, the cards support a real hinge but not a tight local reciprocal pair: `SC020 < SC041 < SC046`, with both edges pointing outward to neighboring production units.

Interpretation This report works because it is deliberately modest. SC041 is not the center of a larger new corridor, and it is not a hidden third member of the `SC039-SC040` report. Its job is architectural: to keep a real final-loss stage explicit between the medial-vowel core and the brightening/restoration region.

The literature supports that move. Campbell and Ringe and Taylor both confirm that final short low-vowel loss belongs to the historical background of the period. CAPR then sharpens that background into one testable rule, and the chronology card shows that the rule sits in a broad but genuine window between the earlier `SC016-SC020` corridor and the later restoration hinge.

Remaining cautions The cautions are mostly structural. SC041 should not be merged back into `SC039-SC040`, because its broad `SC020 < SC041 < SC046` chronology is not the same kind of local reciprocal core that justifies that pair. It should not be merged backward into the pilot `SC016-SC020` corridor or forward into the `SC046-SC048` restoration report, because those links are outward cross-references rather than a larger chapter claim. And its later failures should be described as environment-changing derivational consequences that block restored/fronted outputs, not as a cluster of competing local surface variants inside one narrow chapter.

Surviving bimoric O unrounding context

Sound-change report

Historical formulation SC042 *PWGmcSurvivingBimoricOUnrounding* appears here as a **short singleton context note**. It is not a claim that the handbooks isolate one large independent chapter under exactly this label. The historical point is smaller: CAPR needs an explicit surviving-bimoric ō pathway behind rest, and that pathway belongs immediately on the left edge of the SC043 brightening pivot rather than hidden inside a larger non-contiguous chapter (Campbell 1959, sects 131, 157–158; Hogg 1992, 101, 119; Ringe and Taylor 2014, 157–58, 189–90).

Source tradition The source tradition supports the surrounding fronting and restoration corridor more readily than it isolates SC042 as a textbook chapter in its own right. Campbell treats the early fronting and later restoration developments as part of one intelligible historical region (Campbell 1959, sects 131, 157–158). Hogg is especially useful on the early fronting side and on the broader vocalic environment that later Old English rules continue to read (Hogg 1992, 101, 119). Ringe and Taylor likewise place fronting and later retraction/restoration in a sequence that makes CAPR's narrow rest pathway historically legible even if the precise SC042 label is model-shaped (Ringe and Taylor 2014, 157–58, 189–90).

That is enough support for a modest note. The report should present SC042 as a narrow feeder/context rule inside the post-final-loss seam, not as a coequal chapter center beside SC043, `SC044-SC045`, or `SC046-SC048`.

CAPR implementation CAPR sharpens that broad historical background into one explicit step: SC042 *PWGmcSurvivingBimoricOUnrounding* maps a surviving bimoric ō to the low-vocalic stage that later feeds the fronted/restored rest outcome. This is more specific than the handbook phrasing, but that specificity is exactly why the note is useful. It lets the book show where the model creates the input that SC043 then brightens, instead of leaving the rest pathway implicit.

Place in the cascade This note belongs immediately after the `SC041` **Final bare-a loss note** and immediately before the `SC043` **Anglo-Frisian brightening** report. On its right, the earlier `SC044-SC045` **Breaking and velar-fricative palatalization** report remains a separate neighboring unit rather than part of the same chapter.

That placement is the whole reason to treat SC042 now. It closes the small architectural seam between final-loss and brightening without pretending that the note itself is a major post-brightening chapter.

Order evidence The chronology card gives SC042 a real but very narrow two-sided window: `SC020 < SC042 < SC043`, and both boundaries are carried by rest.

On the earlier side, SC042 must follow SC020. If the unrounding stage is moved earlier across PGmc Final Z Deletion, PGmc rástōz yields rasta rather than expected OE ræste. That earlier edge is historically interpretable, but it should remain a leftward cross-reference to the pilot `SC016-SC020` corridor rather than a larger chapter claim.

On the later side, SC042 must precede SC043. If the same unrounding stage is delayed across Anglo-Frisian Brightening, PGmc rástōz again yields rasta rather than ræste. This later edge is tighter and more local: it is the direct handoff that makes SC042 worth keeping explicit immediately before the brightening report.

Interpretation SC042 belongs here because the modest reading is now the best one. The rest evidence is thin, but it is genuine on both sides, and the book is clearer when the surviving-bimoric feeder to SC043 is stated explicitly. That gives the volume a cleaner local sequence: `SC041 -> SC042 -> SC043`, followed by the already separate `SC044-SC045` and `SC046-SC048` reports.

Remaining cautions The cautions are mostly structural. SC042 should still be narrated as a short context note rather than a major standalone chapter. Its evidence remains witness-limited, so the prose should not overstate independence. It should not be merged non-contiguously with `SC044-SC045` just because that pair belongs to the same broader post-brightening region. And its leftward relation to SC020 should stay a cross-reference rather than a reason to reopen the earlier `SC016-SC020` corridor here.

Anglo-Frisian brightening

Sound-change report

Historical formulation SC043 is the early fronting usually called **Anglo-Frisian Brightening** or **First Fronting**: Germanic low *a* develops to æ-type outputs outside nasal environments. The core rule is stable in the handbook tradition even if the wider English-Frisian shared-innovation framing has to be handled more cautiously than the label itself may suggest (Campbell 1959, sect. 131; Hogg 1992, 101; Ringe and Taylor 2014, 157–58; Fulk 2018, sect. 4.12).

Source tradition Campbell gives the classical formulation of early *a* > æ outside nasal environments and then treats later breaking and restoration as changes that presuppose that fronted stage. Hogg makes the modern label pair explicit and is also the most useful direct support for extending the fronting into unstressed vowels. Ringe and Taylor sharpen the chronology further by arguing that fronting must precede breaking and that later retraction must follow both. Fulk gives the most compact bridge from the basic brightening rule to the later Old English developments that partly conceal it (Campbell 1959, sects 131, 139, 157–158; Hogg 1992, 101, 119, 445; Ringe and Taylor 2014, 157–58, 168–69, 189–90; Fulk 2018, sects 4.12–4.13).

CAPR implementation CAPR formalizes SC043 with three coordinated clauses: an unstressed *a* > æ rule outside nasal environments, a stressed á > æ rule with the same core conditioning, and a long-final ā > ǣ clause. The first two are close to the handbook rule in explicit transducer form. The long-final clause is more model-specific: it is best read as an implementation choice tied to the surviving-bimoric ō > ā pathway inherited from SC042, not as a direct quotation from a single handbook statement.

Place in the cascade SC043 sits in a tight local neighborhood: `SC042` PWGmc Surviving Bimoric O Unrounding, `SC043` Anglo-Frisian Brightening, `SC044` OE Breaking, `SC045` OE Velar Fricative Palatalization, and `SC046` OE A Restoration. That position matters because brightening is both fed and later partly masked inside the same cluster. It receives left-hand input from SC042, creates the front-vowel stage needed by SC044, and is later partly reversed by SC046 in back-vowel environments.

Order evidence The chronology card fixes SC043 in a narrow local window. `SC042 < SC043` is supported by rest: if brightening moves earlier across SC042, the derivation loses the fronted/restored route and yields rasta rather than ræste. `SC043 < SC044` is supported by slay: if brightening moves later than breaking, the derivation yields sleaan | slēaan rather than slēan. The wider local neighborhood also supports `SC043 < SC046`, while earlier links such as `SC034 < SC043` and contextual `SC035 < SC043` are best treated as orientation rather than as reasons to turn this report into a graph chapter.

Interpretation For book purposes, SC043 is a pivot change. The literature establishes the basic fronting rule and its relation to later breaking and restoration. CAPR then makes that relation explicit in rule form, and the chronology card shows that the current local ordering is not arbitrary inside the model. The report should therefore present brightening as an enabling stage: it creates front- vowel input for later Old English developments and is subsequently partly obscured by restoration, so its historical importance often becomes clearest through the later changes that still presuppose it.

Remaining cautions The broader English-Frisian shared-innovation story should be handled carefully: the existence of the fronting rule is well supported, but the subgrouping narrative is less straightforward (Campbell 1959, sect. 131; Ringe and Taylor 2014, 157–58). The unstressed clause in CAPR is defensible from the handbook tradition, yet it remains a modeling choice that should not be mistaken for the whole historical claim (Hogg 1992, 101, 445). Likewise, the long-final clause belongs to CAPR's representation of the SC042-to-SC043 pathway and should be presented as a model-specific approximation rather than as a simple handbook quotation.

Breaking and velar-fricative palatalization

Sound-change report

Historical formulation SC044 OE Breaking is standard handbook material: front vowels develop into broken diphthongal outputs before *h*, *rC*, *lC*, and related local contexts, and it belongs after the fronted stage created by the SC043 Anglo-Frisian Brightening report (Campbell 1959, sect. 139; Ringe and Taylor 2014, sects 6.2.1–6.2.3; Fulk 2018, sect. 4.13). SC045 **OE Velar Fricative Palatalization** is narrower in CAPR's naming, but historically intelligible as the local velar/fricative follow-on: a front-vowel and *j*-side palatalization zone affecting *x* and *ɣ* after the breaking environment is already in place (Campbell 1959, sects 177, 405–406; Ringe and Taylor 2014, sects 6.4.1–6.4.2).

Source tradition The source tradition is clearest on SC044. Campbell, Ringe and Taylor, and Fulk all treat breaking as an ordinary Old English development downstream of earlier fronting (Campbell 1959, sect. 139; Ringe and Taylor 2014, sects 6.2.1–6.2.3; Fulk 2018, sect. 4.13). SC045 is less often given exactly the same label as CAPR's rule name, but the historical material is still recognizable. Ringe and Taylor separate palatalization of velars from the later loss of *w* after non-initial velars, which makes the local chronology legible, and Campbell likewise places fronting of velars after the brightening/breaking corridor rather than inside a single broad undifferentiated palatalization chapter (Campbell 1959, sects 177, 405–406; Ringe and Taylor 2014, sects 6.4.1–6.4.2). The result is a strong adjacent pair, but not a whole post-brightening chapter.

CAPR implementation CAPR makes the local sequence explicit by keeping two adjacent rules. *OEBreaking* creates the broken *ea*/*eo*/*io*-type outputs in the relevant *h*, *rC*, *lC*, *w*, and special velar/fricative contexts. Then *OEVelarFricativePalatalization* palatalizes *x* to *ç* and *ɣ* to *j* beside front vowels or before *j*. That ordering matters: SC044 establishes the local vocalic environment that SC045 then reads. The chapter should therefore present SC045 as the local velar/fricative partner to breaking, not as a broad generic palatalization article.

Place in the cascade This report belongs immediately after the SC043 brightening pivot. On its left, SC042 remains a separate short context note rather than part of this production unit. On its right, the report leads into the `SC046-SC048` post-brightening corridor, while SC060 OE Ws Palatal Umlaut remains only a later rightward cross-reference rather than part of this chapter's architecture.

Order evidence The chronology cards give the report a strong local spine. SC044 must follow SC043: slay shows that if breaking moves earlier than brightening, the model yields sleaan | slēaan rather than expected slēan. SC044 must also precede SC045, and SC045 must follow SC044. That reciprocal local core is carried by fee, fight, flax, knight, and laugh: if the order is reversed, the live broken outputs disappear and the derivations surface with unbroken vowels. SC045 also has a real later boundary at SC060 through six, but that right-hand relation should stay a cross-reference only. It is useful for orientation, not a reason to extend the report beyond its adjacent pair.

Interpretation For book purposes, `SC044-SC045` is the strongest adjacent report immediately to the right of SC043. SC044 is the standard handbook center; SC045 is the local velar/fricative partner that the CAPR implementation makes unusually explicit. That is enough to justify treatment as a full adjacent report. But the chapter should still stay modest: it is not a replacement for the SC043 brightening report, and it is not a whole post-brightening chapter covering everything through SC060.

Remaining cautions The cautions matter as much as the positive result. SC042 should remain visible as a separate context note rather than being pulled into this report non-contiguously. SC045 should not be narrated as a general chapter on all Old English palatalization; here it is the local velar/fricative follow-on to breaking. And although `SC045 < SC060` is a real later relation, it is narrow and should remain a rightward cross-reference rather than a larger chapter claim. The `SC046-SC048` corridor likewise remains a separate neighboring chapter rather than material to absorb into this pair.

Restoration and nasal-tail corridor

Sound-change report

Historical formulation `SC046-SC048` appears here as a **cautious adjacent post-brightening corridor report**, not as a claim that the handbooks present one neat three-change Old English chapter in exactly this form. The internal structure is uneven by design. SC046 OE A Restoration is the source-backed restoration/retraction hinge: it restores fronted low-vowel outcomes to back *a* before following back vowels after the brightening stage has already been created (Campbell 1959, sects 157–159; Ringe and Taylor 2014, sect. 6.3.1; Fulk 2018, sect. 4.13). SC047 OE Heavy Syllable Nasal Apocope and SC048 **OE Secondary Nasalization** form the tighter reciprocal nasal-tail core inside CAPR, but they are less textbook-like as a named pair than SC046 is as a historical restoration step.

Source tradition The source tradition is clearest on SC046. Campbell's restoration of *a* before following back vowels, Ringe and Taylor's general retraction of fronted low vowels, and Fulk's compact overview all describe the same historical corridor that CAPR isolates here (Campbell 1959, sects 157–159; Ringe and Taylor 2014, sect. 6.3.1; Fulk 2018, sect. 4.13). That makes SC046 easy to narrate as a real stage downstream of Anglo-Frisian Brightening.

The nasal-tail side is more mixed. Campbell's sections on loss of nasals and back mutation, together with Ringe and Taylor's discussion of later back umlaut, support the broader historical environment in which SC047 and SC048 operate, but they do not yield one tidy handbook chapter named exactly the way CAPR segments it (Campbell 1959, sect. 403, §§205–206; Ringe and Taylor 2014, sect. 6.9.4). The result is a real historical neighborhood with uneven source support: SC046 is the clearest source-backed hinge, while `SC047-SC048` is a stronger computational core than a classical textbook pair.

CAPR implementation CAPR makes that mixed corridor explicit by keeping three adjacent rules. SC046 *OEARestoration* changes æ back to *a* before the specific class of following back-vowel environments that still trigger restoration. SC047 *OEHeavySyllableNasalApocope* then deletes the final nasalized vowel in heavy syllables. SC048 *OESecondaryNasalization* marks *a* before final *n*, helping separate live -an outcomes from the spurious -en forms that appear if the nasal-tail rules are misordered.

That sequencing is why the report works as one cautious corridor. SC046 is the restoration hinge that still reads cleanly in source terms. SC047 and SC048 are the tighter internal cleanup pair whose broad reciprocal -en failures matter most inside the model. The prose should therefore present one adjacent corridor with a clear internal center of gravity, not three symmetrical coequal laws.

Place in the cascade This report belongs immediately after the `SC044-SC045` **Breaking and velar-fricative palatalization** report and immediately before the `SC049-SC050` **Onset allophony and Sievers-law bridge**. On its left it still depends on the earlier SC043 **Anglo-Frisian Brightening** pivot, because SC046 is one of the stages that partly reverses the fronted outcomes created there. On its right SC048 points forward to SC059 OE Back Mutation, but that later relation should remain a cross-reference only rather than a reason to expand this chapter past its adjacent corridor.

Order evidence The chronology cards justify the grouping precisely because they show one strong hinge plus one stronger internal core.

SC046 must follow SC043 and precede SC048. If it is moved earlier than SC043, bake, fare, flask, grave, and haw keep fronted outcomes such as bæcan and færan instead of the expected restored bacan and faran. If it is delayed beyond SC048, the restored back-vowel outcomes are again lost, with bake, fare, grave, lade, and wade surfacing as fronted forms such as bæcan and wædan. SC046 is therefore not just a local flourish after brightening; it is a real restoration hinge with positive card boundaries on both sides.

SC047 is narrower and less textbook-like. Its earlier positive boundary is only at SC034 OE Aw Long Diphthong, carried by straw, where moving heavy- syllable nasal apocope too far left produces stræw instead of expected strēaw. Its later boundary is much broader and more important inside CAPR: delaying SC047 beyond SC048 creates an 87-row reciprocal -en failure set, so forms such as bake, begin, believe, bind, and bore surface with spurious -en tails.

SC048 is the partner to SC047 inside that nasal-tail core. It must follow SC047, with the same broad reciprocal -en failure set, and it must precede SC059, where steal and weave begin to show back-mutated forms such as steolan and weofan instead of the expected stelan and wefan. That `SC048 < SC059` relation is real, but it should stay a rightward cross-reference only, not a reason to stretch the chapter into a non-local back-mutation report.

Taken together, the cards support `SC043 < SC046 < SC048`, `SC034 < SC047 < SC048,` and `SC047 < SC048 < SC059`. The strongest internal claim is the broad reciprocal `SC047-SC048` core; SC046 is the clearer historical hinge that makes the corridor legible on the left.

Interpretation This report is useful because it keeps three adjacent ordinary FST changes explicit without pretending they are equal members of a classical handbook law. SC046 is the historically clearest member and should carry the source explanation. `SC047-SC048` is where the strongest internal reciprocity lies, and the book should say so plainly. The result is a corridor with one source-backed hinge feeding a more model-internal nasal-tail cleanup zone.

That is enough to justify treatment as full prose. The chapter is no longer just a placeholder, but it remains deliberately cautious about the gap between the handbooks' broader historical region and CAPR's sharper three-rule segmentation. If later narrative shape ever narrows the corridor into SC046 plus a separate `SC047-SC048` pair, that would be a chapter-shape revision rather than proof that the present corridor was unreal.

Remaining cautions The caution is structural as much as descriptive. This report should not absorb the earlier SC043 or `SC044-SC045` material on its left, and it should not absorb the `SC049-SC050` bridge on its right. SC047 and SC048 should not be overstated as if the handbooks routinely present them as a named joint chapter; their strongest support here is the broad reciprocal -en failure core inside the model. And although `SC048 < SC059` is a real later relation, it is narrow and should remain a rightward cross-reference rather than a larger chapter claim. The corridor should stay exactly what it is: an adjacent, chronological, but unevenly source-backed post-brightening report.

Onset allophony and Sievers-law bridge

Sound-change report

Historical formulation This is a short adjacent chronological bridge report, not a major textbook chapter. It keeps two neighboring ordinary FST changes visible between the `SC046-SC048` **Restoration and nasal-tail corridor** and the SC051 and SC052 palatalization reports without pretending that SC049 and SC050 are one historical process.

SC049 PGmc B Allophony is the narrower and more implementation-heavy member. It maps onto the familiar handbook distribution in which *b* surfaces as a stop initially or after nasals and as a voiced bilabial fricative elsewhere (Hogg 1992, 101–2; Ringe and Taylor 2014, 121; Luick 1914--1940, 107). SC050 **Sievers Law Syncope** has stronger traditional backing because the literature treats Sievers-law reflexes, heavy/light stem structure, weak verbs, and gemination as real historical material (Adamczyk 2001; Fulk 2018, 28, sec. 6.15). The pair therefore stays together for practical chronological reasons, not because the sources present a standard `SC049-SC050` chapter.

Source tradition The source tradition for SC049 is thin but real. Hogg gives the clearest Old English formulation: `/b/` is a stop initially, after nasals, and in gemination, but elsewhere it is realized as a bilabial fricative (Hogg 1992, 101–2). Ringe and Taylor support the broader comparative picture by treating Proto-West-Germanic *b* as having stop and fricative allophones by position (Ringe and Taylor 2014, 121), while Luick adds orthographic corroboration for the voiced labial fricative in Old English (Luick 1914--1940, 107). That is enough to support SC049 as a legitimate distributional rule, but not enough to turn it into a large historical chapter of its own.

SC050 is historically more legible. Adamczyk treats Old English reflexes of Sievers' Law as visible in weak-verb and related paradigms, and Fulk gives the compact comparative-grammar cross-check through the familiar biddan, fremman, sellan, and nerian patterning (Adamczyk 2001; Fulk 2018, 28, sec. 6.15). That gives the rule a broader source base than SC049 has. Even so, the local role of SC050 in this part of the book is still modest: it matters here chiefly as the feeder that must stand before SC052 rather than as the core of a separate palatalization chapter.

CAPR implementation CAPR keeps both rules narrower than the handbook categories.

SC049 *PGmcBAllophony* turns *b* into beta after vowels and liquids, then restores stop *b* before geminate *bb*. That is a late, distributional surface-management rule in the model, which is why SC049 reads as implementation-heavy rather than chapter-centered.

SC050 *SieversLawSyncope* deletes *i* before *j* after a consonant or palatal consonant. In the model that makes it a small but important feeder into the following palatalization region. CAPR therefore isolates SC050 cleanly before SC051 OE Sk Palatalization and SC052 OE Velar Palatalization without implying that all three rules should be merged into one unit.

Place in the cascade This unit belongs immediately after the `SC046-SC048` corridor and immediately before the SC051 report. It also stands directly to the left of the SC052 hinge report. That placement is the main reason to treat the pair together: both changes are adjacent, both are ordinary FST changes, and both need explicit prose in the volume.

SC050's forward relation to SC052 should be handled by cross-reference rather than by a noncontiguous `SC050-SC052` chapter. SC052 remains the later hinge report where plain velar palatalization is discussed in its own right; this bridge report only explains why Sievers Law Syncope belongs immediately to its left.

Order evidence The order evidence is outward-facing rather than internally reciprocal. SC049 has one real positive boundary on the left: it must follow SC037 **OE Compound Linking Syncope**, because moving it earlier breaks the voiced compound outcome in rainbow, yielding reġnfoga instead of expected reġnboga. Its later side found no real break before the runner boundary at SC087. That later result must remain a bounded no-break outcome, not a positive claim that SC049 must precede some specific later historical stage.

SC050 shows the opposite profile. Its earlier side found no real break before the bundled *EarlyEnglishLineChanges* runner boundary, so it does not support a positive claim that Sievers Law Syncope must follow any specific earlier historical stage. Its later side is real: SC050 must precede SC052, because delaying it past OE Velar Palatalization breaks the stretch derivation and yields strecċan instead of expected streċċan.

Taken together, the cards support `SC037 < SC049` and `SC050 < SC052`, but they do not support an internal reciprocal chronology claim for `SC049-SC050` as a pair.

Interpretation This report is useful precisely because it stays modest. SC049 is present because every ordinary FST change needs explicit prose somewhere in the book, and because its narrow allophonic role still has one live chronology witness in rainbow. SC050 is present because it has stronger source backing and because its local function as a feeder into SC052 is real and testable.

That makes the pairing practical and chronological rather than doctrinal. The book does not need to pretend that SC049 and SC050 form a traditional joint chapter. It only needs to keep them visible, in order, and in proportion to the evidence they actually have.

Remaining cautions SC049 should not be overstated. The handbooks support the stop/fricative distribution of *b*, but the current chronology evidence is still one-sided and compound-specific. SC050 should not be allowed to duplicate the already SC052 report. Its role here is to mark the Sievers-law bridge and the stretch dependency, not to absorb the later plain-velar palatalization discussion into an out-of-order unit.

More broadly, neither runner-bounded side should be rewritten as positive chronology evidence. SC049's later no-break result and SC050's earlier no-break result are both important cautions, not hidden boundaries. The report should remain what it is: a short adjacent bridge that keeps both ordinary changes visible without inflating weak or one-sided evidence into a major chapter.

OE Sk Palatalization

Sound-change report

Historical formulation SC051 isolates the Old English palatalization and assibilation of *sk to palatal* sc. That is a real and recognizable historical development in the handbook tradition, even though it is usually presented as part of the broader Old English palatalization complex rather than as a wholly independent headline chapter (Campbell 1959, sect. 170; 1959, sects 440–441; Hogg 1992, 106–7; Ringe and Taylor 2014, sect. 6.4.1; Luick 1914--1940, sect. 168; Fulk 2018, 28; Brunner 1965, sect. 91.a).

This report is therefore deliberately narrow. It treats SC051 as its own concise singleton chapter between the `SC049-SC050` bridge report on the left and the SC052 standalone hinge report on the right. The historical claim is that OE sk > sc deserves its own chapter shape inside the broader palatalization region without collapsing those neighboring reports into one larger unit.

Source tradition The source tradition supports both the rule and the need for restraint in how it is framed. Campbell treats *sk as especially prone to palatalization and assibilation, distinguishing the broad initial development from medial and final environments that still resist palatalization after back vowels (Campbell 1959, sect. 170; 1959, sects 440–441). Hogg is especially clear that the cluster sk undergoes a parallel change to palatal sc and that this should be kept separate from later palatal diphthongization (Hogg 1992, 106–7, 111–12). Ringe and Taylor are the sharpest guide to the chapter boundary: they distinguish* sk palatalization from later West Saxon diphthongization after initial palatals, which is exactly the distinction this report needs (Ringe and Taylor 2014, sects 6.4.1, 6.5.1).

Luick is most useful for the broader historical framing, since he treats the whole area as an early movement in a palatal direction and places the diphthongal West Saxon effects later (Luick 1914--1940, sect. 168; 1914--1940, sects 171–174). Fulk gives the most compact prose version of the core conditioning: OE *sc is palatal except internally before or finally after a back vowel (Fulk 2018, 28). Sievers-Brunner offers older corroboration for the same layered picture: broad initial palatalization of* sc, but preserved back-vowel contrasts and later vocalic effects after palatals [Brunner (1965), §91.a; §50; §105 Anm. 2].

CAPR implementation CAPR makes this handbook material more explicit by isolating *OESkPalatalization* as its own rule. That is a modeling choice, but it is a disciplined one. The rule captures the *sk palatalization itself, while the adjacent SC052 OE Velar Palatalization and SC056 **OE Ws Palatal Diphthongization** handle neighboring parts of the broader palatalization complex.

That separation is analytically useful because the handbooks often discuss the whole region together, whereas CAPR needs narrower rule boundaries to test local ordering claims. The report should therefore present SC051 as a legitimate historical rule isolated by the model, not as proof that the traditional literature itself always treated *sk palatalization as a wholly self-standing chapter.

Place in the cascade SC051 follows SC046 OE A Restoration, sits inside the broader `SC049-SC052` palatalization/fronting region, and precedes SC056 **OE Ws Palatal Diphthongization**. In the new assembly layout it now stands between the *049-050* bridge report on the left and the SC052 hinge report on the right, rather than remaining buried inside the old four-change placeholder.

That placement matters for book structure as well as for local order. On its right, SC051 belongs near the earlier `SC055-SC056` umlaut-core report, but it must only cross-reference that chapter rather than duplicate it. The point of this report is to clarify one narrow palatalization rule before the book returns to the neighboring velar-palatalization and umlaut material.

Order evidence The chronology card gives SC051 the strongest local standalone evidence in the old mixed cluster. It must follow SC046: the key earlier-side witnesses are flask and wash, where moving SC051 earlier than restoration yields fronted outputs such as flæsce and wæscan instead of the live back-vowel forms flasce and wascan. It must also precede SC056: the key later-side witnesses are shaft, shear, sheath, sheep, and *shield, where delaying SC051 beyond West Saxon palatal diphthongization removes the live* sċea- / *sċie- outcomes (Campbell 1959, sect. 170; 1959, sects 440–441; Hogg 1992, 106–7; 1992, 111–12; Ringe and Taylor 2014, sect. 6.4.1; 2014, sect. 6.5.1; Fulk 2018, 28; 2018, sect. 4.13).

Both boundaries are historically interpretable and fairly local. That is the main reason SC051 is promotable as a narrow report: it has stronger local two-sided evidence than the parent `SC049-SC052` cluster, and it does not depend on overreading a runner-bounded side as a positive historical boundary.

Interpretation This report isolates one clean palatalization chapter from the broader region. Its value is architectural as much as historical. The neighboring reports are now clearer because they no longer have to share one placeholder: `SC049-SC050` forms the left-hand bridge report, SC052 stands as the standalone hinge into the umlaut region, and the SC057 note remains a later palatalization-side note rather than part of this chapter.

That makes SC051 a good example of how the production layer should mature. The book does not need to solve the whole palatalization/fronting region at once. It can treat the rule that is already historically legible, source-backed, and locally well bounded, while leaving the rest of the structure visible for later review.

Remaining cautions This report should stay narrow. It does not exhaust Old English palatalization, and it should not absorb SC052 or SC056 into a larger chapter by implication. The handbooks support SC051 as a real process, but they often discuss it inside the wider palatalization complex rather than as a fully isolated law (Campbell 1959, sect. 170; 1959, sects 440–441; Hogg 1992, 106–7; Ringe and Taylor 2014, sect. 6.4.1; Luick 1914--1940, sect. 168).

The chapter must also avoid duplicating the earlier SC052 hinge report or the `SC055-SC056` umlaut-core report. SC056 belongs here only as SC051's right-hand local boundary, not as coequal chapter content. SC052 should be cross-referenced as the neighboring hinge chapter, and the SC057 note remains separate later palatalization-side material. This report settles only the narrow case for SC051.

Velar palatalization hinge

Sound-change report

Historical formulation SC052 isolates the Old English palatalization of plain velars *k* and *g* in front-vocalic and *j*-adjacent environments. That is a real Old English process in the handbook tradition, even though the same sources usually discuss plain velars and *sk* inside a broader palatalization complex rather than as wholly independent headline chapters (Campbell 1959, sect. 170; Hogg 1992, 106–7; Ringe and Taylor 2014, sect. 6.4.1; Luick 1914--1940, sects 168–176; Fulk 2018, 28).

This report therefore treats SC052 as a standalone chronological hinge, not as part of a non-contiguous `SC050-SC052` pair. On its left stands the already SC051 OE Sk Palatalization report; on its right stands the `SC055-SC056` umlaut-core report. The historical claim is narrow but important: plain velar palatalization deserves its own concise report between those neighboring chapters.

Source tradition The source tradition supports that framing. Campbell, Hogg, Ringe and Taylor, Luick, and Fulk all treat plain velar palatalization as a real Old English development, and all place it in the same broad historical neighborhood as *sk*-palatalization and later front-mutation material (Campbell 1959, sect. 170; Hogg 1992, 106–7, 111–14; Ringe and Taylor 2014, sects 6.4.1, 6.6.1–6.6.4; Luick 1914--1940, sects 168–176, 181–183; Fulk 2018, sects 4.7, 4.13). The handbooks usually describe SC051 and SC052 inside that wider palatalization complex, but the book can still treat them as adjacent chronological reports so long as it does not pretend they are historically unrelated.

The strongest source support for SC052's right edge comes from Ringe and Taylor and from the broader handbook sequence in Campbell, Hogg, Luick, and Fulk: palatalization belongs to an earlier consonantal zone, while i-umlaut or front mutation is later enough to create contrasts with non-palatalized velars (Campbell 1959, sects 190–197; Hogg 1992, 111–14; Ringe and Taylor 2014, sects 6.4.1, 6.6.1–6.6.4; Luick 1914--1940, sects 181–183; Fulk 2018, sect. 4.7). The left edge is narrower. Adamczyk is useful precisely because she supports treating SC050 as a real Sievers-law and prosodic feeder issue without turning it into a natural coequal member of a velar-palatalization chapter (Adamczyk 2001).

CAPR implementation CAPR isolates *OEVelarPalatalization* more sharply than many handbook presentations do. In the model, plain *k* is palatalized before front vowels, before *j*, and in a few front-vowel-adjacent positions that preserve the local contrast system. Plain *g* is likewise palatalized before front vowels, after front vowels in certain edge positions, and before *j*. That narrower rule boundary is analytically useful because it lets CAPR distinguish four adjacent functions that the handbook tradition often discusses together:

1. SC050 **Sievers Law Syncope** as a left-edge feeder;
2. SC051 OE Sk Palatalization as the cluster rule;
3. SC052 OE Velar Palatalization as the plain-velar hinge;
4. SC055 OE I Umlaut as the next major vowel chapter.

That segmentation is a modeling choice, but it is a disciplined one. The sources support the larger historical zone, while CAPR makes the internal order testable without treating every neighboring process as one mixed chapter (Campbell 1959, sect. 170; Hogg 1992, 106–7, 111–14; Ringe and Taylor 2014, sects 6.4.1, 6.6.1–6.6.4; Luick 1914--1940, sects 168–176, 181–183; Fulk 2018, sects 4.7, 4.13).

Place in the cascade SC052 sits at current order 52. SC050 remains its feeder/context on the left: the stretch relation matters because Sievers Law Syncope creates the shape that the palatalization rule then acts on, not because SC050 and SC052 should be merged into one out-of-order chapter. In the assembled book, SC051 is the immediately preceding report, and SC052 is the immediately following standalone hinge report.

To the right, SC052 now precedes the `SC053-SC054` pre-umlaut bridge report and the `SC055-SC056` umlaut-core report. Farther right, the SC057 note remains a later palatalization-side development. That placement is the main architectural gain of this report: the book can now discuss the plain-velar hinge explicitly while keeping the assembled order strictly chronological.

Order evidence The chronology cards give SC052 a small but real local network. It must follow SC050: if SC052 is moved earlier than Sievers Law Syncope, stretch yields the wrong palatal and geminate outcome. It must also precede SC055: if SC052 is moved later than OE I Umlaut, cow and lung become over-palatalized. Those results show `SC050 < SC052 < SC055`.

The left edge still requires restraint. SC050's earlier side is runner-limited, not a positive historical boundary, so the stretch relation must not be inflated into a whole `SC050-SC052` chapter. The right edge is narrow as well, but it is historically interpretable because the source tradition broadly places palatalization before i-umlaut or front mutation (Campbell 1959, sects 190–197; Hogg 1992, 111–14; Ringe and Taylor 2014, sects 6.4.1, 6.6.1–6.6.4; Luick 1914--1940, sects 181–183; Fulk 2018, sect. 4.7).

Interpretation SC052 is stronger than residual placeholder material. Its value is not just that it has one witness on the left and two on the right, but that those witnesses make historical sense inside the broader source tradition. This is therefore the kind of central hinge rule that can be without pretending the whole palatalization region has been fully resolved.

The report also shows how the sound-change half can mature while preserving strict chronological order. SC050 remains visible as feeder/context in the earlier *049-050* bridge report, SC051 remains a separate cluster chapter, SC052 now receives its own prose immediately after SC051, and `SC055-SC056` remains the umlaut-core chapter to the right. The neighboring reports should be cross-referenced, not duplicated.

Remaining cautions This report should stay narrow. It does not exhaust Old English palatalization, and it should not be read as if SC051 and SC052 were historically unrelated just because the book now treats them as adjacent reports. It must not duplicate the earlier SC051 OE Sk Palatalization report or the `SC055-SC056` umlaut-core report.

The chapter must also keep the left edge in proportion. stretch is enough to show that SC050 matters as feeder/context, but not enough to justify a non-contiguous `SC050-SC052` production unit. The SC057 note should remain separate rather than being absorbed here by implication.

Pre-umlaut bridge and W-loss

Sound-change report

Historical formulation This is a short adjacent chronological bridge report, not a major textbook chapter. It keeps two neighboring ordinary FST changes visible between the SC052 OE Velar Palatalization hinge report and the `SC055-SC056` umlaut-core report without pretending that SC053 and SC054 form one strong traditional chapter.

SC053 OE Post Velar W Loss is the weaker and more residual member. It is a narrow ngw > *ng* cleanup rule with only thin comparative anchoring in derivations such as singwan > singan (Ringe and Taylor 2014, sect. 6.4.2). SC054 OE W Loss Before I is the stronger positive member. The handbook tradition does support loss of *w* before unstressed *i*, especially in the sea derivation from earlier saiwi- / sawi- to OE sǣ (Campbell 1959, sect. 406; Ringe and Taylor 2014, sect. 6.7.1; Luick 1914--1940, sect. 187). The pair therefore stays together for practical chronological reasons, not because the sources present a standard `SC053-SC054` chapter.

Source tradition The source support for SC053 is thin but usable. Ringe and Taylor explicitly derive PGmc singwan to OE singan, which is enough to support CAPR's narrow ngw > *ng* simplification (Ringe and Taylor 2014, sect. 6.4.2). That is a legitimate comparative anchor, but it does not make SC053 a large handbook chapter in its own right. The rule reads more naturally as residual bridge material inside the pre-umlaut zone.

SC054 is historically more legible. Campbell gives the classic handbook warning that Old English often loses *w* before *i*, while analogy can restore the glide in parts of the paradigm (Campbell 1959, sect. 406). Ringe and Taylor present the same process in a cleaner derivational sequence, deriving sea from earlier saiwiz > sawi > sei > sǣ (Ringe and Taylor 2014, sect. 6.7.1). Luick supports the same trajectory with *sa* / sǣ from sāwi- < saiwi- (Luick 1914--1940, sect. 187). That gives SC054 a real handbook footing, though still a narrow one.

The pair as a pair remains practical rather than doctrinal. The sources support SC054 far more strongly than SC053, and none of them singles out exactly this adjacent two-rule unit as a major chapter. The honest historical claim is therefore modest: SC053 belongs here because every ordinary FST change needs explicit prose, while SC054 belongs here because it has real source-backed chronology within the same local stretch of the cascade.

CAPR implementation CAPR keeps both rules sharper than the handbook categories.

SC053 *OEPostVelarWLoss* deletes *w* in ngw clusters:

```
*w -> 0 || *n *g _
```

That is exactly the kind of narrow cleanup rule that the model needs but the handbooks do not usually headline separately. The internal rule comments also make the restriction explicit by excluding post-vocalic *gw* material such as the etyma behind snow and swallow.

SC054 *OEWLossBeforeI* deletes non-word-initial *w* before unstressed final *i*:

```
*w -> 0 || EnglishStarVocalic _ *i .#.
```

Here the model aligns more directly with the handbook tradition. The sea derivation is the clearest example, and the ordering comment in the live rule is historically useful: this loss must remain before the umlaut core so that the preceding vowel can continue into the later ǣ outcome rather than preserving a too-late glide.

Place in the cascade This unit belongs immediately after SC052 OE Velar Palatalization and immediately before the `SC055-SC056` umlaut-core report. That is the main reason to treat the pair together: both changes are adjacent, both are ordinary FST changes, and both need explicit prose in the volume.

SC054's later relation to SC063 OE High Vowel Apocope should be handled by cross-reference only. SC063 remains the later report where high-vowel apocope is discussed in its own right; this bridge report only explains why the glide-loss rule must already be in place to the left of that later chapter.

Order evidence The order evidence is sharply asymmetric. SC053 has **no positive first-break boundary in either tested direction**. On the earlier side, the runner reaches order *13* and then stops at bundled *EarlyEnglishLineChanges* with no real break. On the later side, it reaches the current safe boundary at SC087 with no real break. Those are boundary-limited negative results, not hidden chronology claims. This report therefore does not claim that SC053 must follow any specific earlier historical stage or precede any specific later one.

SC054 is the stronger member. It must follow SC020 PGmc Final Z Deletion: if the rule is moved earlier, the sea derivation leaves sǣw instead of expected sǣ. It must also precede SC063 OE High Vowel Apocope: if the rule is moved later than SC063, the same witness again yields sǣw rather than sǣ. Both positive boundaries are real, but both are narrow because they depend on the same witness.

Taken together, the cards support `SC020 < SC054 < SC063`, while SC053 remains card-negative residual bridge material. They do not support an internal reciprocal chronology claim for `SC053-SC054` as a pair.

Interpretation This report is useful because it stays in proportion to the evidence. SC053 is present because every ordinary FST change needs explicit prose somewhere in the book, and because its narrow ngw > *ng* cleanup still has a plausible comparative anchor in singan. SC054 is present because it has stronger source backing and a real, if witness-limited, two-sided chronology profile centered on sea.

That makes the pairing practical and chronological rather than traditional. The book does not need to pretend that SC053 and SC054 form a standard handbook chapter. It only needs to keep them visible, in order, and properly situated between the SC052 hinge report and the `SC055-SC056` umlaut core.

Remaining cautions SC053 should not be made stronger than it is. Its earlier and later card results are boundary-limited negative evidence, not positive chronology claims. SC054 should not be allowed to sprawl into a non-contiguous `SC054-SC063` chapter just because its later boundary happens to be SC063. That rightward relation belongs in cross-reference, not in a larger chapter claim.

More broadly, this report should not duplicate the `SC055-SC056` umlaut-core report on the right or the SC063 **High-vowel apocope** report farther right. Its purpose is narrower: to keep a weak residual member and a narrow positive member visible in strict chronological order without inflating either one into a major chapter.

Umlaut core and palatal-diphthongization follower

Sound-change report

Historical formulation SC055 OE I Umlaut is one of the major handbook sound changes of Old English: a following i or j fronts back vowels and raises parts of the front-vowel system, leaving a large morphological and lexical footprint (Campbell 1959, sects 190–197; Hogg 1992, 111–14; Luick 1914--1940, sects 181–183; Ringe and Taylor 2014, sects 6.6.1–6.6.4; Fulk 2018, sect. 4.7). SC056 **OE Ws Palatal Diphthongization** is real as well, but it is narrower: a West Saxon palatal-triggered diphthongization zone on the right edge of the umlaut neighborhood rather than a second coequal handbook headline (Campbell 1959, sect. 39; Hogg 1992, 104–5; Ringe and Taylor 2014, sect. 6.5.1; Fulk 2018, sect. 4.13).

Source tradition The source tradition is clear about the chapter's center. Campbell, Hogg, Luick, Ringe and Taylor, and Fulk all treat i-umlaut as a large and central Old English development (Campbell 1959, sects 190–197; Hogg 1992, 111–14; Luick 1914--1940, sects 181–183; Ringe and Taylor 2014, sects 6.6.1–6.6.4; Fulk 2018, sect. 4.7). The same sources also recognize West Saxon palatal diphthongization, but they do not present it in the same way. Campbell and Hogg treat it in separate palatal-vowel discussions rather than folding it into the main i-mutation chapter (Campbell 1959, sect. 39; Hogg 1992, 104–5). Ringe and Taylor and Fulk are the most important cautions here: both place palatal-triggered diphthongization differently in the broader textbook chronology, so SC056 should not be narrated as if the handbook tradition simply states "after i-umlaut comes palatal diphthongization" (Ringe and Taylor 2014, sects 6.5.1, 6.6.1–6.6.4; Fulk 2018, sects 4.7, 4.13).

CAPR implementation CAPR makes that distinction explicit by keeping two adjacent rules. *OEIUmlaut* handles the broad fronting and raising effects conditioned by following *i/j.* OEWsPalatalDiphthongization* then handles a narrower West Saxon palatal-triggered diphthongization zone. This is useful book material because it lets the model isolate the local right edge of the umlaut core. But the local CAPR relation `SC055 < SC056` should still be presented as a technical ordering result, not as a verbatim restatement of the broad handbook chronology.

Place in the cascade This report belongs immediately after SC052 OE Velar Palatalization and, in the present assembly, after the `SC053-SC054` pre-umlaut bridge report. On its right, it precedes later material, especially SC063 **OE High Vowel Apocope** and the `SC066-SC068` syncope corridor, both of which depend on the earlier umlaut setting already being in place. That position gives the chapter a useful book role: it is the early Old English vowel core that stands between palatalization and the later weak-tail reductions.

Order evidence The chronology cards give this report a tight local spine. SC055 must follow SC052: the key witnesses are cow and lung, where moving umlaut earlier than velar palatalization yields over-palatalized outcomes. SC055 must also precede SC056, and SC056 must follow SC055: on both sides, the key witnesses are *gift and* sheath, whose derivations collapse into the wrong diphthongal forms if the order is reversed. That reciprocal local evidence is the main reason the pair is promotable as a chapter. The important limit is on SC056's later side: the card finds no real later break before the current runner boundary at SC087, and that negative result must not be rewritten as a positive claim that SC056 must precede SC087.

Interpretation This is the first central Old English vowel-change report in the production layer. Its value is not merely that i-umlaut is famous, but that CAPR can now anchor that major handbook rule with a small but real local chronology network. The chapter should therefore keep the asymmetry visible: SC055 is the historical center, while SC056 is a narrower West Saxon right-edge follower. That is exactly the kind of case where the book can show how CAPR's local evidence sharpens a standard historical chapter without reducing the whole chapter to a handful of witness forms.

Remaining cautions SC056 is still narrower and more dialect-specific than SC055, so the prose should not grant it equal historical weight. More broadly, the broad textbook chronology and CAPR's local chronology do not coincide perfectly, especially in sources such as Ringe and Taylor and Fulk that place palatal-triggered diphthongization differently in the wider sequence (Ringe and Taylor 2014, sects 6.5.1, 6.6.1–6.6.4; Fulk 2018, sects 4.7, 4.13). `SC053-SC054` should remain a separate bridge report rather than being absorbed into the umlaut core, and the SC057 note should remain separate from both this chapter and the `SC049-SC052` palatalization/fronting cluster. The chapter should also stay narrow: it is a report on the umlaut core and its West Saxon right edge, not a general handbook article on every aspect of Old English i-mutation.

OE J Cluster Coalescence note

Sound-change report

Historical formulation SC057 *OEJClusterCoalescence* appears here as a **short finished singleton note**. It belongs to the broader palatalization and fronting neighborhood, but it is not the center of that region and should not be merged back into the `SC049-SC052` or `SC055-SC056` reports. The honest final form is a short one-sided note: historically legible, explicitly placed, but narrower than the major palatalization and umlaut chapters around it.

Source tradition The source tradition supports SC057 only as part of the broader palatalization/fronting zone rather than as a major handbook chapter of its own. Campbell, Ringe and Taylor, and Fulk all treat the same neighborhood of palatalized and fronted outcomes that underlies forms such as bīeġan and sēċan (Campbell 1959, sect. 120, §170; Ringe and Taylor 2014, sects 6.4.1, 6.5.1, 6.6.1–6.6.4; Fulk 2018, sects 4.7, 4.13).

That is enough support for a short finished note. It is not enough to make SC057 a coequal companion to the SC051, SC052, or `SC055-SC056` reports, which is exactly why the prose should remain brief and structurally restrained.

CAPR implementation CAPR isolates this material as one explicit step: SC057 *OEJClusterCoalescence*. That narrower formulation is useful because it keeps a real later palatalization-side rule visible in strict order without pretending that the whole broader palatalization/fronting region forms one chapter.

Place in the cascade This note belongs immediately after the `SC055-SC056` **Umlaut core and palatal-diphthongization follower** report and immediately before the SC058 OE Nasal Dissimilation residual note. Its meaningful positive chronology link points outward on the left to the SC052 **Velar palatalization hinge**, but that relation should remain a cross-reference rather than a reason to merge non-adjacent chapters.

Farther right, the next stronger seam belongs to SC059 back mutation. That broader context matters for orientation, but SC057 itself stays a short palatalization-side note rather than a bridge chapter.

Order evidence SC057 has one real positive chronology edge. It must follow SC052. If OE J Cluster Coalescence is moved before the velar-palatalization hinge, PGmc báugijaną yields bēaġan rather than expected OE bīeġan, and PGmc sōkijaną yields sōċan rather than expected sēċan. Related forms such as follow, hedge, and singe fail in the same broader palatalization/fronting way.

The later side is different. The current runner finds no real later break for SC057 before the SC087 boundary. That result is runner-bounded and must not be rewritten into a claim that SC057 must precede SC087.

Interpretation SC057 belongs here because the final volume should not leave ordinary sound changes in placeholder form where a short honest note will do. The right final treatment is therefore explicit but modest: keep the real SC052 anchor visible, keep the later side non-positive, and let the note stand on its own instead of forcing it back into a larger palatalization chapter.

Remaining cautions The note should stay narrow. SC052 remains a leftward cross-reference only, not a reason to recreate a non-contiguous `SC049-SC052` chapter. the `SC055-SC056` umlaut core likewise remains separate. And the runner-bounded later side must not be turned into a positive historical boundary.

OE Nasal Dissimilation residual note

Sound-change report

Historical formulation SC058 *OENasalDissimilation* appears here as a **short finished residual note**. It is included because every ordinary sound change needs final prose in the final volume, not because the current evidence makes it a strong chronological hinge. The right final form is therefore explicitly residual and contextual.

Source tradition The source tradition is thinner here than for the neighboring palatalization and umlaut rules. That is itself part of the reason the prose should stay short. SC058 belongs in the book because it is an ordinary modeled sound change, but the current historical profile does not justify a larger chapter claim.

CAPR implementation CAPR isolates this residual material as one explicit step: SC058 *OENasalDissimilation*. That explicitness is enough to justify a short finished note even though the rule does not currently recover any positive first-break chronology of its own.

Place in the cascade This note belongs immediately after the SC057 **OE J Cluster Coalescence note** and immediately before the SC059 **OE back mutation** report. It remains a short residual/context note in that position rather than a coequal partner to either neighboring chapter.

Order evidence Current testing does not identify a positive historical first-break boundary for SC058 in either direction. On the earlier side, the search reaches bundled *EarlyEnglishLineChanges*; that is a methodological runner limit, not a detected historical boundary. On the later side, the search reaches the current SC087 boundary with no real break; that is a no-break-before-boundary result bounded by the present search space, not a claim that SC058 must precede SC087.

This means the note records a boundary-limited computational result rather than a positive local chronology claim.

Interpretation SC058 belongs here because residual completion requires final prose even for the weakest ordinary rules. The honest final treatment is therefore very short: keep the rule explicit, state clearly that the current chronology is boundary-limited in both directions, and avoid inflating that limited evidence into a larger chapter.

Remaining cautions The methodological boundaries matter here. Bundled *EarlyEnglishLineChanges* and the current SC087 boundary are search limits, not historical anchors for SC058. This note should therefore stay brief and should not be read as a coequal partner to the SC057 note or the SC059 back-mutation report.

OE back mutation

Sound-change report

Historical formulation SC059 *OEBackMutation* appears here as a **short singleton back-mutation report**. It stands apart from the older grouped `SC059-SC061` version because the evidence now shows that SC059 is the clear source-backed center of that seam. The historical point is narrower than the old grouped bridge: back mutation or back umlaut is a real Old English vowel-development zone in the handbooks, but that does not justify pulling SC060, SC061, or the later weak-tail cluster into one shared chapter (Campbell 1959, sect. 207; Hogg 1992; Ringe and Taylor 2014, sect. 6.9.4; Fulk 2018, sect. 4.8).

Source tradition The source tradition is strong enough to support a short standalone note. Campbell treats back mutation directly as a later diphthongizing development before following back vowels and illustrates why forms such as heofon are historically legible outcomes rather than model artifacts (Campbell 1959, sect. 207). Hogg likewise treats back mutation as a real later change parallel in type to breaking, even if narrower and more restricted in West Saxon (Hogg 1992). Ringe and Taylor sharpen the comparative picture by discussing back umlaut as a distinct later zone and by citing non-West-Saxon forms such as geofad beside West Saxon giefan and weofan beside West Saxon wefan (Ringe and Taylor 2014, sect. 6.9.4). Fulk reinforces the same point by treating back mutation as a separate historical phenomenon rather than a mere side effect of earlier umlaut (Fulk 2018, sect. 4.8).

That is enough support for explicit prose. The report should stay focused on back mutation itself, not inflate SC059 into a general weak-tail chapter and not re-create the old grouped `SC059-SC061` bridge under another name.

CAPR implementation CAPR isolates this historical zone as one explicit step: SC059 *OEBackMutation*. That rule introduces the back-mutated diphthongal outcomes that appear in forms such as ġiefan, stelan, and wefan when the rule is misordered. This is more specific than the handbook phrasing, but that specificity is exactly why the singleton report is useful. It lets the book mark the real center of the seam between the post-brightening corridor on the left and the later weak-tail region on the right without pretending those regions belong to the same chapter.

Place in the cascade This report belongs immediately after the SC058 OE Nasal Dissimilation residual note and immediately before the SC060 **WS palatal umlaut note**. Its real chronology edges point outward rather than architecturally inward: leftward to the earlier `SC046-SC048` **Restoration and nasal-tail corridor** through SC048, and rightward to the split late-tail region through the standalone `SC078` **Weak-tail reduction right-edge note**.

That placement matters because SC059 now serves as the real center of the seam while leaving SC060 and SC061 visible as separate short notes in strict chronological order.

Order evidence The chronology card gives SC059 a real two-sided window: `SC048 < SC059 < SC078`.

On the earlier side, SC059 must follow SC048. If OE Back Mutation is moved before the restoration and nasal-tail corridor reaches SC048, PGmc gébaną yields ġeofan rather than expected OE ġiefan, and PGmc stélaną likewise yields steolan rather than stelan. That earlier edge is historically real, but it belongs as a leftward cross-reference to the `SC046-SC048` report rather than as a reason to merge the two units.

On the later side, SC059 must also precede SC078. If OE Back Mutation is delayed into the later weak-tail region, PGmc stélaną again yields steolan instead of stelan, and PGmc wébaną yields weofan instead of wefan. That later edge is also historically real, but it is broad/far and should remain a rightward cross-reference to the split late-tail region rather than a reason to expand this note into `SC078` or the larger late weak-tail sequence.

Taken together, the cards support a genuine singleton center with two outward links: `SC048 < SC059 < SC078`.

Interpretation SC059 belongs here because the modest reading is now the best one. The evidence makes clear that back mutation is the only member of the old bridge with both strong source support and real two-sided chronology. That makes it worth stating explicitly in prose, while SC060 and SC061 remain as one-sided companion notes rather than coequal chapter centers.

This is therefore an architectural singleton. It marks the true center of the late vowel seam without turning neighboring cross-references into a larger chapter claim and without claiming that the whole weak-tail region has already been resolved into final prose.

Remaining cautions The cautions are structural as much as descriptive. SC059 should not be merged backward into the `SC046-SC048` corridor just because `SC048 < SC059` is real. It should not be expanded forward into SC078 or the larger split late-tail region just because `SC059 < SC078` is real. And the report should stay focused on back mutation itself rather than broadening into a general article on the whole later weak-tail region. The point is precise: isolate the source-backed center, keep the outward links as cross-references, and leave SC060 and SC061 as separate notes.

WS palatal umlaut note

Sound-change report

Historical formulation SC060 *OEWsPalatalUmlaut* appears here as a **short finished singleton note**. It is historically legible through the might / night material, but its chronology remains one-sided. The right final form is therefore a short singleton note in strict order, not a reason to merge the rule back into the umlaut core.

Source tradition The source tradition is narrower than for the main umlaut chapter but still real enough for final prose. Campbell and Ringe and Taylor both support the historical development behind forms such as miht and niht, while Fulk's broader chronology cautions help keep the rule tied to the umlaut and palatal-vowel region rather than treated as a coequal chapter center (Campbell 1959, sects 248–251; Ringe and Taylor 2014, sects 6.5.1, 6.6.1–6.6.4; Fulk 2018, sects 4.7, 4.13).

That support is enough for a short finished note. It is not enough to justify folding SC060 back into the `SC055-SC056` report.

CAPR implementation CAPR isolates this narrower West Saxon material as one explicit step: SC060 *OEWsPalatalUmlaut*. That explicit rule is analytically useful because it keeps a real one-sided later note visible after back mutation without pretending it anchors a larger chapter.

Place in the cascade This note belongs immediately after the SC059 OE back mutation report and immediately before the SC061 **Weak-tail nasal-loss note**. Its meaningful positive chronology link points outward on the left to the `SC055-SC056` umlaut core, but that relation should remain a cross-reference rather than a reason to merge chapters.

Order evidence SC060 has one real positive historical boundary. It must follow SC055. If OE WS Palatal Umlaut is moved before the umlaut core, PGmc máxtiz yields mieht rather than expected OE miht, and PGmc náxti yields nieht rather than expected niht.

The later side is different. The current runner finds no real later break for SC060 before the SC087 boundary. That result is runner-bounded and must not be rewritten into a claim that SC060 must precede SC087.

Interpretation SC060 belongs here because the final volume should not leave one-sided but historically legible ordinary rules in placeholder form when a short finished note will do. The correct final treatment is modest: keep the real leftward umlaut anchor explicit, keep the later side non-positive, and leave the main umlaut chapter where it already belongs.

Remaining cautions The note should stay narrow. The `SC055-SC056` umlaut core remains a leftward cross-reference only, not a destination for merger. And the runner-bounded later side must remain a methodological limit rather than a positive historical boundary.

Weak-tail nasal-loss note

Sound-change report

Historical formulation SC061 *OEWeakTailNasalLoss* appears here as a **short finished singleton note**. It is historically legible through the *do* / dōn pathway, but its chronology remains one-sided. The right final form is therefore a short singleton note in strict order, not a reason to merge the rule back into SC023 or to widen it into a broader weak-tail chapter.

Source tradition The source tradition is modest but real. Ringe and Taylor discuss the development of dōn and the surrounding reduction of older morphological material, which is enough to keep the rule historically legible even though it does not rise to a larger textbook chapter of its own (Ringe and Taylor 2014).

That is enough support for a short finished note. It is not enough to justify merging SC061 backward into the SC023 singleton or outward into a larger weak-tail chapter.

CAPR implementation CAPR isolates this material as one explicit step: SC061 *OEWeakTailNasalLoss*. That narrower implementation is useful because it keeps the dōn pathway explicit in strict order even though the rule remains one-sided in current chronology testing.

Place in the cascade This note belongs immediately after the SC060 **WS palatal umlaut note** and immediately before the SC063 **OE high vowel apocope** report. Its meaningful positive chronology link points outward on the left to the SC023 **N-stem n-loss note**, but that relation should remain a cross-reference rather than a reason to merge non-adjacent chapters.

Order evidence SC061 has one real positive historical boundary. It must follow SC023. If OE Weak Tail Nasal Loss is moved before the SC023 singleton note, PGmc dōną no longer yields expected OE dōn; the earlier-shifted derivation records no output at all.

The later side is different. The current runner finds no real later break for SC061 before the SC087 boundary. That result is runner-bounded and must not be rewritten into a claim that SC061 must precede SC087.

Interpretation SC061 belongs here because residual completion requires final prose even for one-sided ordinary rules. The honest final treatment is therefore a short note: keep the real SC023 anchor explicit, keep the later side non-positive, and let the rule remain a small but visible part of the weak-tail sequence.

Remaining cautions The note should stay narrow. The SC023 report remains a leftward cross-reference only, not a merger target. The later side remains runner-bounded and must not be turned into a positive historical boundary. And the note should not be broadened into a larger weak-tail chapter.

High-vowel apocope

Sound-change report

Historical formulation SC063 is OE High Vowel Apocope: final high vowels *i*, *u*, and *ų* are lost after heavy syllables and in the relevant trisyllabic configurations, but not simply after every short stressed syllable. The basic historical claim is stable across the handbook tradition even though older scholarship often folds it into a broader account of unstressed-vowel loss rather than treating it as a single named law (Luick 1914--1940, sects 304–308; Campbell 1959, sects 345–349; Hogg 1992, 120; Ringe and Taylor 2014, sects 6.8.1, 6.8.4; Fulk 2018, sect. 5.6).

Source tradition Luick gives the clearest older formulation: final `i/u` disappear after heavy syllables and in the relevant trisyllabic environments, but not immediately after a short stressed syllable. Campbell offers the most compact section-safe handbook description of final unaccented high-vowel loss, while Hogg keeps apocope and medial syncope conceptually adjacent without collapsing them into one process. Ringe and Taylor sharpen the chronology by placing apocope after general syncope and among the last prehistoric Old English sound changes, and Fulk adds a useful caution about inflectional exceptions such as Mercian *-u* retention (Luick 1914--1940, sects 304–308; Campbell 1959, sects 345–349; Hogg 1992, 120; Ringe and Taylor 2014, sects 6.8.1, 6.8.4; Fulk 2018, sect. 5.6).

CAPR implementation CAPR makes the rule far more explicit than a handbook formulation. The live implementation separates heavy disyllabic branches, long- and short-diphthong handling, heavy-by-position branches, trisyllabic branches, vowel-hiatus after long vowels, and final *x*/*h* environments. The historical core is still the classical one: weight-sensitive and trisyllabic loss of final high vowels. The special hiatus and final-*x* clauses are better read as technical refinements inside the model than as the central historical claim (Campbell 1959, sects 238, 345–346; Ringe and Taylor 2014, sect. 6.6.1).

Place in the cascade SC063 belongs to a late weak-tail neighborhood that includes `SC054` OE W Loss Before I, `SC055` OE I Umlaut, `SC063` OE High Vowel Apocope, and `SC072` OE Unstressed Long Vowel Shortening. This placement matters because the rule must come late enough to preserve umlaut-sensitive developments on the left, yet still early enough to precede the last cleanup of unstressed long vowels on the right. It is therefore one decisive weak-tail reduction, but not the whole late weak-tail story by itself.

Order evidence The chronology card gives SC063 a safe window of *56-71* and fixes the main anchors on both sides. `SC055 < SC063` is the most important left boundary: if apocope is moved earlier than i-umlaut, forms such as cȳ and brȳd collapse to cū and brūd because the final high vowel disappears before it can condition umlaut. `SC054 < SC063` is a narrower left-side anchor in the same late cluster. On the right, `SC063 < SC072` is enforced by forms such as fyrhte: if later unstressed-long-vowel shortening moves ahead of apocope, the derivation yields fyrht instead. Taken together, the evidence shows that SC063 is late, but not final, inside the weak-tail sequence.

Interpretation For book purposes, SC063 should be presented as a decisive but not terminal late weak-tail reduction. The literature strongly supports the classical rule: final high vowels are lost after heavy syllables and in the relevant trisyllabic patterns, and this process is closely related to but not identical with medial syncope. CAPR then shows how much technical branching is needed to implement that apparently simple rule across a large derivational corpus. The historical chapter should therefore keep the core claim central and treat the detailed formal branches as model-level elaboration rather than as separate historical laws.

Remaining cautions The line between apocope and medial syncope still needs careful handling in the prose, because the neighboring processes are related but not identical (Hogg 1992, 120; Ringe and Taylor 2014, sect. 6.8.1). The trisyllabic branch is the most delicate part of the rule, and Mercian *-u* retentions such as lytelu and nētenu warn against overgeneralizing every branch of the implementation (Fulk 2018, sect. 5.6). The special final-*x* and vowel-hiatus clauses are useful model refinements, but they should not be mistaken for the whole historical claim (Campbell 1959, sects 238, 345–346; Ringe and Taylor 2014, sect. 6.6.1). More broadly, the chapter should avoid turning every FOMA branch into a separate historical assertion.

Post-apocope tail

Sound-change report

Historical formulation This is a short adjacent chronological bridge report, not a major symmetrical chapter. It keeps two immediate post-apocope rules visible in final book prose without pretending that both are equally strong or equally chapter-like.

SC064 NWGmc In Stem N Loss is the narrower and stronger member. In CAPR it is centered on the fright / furht- witness family, where the weak-tail derivation yields OE fyrhte. SC065 OE Medial Syncope is different in kind: it represents a narrow CAPR slice of the broader medial-syncope material that follows the main apocope zone. The pair stays together because the two changes are adjacent and occupy the immediate post-SC063 tail before the `SC066-SC068` syncope/degemination corridor.

Source tradition The source tradition strongly supports the **broader late weak-tail environment** around this unit. Campbell, Hogg, Ringe and Taylor, Luick, and Fulk all support a post-apocope zone in which additional medial reduction, cluster pressure, and weak-tail cleanup remain active (Campbell 1959, sects 345–349, 388–389; Hogg 1992, 120–21; Ringe and Taylor 2014, sects 6.7.3–6.8.4; Luick 1914--1940, sects 304–306; Fulk 2018, sect. 5.6). Kroonen adds the key lexical support for the inherited furht- family behind the fright witness that carries SC064's current positive evidence (Kroonen 2013, 201).

What the sources do not strongly support is a standard textbook pair made of exactly SC064 plus SC065. SC064 looks more like a narrow lexical bridge note than a handbook headline, while SC065 maps more naturally onto broad discussions of medial syncope than onto CAPR's precise pre-dental rule. The honest source claim is therefore asymmetrical: the post-apocope setting is well supported, the fright family is real, but the pair itself is a practical chronological bridge report rather than a traditional two-rule chapter (Campbell 1959, sects 345–349, 388–389; Hogg 1992, 120–21; Ringe and Taylor 2014, sects 6.7.3–6.8.4; Luick 1914--1940, sects 304–306; Fulk 2018, sect. 5.6; Kroonen 2013, 201).

CAPR implementation CAPR makes both rules more specific than the handbook categories.

SC064 *NWGmcInStemNLoss* deletes stem-final *n* after ī in the narrow environment represented by the fright derivation. SC065 *OEMedialSyncope* deletes medial *i* before a following dental after a heavy syllable. Those are useful explicit rules in the model, but they are not simple one-to-one matches for ordinary grammar chapter labels. SC064 is especially narrow and witness-driven. SC065 is better read as one carved-out formal slice of the broader late medial-syncope tradition than as the whole historical phenomenon (Hogg 1992, 120–21; Ringe and Taylor 2014, sects 6.7.3–6.8.4).

Place in the cascade This unit belongs immediately after SC063 OE High Vowel Apocope and immediately before the `SC066-SC068` **Syncope and Degemination Corridor**. That is the main reason to keep the pair together: it supplies a short post-apocope bridge in strict chronological order rather than leaving the weak-tail aftermath undescribed.

SC072 OE Unstressed Long Vowel Shortening matters here only as a later cross-link for SC064. It is not the next assembled section, and it should be handled by cross-reference rather than by stretching this bridge report beyond its local chronological place.

Order evidence The order evidence is real but sharply uneven. SC064 has positive boundaries on both sides: it must follow SC041 PWGmc Final Bare A Loss and precede SC072 OE Unstressed

Long Vowel Shortening. Both of those boundaries are carried by the same representative failure, fright, where shifting the rule yields fyrhten instead of expected fyrhte. That makes SC064 historically meaningful, but narrow.

SC065 is different. Its card has **no positive first-break boundary in either direction**. The earlier side stops only at the bundled *EarlyEnglishLineChanges* runner boundary, and the later side reaches the current SC087 runner boundary with no real break. Those runner-boundary results must not be treated as historical constraints. This report therefore does not claim that SC065 must follow a specific earlier rule or must precede SC087.

Interpretation This report is valuable because it gives explicit book prose to two adjacent FST changes without inflating their status. SC064 is the stronger, narrower member: it has real chronology evidence, but that evidence is concentrated in one inherited witness family. SC065 is the weaker, more contextual member: it is included because every ordinary FST change needs explicit prose somewhere in the book, and because it belongs in the immediate post-apocope weak-tail region even though its current chronology card is negative.

That is exactly what a cautious bridge report is for. It shows how weak, narrow, or model-specific rules can still be described honestly inside the chronological cascade: not omitted, not overstated, and not forced into a larger neighboring chapter that would blur their actual evidence profile.

Remaining cautions The prose must not overstate the fright evidence for SC064. Both positive boundaries are real, but both are carried by the same witness family. SC065 must not be turned into a positive chronology anchor; its card remains negative and boundary-limited. The chapter should not duplicate the SC063 **High-vowel apocope** report on the left or the `SC066-SC068` **Syncope and degemination corridor** on the right.

More broadly, the report should not imply that the handbooks present `SC064-SC065` as a standard paired chapter. The point is narrower: both ordinary FST changes now receive explicit prose in their proper chronological place, with SC064 treated as the narrow positive-evidence member and SC065 treated as card-negative contextual syncope material.

Syncope and degemination corridor

Sound-change report

Historical formulation SC066 OE L Adjacent Syncope, SC067 OE Dental Assimilation, and SC068 OE Preconsonantal Degemination belong to a late Old English corridor in which medial syncope creates new consonant clusters and later cleanup reduces their weight. The historical core is therefore not three equally standard handbook laws, but one narrower sequence: medial syncope is strongly supported, downstream cluster simplification is moderately supported, and CAPR's exact three-rule segmentation is supported only unevenly (Campbell 1959, sects 345–349, 388–389; Hogg 1992, 120–21; Ringe and Taylor 2014, sects 6.7.3–6.7.5, 6.8.1–6.8.2; Luick 1914--1940, sects 304–307; Brunner 1965, sects 157–159).

Source tradition Campbell gives the most compact Old English statement of the main historical core: medial unaccented vowels are lost after short syllables, especially where the result is a consonant + `l`/`r` cluster, which makes him especially useful for SC066 (Campbell 1959, sects 388–389). Hogg is valuable at the broader structural level, since he keeps apocope, syncope, and the simplification of over-heavy consonant groups in one connected late weak-tail story (Hogg 1992, 120–21). Ringe and Taylor provide the clearest staged formulation for this report: they distinguish general syncope from later apocope, treat nettle as a syncope type, and state explicitly that geminates are simplified next to another consonant, which makes them especially useful for spindle (Ringe and Taylor 2014, sects 6.7.3–6.7.5, 6.8.1–6.8.2). Luick supports the broader late weak-tail chronology by separating final high-vowel loss from later medial syncope (Luick 1914--1940, sects 304–307). Sievers-Brunner is particularly useful for the netle / netele type and for the idea that fuller forms can reflect later secondary-vowel restoration rather than the primary older stage (Brunner 1965, sects 157–159). Fulk supports the narrower chronology claim that this syncope is later than i-umlaut, but it is better used as background for timing than as the main descriptive source for the corridor itself (Fulk 2018, sect. 5.6).

CAPR implementation CAPR sharpens this broader historical sequence into three adjacent rules. SC066 deletes medial *i* before *l* after a stressed syllable; SC067 deletes *θ* after *t* in a local dental-cleanup environment; SC068 simplifies *tt* or *nn* before a following sonorant. That articulation is useful in the model because it makes the cleanup logic explicit, but the book should not imply that the handbooks routinely isolate SC067 as a coequal sound law. SC067 is better read as a model-internal bridge or cleanup step between the stronger historical poles of syncope and post-syncope cluster simplification (Hogg 1992, 120–21; Ringe and Taylor 2014, sects 6.7.3–6.7.5, 6.8.1–6.8.2).

Place in the cascade This corridor belongs to the late weak-tail zone. It follows SC063 **OE High Vowel Apocope** and the immediate post-apocope tail `SC064-SC065`, and it precedes the split late-tail region (`SC069`, `SC070-SC071`, `SC072-SC073`, `SC074-SC075`, `SC076`, `SC078`). That placement is important because the report should remain a narrow corridor rather than a proxy for the whole late-tail zone: it isolates one specific sequence in which later weak-tail reduction produces clusters that then need local cleanup.

Order evidence The chronology cards support the corridor most clearly at its edges. SC066 must follow SC055 OE I Umlaut; the key failures are nettle and spindle, because moving syncope earlier destroys the umlauted vocalism. SC066 must also precede SC068; here spindle is again the crucial witness, since delaying syncope past degemination leaves spinnl instead of spinl. SC067 currently has no positive first-break boundary in either direction, so it should not be presented as a strong chronology anchor in its own right. The clearest positive local claim is therefore `SC066 < SC068`, not a perfect three-link reciprocal chain.

Interpretation For book purposes, this report shows that late weak-tail reduction is not only a story of vowel loss. Once syncope removes medial material, the derivation can inherit clusters that require cleanup, and CAPR makes that logic visible by keeping the cleanup stages explicit. The useful historical claim is therefore two-layered: the literature strongly supports late medial syncope and gives moderate support to downstream simplification of the clusters it creates, while CAPR adds a more segmented internal articulation so the local ordering can be tested. The chapter should make that distinction explicit rather than blurring literature-backed phonology and model-specific segmentation.

Remaining cautions The narrative should not give all three rules equal weight. SC067 may remain no more than a bridge note inside the corridor, and SC068 is probably best treated as subordinate to the syncope narrative rather than as an independent headline. *Spindle* is an excellent local witness, but it currently carries much of the right-edge order argument by itself, so the prose should not overclaim a wider evidential base than the cards actually provide. More broadly, the final title should stay modest: this is a narrow late weak-tail syncope-and-cleanup report, not proof that the handbook tradition recognizes three coequal named laws.

Early o-shortening context note

Sound-change report

Historical formulation SC069 *OEEarlyOShortening* appears here as a **short finished late-tail opener/context note**. The change is historically legible as part of the larger late unstressed-vowel cleanup, but it is not the beginning of the real local late-tail core. The honest final treatment is therefore contextual and architectural: keep the rule visible in strict order without inflating it into a chapter center.

Source tradition The source tradition supports the broader late unstressed-vowel region more strongly than it supports SC069 as a major handbook center. Campbell, Hogg, Ringe and Taylor, Luick, and Fulk all treat the larger zone of apocope, syncope, shortening, and later weak-tail cleanup as historically real (Campbell 1959, sects 345–354, 388–389; Hogg 1992, 120–21; Ringe and Taylor 2014, sects 6.7.3–6.9.3; Luick 1914--1940, sects 304–307; Fulk 2018, sects 5.6–5.7).

That is enough to justify final prose for SC069. It is not enough to turn the rule into a new local late-tail core alongside the earlier bridges and center of gravity to its right.

CAPR implementation CAPR isolates this broad background as one explicit step: SC069 *OEEarlyOShortening*. That explicit rule is useful because it gives the late-tail region a visible left edge in the assembled book instead of leaving the rule implicit inside neighboring reductions.

Place in the cascade This note belongs immediately after the `SC066-SC068` **Syncope and degemination corridor** and immediately before the `SC070-SC071` **Early unstressed fronting and shortening bridge**. Its only positive outward chronology relation points far left to the SC023 **N-stem n-loss note**, but that broad/far relation should remain a distant cross-reference rather than a larger chapter claim.

The local late-tail architecture begins more credibly with the `SC070-SC071` bridge and is strongest in the `SC072-SC073` core. SC069 therefore remains the opener/context note of the region, not its hidden main chapter.

Order evidence SC069 has one real positive chronology relation, and it is broad/far. If OE Early O Shortening is moved before SC023, PGmc nḗdrōn yields nǣdran rather than expected OE nǣdre, PGmc érθōn yields eorþan rather than eorþe, and PGmc fláskōn yields flascan rather than flasce. The card records 17 failures overall, with representative examples including adder, earth, flask, heart, and line.

The later side is different. The current runner finds no real later break for SC069 before the SC087 boundary. That result is runner-bounded and must not be rewritten as an SC087 boundary claim.

Interpretation SC069 is now final prose because every ordinary sound change should appear in the volume, not because the rule has become a local chronology hinge. The right final treatment is therefore a modest context note: keep the broad/far `SC023 < SC069` relation explicit, keep the later side non-positive, and leave the real late-tail architecture to the earlier bridge and core units to the right.

Remaining cautions This note should stay contextual. Its positive relation to SC023 is too distant to organize local prose, so the book should not build a non-contiguous chapter around that edge. The runner-bounded later side must remain a methodological limit rather than a must-precede claim about SC087. And the note should not be allowed to reopen or rival the late-tail bridge/core structure.

Early unstressed fronting and shortening bridge

Sound-change report

Historical formulation This is a cautious early late-tail bridge full report centred on SC070 *OEUnstressedFrontingEarly* and SC071 *OELateOShortening*. The pair is real and adjacent, but it is not the main late-tail core. SC070 is the stronger hinge: it has a real leftward relation to the SC052 hinge and a local right edge to SC071. SC071 is the narrower follower.

That narrower reading is the honest one. The report discusses only SC070 and SC071. It does not merge backward into the SC069 opener/context note, does not merge forward into the `SC072-SC073` core, and does not treat this early bridge as a reason to reopen any of the later closing reports.

Source tradition The source tradition supports the pair as part of the broader late unstressed-vowel leveling and reduction environment, but not as one of the major handbook centers in that region. Hogg is useful because he keeps apocope, syncope, and later consonant-cluster simplification in the same late zone while still distinguishing them as separate processes (Hogg 1992, 120–21). Ringe and Taylor give the clearest staged framework for that zone, moving from general syncope and apocope into shortening of unstressed long vowels and later post-apocope developments (Ringe and Taylor 2014, sects 6.7.3–6.8.4, 6.9.1–6.9.3). Luick likewise distinguishes final loss from open-medial syncope while keeping both inside the same late unstressed-vowel environment (Luick 1914--1940, sects 304–307).

Together those sources justify a modest bridge report. They support the broader late-tail environment more strongly than they support this exact pair as a handbook chapter, which is why the prose stays smaller and more technical than the `SC072-SC073` core or the SC078 right-edge report.

CAPR implementation CAPR makes the early late-tail bridge explicit as two adjacent rules. SC070 *OEUnstressedFrontingEarly* handles the earlier unstressed fronting step, and SC071 *OELateOShortening* then resolves the later shortening outcome. That segmentation is finer-grained than most handbook presentations, but the chronology cards show why it is useful: the local center of the report is the real adjacent seam `SC070 < SC071`.

Place in the cascade This report belongs immediately after the SC069 **Early o-shortening context note** and immediately before the `SC072-SC073` **Unstressed long-vowel shortening and ae-merger core**. Its meaningful outward chronology relation points left to the SC052 **Velar palatalization hinge**, but that leftward relation should remain a cross-reference rather than a larger chapter claim.

That placement matters because `SC070-SC071` now gives the split late-tail region an explicit early bridge without displacing the stronger middle core on the right. Farther right, the `SC074-SC075` bridge and the SC076 note remain distinct, while SC078 remains the treated right-edge report rather than a co-member of this chapter.

Order evidence The local center of the report is `SC070 < SC071`.

SC070 must follow SC052 and precede SC071. If OE Unstressed Fronting Early is moved before SC052, PGmc lúnganjō yields lunġen rather than expected OE lungen. That leftward relation is real, but it should remain only a cross-reference to the SC052 report rather than a reason to merge non-adjacent chapters.

SC070 must also precede SC071. If it is moved later than SC071, PGmc búrōθi yields boreþ rather than expected OE boraþ, PGmc mḗnōθz yields mōneþ rather than expected mōnaþ, and related verbal and nominal endings fail in the same unstressed-vowel way.

SC071 directly reciprocates that left edge. If OE Late O Shortening is moved before SC070, PGmc búrōθi again yields boreþ rather than expected boraþ, and PGmc líznōθi yields liorneþ rather than expected liornaþ. The current runner finds no later real break for SC071 before the SC087 boundary. That result is runner-bounded and must not be rewritten into a positive claim that SC071 must precede SC087.

Interpretation `SC070-SC071` belongs here because it is now the strongest remaining local pair on the early side of the split late-tail region. The report remains cautious on purpose. SC070 is the stronger hinge because it has positive evidence on both sides; SC071 is the local follower and remains one-sided on the right.

This gives the late-tail architecture a clearer shape in book form: SC069 as the opener/context note, `SC070-SC071` as the early bridge, `SC072-SC073` as the compact late-tail core, `SC074-SC075` as the narrow medial bridge, SC076 as the remaining weaker context note, and SC078 as the right-edge report.

Remaining cautions The report should stay narrow. It should not absorb SC069 just because the two units are adjacent, and it should not be absorbed into the `SC072-SC073` core just because that stronger chapter follows it immediately. Its leftward `SC052 < SC070` relation should remain a cross-reference only, not a license to build a non-adjacent chapter. And SC071's runner-bounded later side must not be turned into a positive historical boundary.

Unstressed long-vowel shortening and ae-merger core

Sound-change report

Historical formulation This is a compact adjacent late-tail report centred on SC072 *OEUnstressedLongVowelShortening* and SC073 *OEUnstressedAEMerger*. The evidence shows that this pair is the strongest internal seam in the split late-tail region: it has the clearest middle core, real local reciprocity, and source support that is stronger than the flanking bridges on either side.

That stronger reading is also narrower than the broader late-tail region as a whole. The report discusses only SC072 and SC073. It does not absorb the leftward `SC064 < SC072` relation into the `SC064-SC065` bridge, does not pull the rightward `SC073 < SC085` relation into the `SC085-SC086` closing report, and does not expand into the `SC070-SC071` bridge, the `SC074-SC075` bridge, the SC076 note, or the SC078 report.

Source tradition The source tradition supports this pair as a compact late weak-tail core. Campbell, Hogg, Ringe and Taylor, Luick, and Fulk all support the broader late weak-tail environment in which syncope, apocope, unstressed long-vowel shortening, and later cleanup remain active (Campbell 1959, sects 345–354, 388–389; Hogg 1992, 120–21; Ringe and Taylor 2014, sects 6.7.3–6.8.4; Luick 1914--1940, sects 304–307; Fulk 2018, sect. 5.6). Within that broader environment, Ringe and Taylor give SC072 the clearest handbook footing by explicitly placing shortening of unstressed long vowels among the last prehistoric Old English sound changes and tying it directly to post-apocope developments (Ringe and Taylor 2014, sects 6.8.3–6.8.4).

SC073 is slightly different in source profile. It is historically real and book-useful, but it reads more as the immediate follower to SC072 than as a separate chapter center. The pair therefore works best as one compact adjacent report: SC072 carries the historical center of gravity, while SC073 sharpens the right edge and hands forward by cross-reference into the later closing region.

CAPR implementation CAPR sharpens this late weak-tail history into two adjacent rules. SC072 *OEUnstressedLongVowelShortening* shortens the relevant unstressed long vowels before the final weak-vowel outcomes settle, while SC073 *OEUnstressedAEMerger* then merges the resulting unstressed æ / *e* material. That segmentation is more explicit than most handbook prose, but the chronology cards show why both steps are needed: the local center of the report is the real adjacent seam `SC072 < SC073`.

Place in the cascade This report belongs immediately after the `SC070-SC071` **Early unstressed fronting and shortening bridge** and immediately before the `SC074-SC075` **Medial unstressed-i lowering bridge**. Its meaningful outward chronology links point left to the `SC064-SC065` **Post-apocope tail** through SC064 and right to the `SC085-SC086` **H-loss and contraction core** through SC085.

Those outward relations matter, but they remain cross-references only. `SC064 < SC072` should not pull the post-apocope tail forward into this chapter, and `SC073 < SC085` should not pull this report rightward into the closing h-loss/contraction core. The report stays strictly adjacent and chronological: only SC072 and SC073 are here.

Order evidence The local center of the report is `SC072 < SC073`.

SC072 must follow SC064 and precede SC073. If OE Unstressed Long Vowel Shortening is moved earlier than SC064, PGmc fúrxtīnaz yields fyrhten rather than expected OE fyrhte, so the leftward SC064 relation is real but should remain a cross-reference to the `SC064-SC065` report. If SC072 is moved later than SC073, broad final *-e* outcomes merge too early to -æ: PGmc nḗdrōn yields nǣdræ instead of nǣdre, and PGmc fádēr yields fædær instead of fæder, alongside the wider adder / father failure set.

SC073 then directly reciprocates the left edge. If OE Unstressed AE Merger is moved earlier than SC072, the same broad adder / father / related final- vowel set merges too soon. Its later side is real but narrower: if SC073 is moved later than SC085, PGmc táixōn yields tāæ rather than expected OE tā. That rightward `SC073 < SC085` edge remains a forward cross-reference only to the `SC085-SC086` report rather than a reason to enlarge this chapter beyond its adjacent pair.

Interpretation `SC072-SC073` belongs here because it is the clearest remaining compact late-tail core. The pair has stronger source support and cleaner internal chronology than the flanking `SC070-SC071` and `SC074-SC075` bridges, while still remaining strictly adjacent and modest in scope. This is the same editorial move that has worked elsewhere in the half: keep the compact source-backed center together, keep the real but outward relations as cross-references, and leave the weaker neighboring material separate.

Remaining cautions The report should stay focused. It should not expand backward into `SC064-SC065` just because `SC064 < SC072` is real. It should not expand forward into the `SC085-SC086` report just because `SC073 < SC085` is real. It should not absorb the `SC070-SC071` bridge, the `SC074-SC075` bridge, the SC076 note, or the SC078 report, just because they belong to the same broader late-tail region. And although SC073 has a real rightward edge to SC085, that narrower handoff should remain only a cross-reference rather than a claim that this pair and the closing core belong to one larger adjacent report.

Medial unstressed-i lowering bridge

Sound-change report

Historical formulation This is a narrow medial unstressed-i lowering bridge full report centred on SC074 *OEMedUnstressedILowering1* and SC075 *OEMedUnstressedILowering*. The pair is real and adjacent, but it is not the main late-tail core. SC074 is the stronger hinge because it has a real leftward relation to the `SC072-SC073` core and a local right edge to SC075. SC075 is the narrower follower.

That smaller reading is the honest one. The report discusses only SC074 and SC075. It does not merge backward into the `SC072-SC073` core, does not absorb the SC076 context note, and does not turn this narrow bridge into a reason to reopen the SC078 right-edge report.

Source tradition The source tradition supports the pair as part of the broader late unstressed-vowel and weak-tail environment, but not as one of its major handbook centers. Hogg is useful because he keeps apocope, syncope, and later cluster simplification in the same late zone while warning against turning every internal CAPR step into a separate large historical law (Hogg 1992, 120–21). Ringe and Taylor provide the clearest staged late framework, but their strongest emphasis falls on the shortening/apocope core rather than on this narrower medial-lowering bridge (Ringe and Taylor 2014, sects 6.7.3–6.8.4, 6.9.1–6.9.3). Luick likewise supports the broader late unstressed-vowel environment without making this pair a standalone chapter center (Luick 1914--1940, sects 304–307).

Together those sources justify keeping `SC074-SC075` explicit in prose, but only as a modest, witness-limited bridge. The broader environment is historically real; this specific pair remains smaller and less source-central than the `SC072-SC073` core or the SC078 right-edge report.

CAPR implementation CAPR makes this narrow bridge explicit as two adjacent lowering rules. SC074 *OEMedUnstressedILowering1* handles the first medial unstressed-*i* lowering step, and SC075 *OEMedUnstressedILowering* then completes the local sequence. That segmentation is finer-grained than handbook prose, but the chronology cards show why it is useful: the local center of the report is the real adjacent seam `SC074 < SC075`.

Place in the cascade This report belongs immediately after the SC072-SC073 **Unstressed long-vowel shortening and ae-merger core** and immediately before the SC076 **Prefix i-reduction note**. Its meaningful outward chronology relation points left to the late-tail core through SC072, but that relation should remain a cross-reference rather than a reason to merge chapters.

That placement matters because SC074-SC075 now gives the split late-tail region an explicit narrow middle bridge while leaving the stronger core on the left and the weaker context note on the right in their own proper shapes. The SC070-SC071 early bridge remains farther left as regional context, and the SC078 right-edge report remains distinct farther right.

Order evidence The local center of the report is SC074 < SC075.

SC074 must follow SC072 and precede SC075. If OE Med Unstressed I Lowering1 is moved before SC072, PGmc fúrxtīnaz yields fyrhti rather than expected OE fyrhte. That leftward relation is real, but it should remain only a cross-reference to the SC072-SC073 report rather than a reason to merge chapters.

SC074 must also precede SC075. If it is moved later than SC075, PGmc skíllingaz yields sċilleng instead of expected sċilling.

SC075 directly reciprocates that left edge. If OE Med Unstressed I Lowering is moved before SC074, PGmc skíllingaz again yields sċilleng rather than expected sċilling. The current runner finds no later real break for SC075 before the SC087 boundary. That result is runner-bounded and must not be rewritten as an SC087 boundary claim.

Interpretation SC074-SC075 belongs here because it is the strongest remaining local late-tail pair after the earlier bridge and the compact core were already made explicit. The report remains narrow on purpose. SC074 is the stronger hinge because it has positive evidence on both sides; SC075 is the local follower and remains one-sided on the right.

This gives the late-tail architecture a cleaner middle shape in book form: SC070-SC071 as the early bridge, SC072-SC073 as the main compact core, SC074-SC075 as the narrow witness-limited middle bridge, SC076 as the context note to the right, and SC078 as the late-tail right-edge report.

Remaining cautions The report should stay narrow. It should not be absorbed into the SC072-SC073 core just because the two units are adjacent. It should not absorb SC076 just because the note follows it. And it should not be treated as if it were a second late-tail core merely because its local chronology is real. Its leftward SC072 < SC074 relation should remain a cross-reference only, and SC075's runner-bounded later side must not be turned into a positive historical boundary.

Prefix i-reduction note

Sound-change report

Historical formulation SC076 *OEPrefixIReduction* appears here as a **short finished late-tail residual/context note**. The source support outruns the chronology evidence: the rule is historically legible enough to deserve final prose, but it has not become a local chronology hinge. The right final form is therefore brief and explicit rather than chapter-sized.

Source tradition Campbell and Fulk give the clearest support for late prefix-vowel weakening inside the larger late unstressed-vowel environment (Campbell 1959, sects 345–354, 388–389; Fulk 2018, sects 5.6–5.7). Hogg, Ringe and Taylor, and Luick support the same broader late weak-tail setting even though they do not isolate SC076 as a major handbook center (Hogg 1992, 120–21; Ringe and Taylor 2014, sects 6.7.3–6.9.3; Luick 1914--1940, sects 304–307).

That is enough to justify final prose. It is not enough to turn SC076 into a larger bridge or to reopen the late-tail structure that has already been around it.

CAPR implementation CAPR isolates this background as one explicit step: SC076 *OEPrefixIReduction*. That explicit rule is analytically useful because it keeps the prefix-vowel note visible in strict order even though current first-break testing does not recover any positive historical boundary for it.

Place in the cascade This note belongs immediately after the `SC074-SC075` **Medial unstressed-i lowering bridge** and immediately before the SC078 **Weak-tail reduction right-edge note**. It should remain a short residual note in that position rather than being folded into either neighboring report.

Order evidence Current testing does not identify a positive historical first-break boundary for SC076 in either direction. On the earlier side, the search moves safely down to order *13* before reaching bundled *EarlyEnglishLineChanges*; that is a methodological runner limit, not a detected historical boundary. On the later side, the search runs safely through order *86* before reaching the current SC087 boundary with no real break; that is a runner-bounded no-break result, not a claim that SC076 must precede SC087.

Failure counts are zero in both directions. The chronology statement is therefore negative and boundary-limited rather than positive.

Interpretation SC076 is now final prose because every ordinary sound change should be represented by final prose, not because the rule has become a local chronology anchor. The correct final treatment is a short residual/context note: keep the historical legitimacy of late prefix-vowel weakening explicit, but keep the chronology language strictly methodological.

Remaining cautions This note should stay brief. It should not be folded into the `SC074-SC075` bridge, the SC078 right-edge report, or any larger late-tail chapter. Bundled *EarlyEnglishLineChanges* and the current SC087 boundary remain search limits, not historical boundaries for SC076.

Weak-tail reduction right-edge note

Sound-change report

Historical formulation SC078 *OEWeakTailReduction* appears here as a **singleton right-edge late-tail report**. The evidence shows that it is stronger than a mere context note because it has real chronology on both sides, but the modest reading is still the right one: this is a singleton report, not a bridge into the `SC085-SC086` closing core and not a license for a non-contiguous `SC078-SC086` chapter.

Source tradition The source tradition supports late weak-tail reduction and the broader post-apocope cleanup environment, but not a chapter that stretches out of chronological order to join SC078 to `SC085-SC086`. Campbell, Hogg, Ringe and Taylor, Luick, and Fulk all support a late weak-tail region in which apocope, syncope, shortening, contraction, and further cleanup remain active (Campbell 1959, sects 345–354, 388–389; Hogg 1992, 120–21; Ringe and Taylor 2014, sects 6.7.3–6.9.3; Luick 1914--1940, sects 304–307; Fulk 2018, sect. 5.6).

Within that broader environment, SC078 is best read as the right-edge weak-tail hinge. The source tradition supports the late reduction setting that makes forms such as flēon and slēan historically legible, but it does not require that this report absorb the later h-loss/contraction core. The honest source claim is therefore narrower: late weak-tail reduction is real, its seam with SC086 is real, and that seam should remain a cross-reference only.

CAPR implementation CAPR isolates this late weak-tail hinge as one explicit step: SC078 *OEWeakTailReduction*. That formalization is narrower than the handbook tradition, but it is useful in the book because it gives the right edge of the split late-tail region a precise local place instead of leaving it buried inside a broader grouping.

Place in the cascade This report belongs immediately after the SC076 **Prefix i-reduction note** and immediately before the `SC079-SC080` **Final-j loss and final-geminate simplification bridge**. Its meaningful rightward chronology link points through the opening bridge and into the `SC081-SC083` technical middle bridge before reaching the `SC085-SC086` **H-loss and contraction core** through SC086.

That rightward link is real, but it remains a cross-reference only. SC078 should not be pulled forward into a non-contiguous closing report, and `SC079-SC080` or `SC081-SC083` should not be erased from the chronology just because SC086 is the more interpretable right-edge target.

The report also belongs to the right of the `SC072-SC073` **Unstressed long-vowel shortening and ae-merger core** as part of the same late-tail region, but that core remains a left-side regional context, not a co-member of this report.

Order evidence SC078 has real chronology on both sides, but the two sides are not equally local.

On the earlier side, SC078 must follow SC070. If OE Weak Tail Reduction is moved before SC070, PGmc bákaną yields bacen rather than expected OE bacan, and PGmc bíndaną yields binden rather than expected bindan, alongside a very broad set of comparable failures. That earlier side has 87 failing rows, so it should be narrated as a large computational boundary rather than as a tight local adjacency claim.

On the later side, SC078 must precede SC086. If OE Weak Tail Reduction is moved later than the h-loss/contraction core, PGmc fléuxaną yields flēoan instead of expected OE flēon, and PGmc sláxaną yields sleaan instead of slēan. This later side is much narrower and is the more interpretable right-edge seam.

That difference in profile is exactly why the report should stay singleton. The left edge is broad and computationally heavy; the right edge is sharper and book-legible. Together they make SC078 worth its own report, but not worth merging into any neighboring report.

Interpretation SC078 belongs here because it is the strongest remaining singleton candidate in the late-tail region. The report makes the right edge of that region explicit without pretending that the broader closing architecture has collapsed into one chapter. This follows the same pattern used elsewhere in the half: keep the adjacent source-backed center or hinge, keep real but outward relations as cross-references, and keep weaker neighboring notes separate.

Remaining cautions The cautions are mainly architectural. The earlier `SC070 < SC078` boundary is real, but it should remain a broad computational limit rather than a tight local adjacency claim. The later `SC078 < SC086` edge is also real, but it should remain a rightward cross-reference to the `SC085-SC086` report through SC086 only. The report should not absorb the SC076 note, the `SC079-SC080` or `SC081-SC083` bridges, and it should not treat the `SC072-SC073` late-tail core as a co-member rather than as left-side context.

Final-j loss and final-geminate simplification bridge

Sound-change report

Historical formulation This is a modest opening closing-bridge full report centred on SC079 *OEJLossAfterHeavy* and SC080 *OEFinalGeminateSimplification*. The pair is historically real and adjacent, but it is not the main closing core. SC079 is the stronger hinge, with a broad/far leftward tie to the earlier umlaut region and a real local right edge to SC080. SC080 is the narrower one-sided follower.

That narrower reading is also the honest one. The report discusses only SC079 and SC080. It does not merge backward into the SC078 singleton report, does not stretch forward into the `SC081-SC083` technical bridge, and does not pretend that this opening seam is as book-style as the earlier `SC085-SC086` and SC087 reports farther right.

Source tradition The source tradition gives the pair enough technical support for explicit prose, but not enough to inflate it into a major handbook chapter. The best support for SC079 comes from loss of *j* between unstressed vowels and the broader contracted-verb background that also feeds the rest of the closing zone (Fulk 2018, sect. 6.11, §12.21). Campbell's vocalization material and final consonant discussions give the technical background for both the *j*-loss material and the final-geminate issue (Campbell 1959, sects 266–274, §401, §731). Hogg's account of late final degemination gives SC080 its clearest handbook support (Hogg 1992).

That support is enough for a short adjacent report, but the hierarchy remains uneven. SC079 carries the larger historical load; SC080 is best read as the local follower that resolves the final-geminate issue after SC079.

CAPR implementation CAPR sharpens this opening closing sequence into two adjacent steps. SC079 *OEJLossAfterHeavy* removes *j* after the relevant heavy-syllable patterns, and SC080 *OEFinalGeminateSimplification* then removes the leftover final geminate material. That segmentation is more explicit than most handbook prose, but the chronology cards show why the two-step articulation is useful: the local seam is the real adjacent pair `SC079 < SC080`.

Place in the cascade This report belongs immediately after the SC078 **Weak-tail reduction right-edge note** and immediately before the `SC081-SC083` **J-strengthening, vocalization, and ei-contraction bridge**. Its meaningful outward chronology relation points far left to the `SC055-SC056` **Umlaut core and palatal-diphthongization follower**, but that broad/far link should remain a cross-reference rather than a reason to merge non-adjacent chapters.

That placement matters because The report fills the opening gap in the split closing region while remaining clearly subordinate to the stronger technical middle bridge and the later closing core.

Order evidence The local center of the report is `SC079 < SC080`.

SC079 must follow SC055 and precede SC080. If OE J Loss After Heavy is moved before SC055, forms such as believe, bow, and follow undo the live umlaut-sensitive vowel development: PGmc galáubijaną yields ġelēafan instead of expected OE ġelīefan, PGmc báugijaną yields bēaġan instead of expected bīeġan, and PGmc fúlgijaną yields fulġan instead of fylġan, alongside a wider failure set. That left edge is historically real, but it is broad/far across SC055 rather than a tight local adjacency claim.

SC079 must also precede SC080. If it is moved later than SC080, PGmc lúnganjō yields lungenn rather than expected OE lungen. SC080 directly reciprocates that same left edge: if OE Final Geminate Simplification is moved before SC079, the lung derivation again yields lungenn instead of lungen.

SC080 is different on the right. The current runner finds no real later break for it before the SC087 boundary. That no-break result is runner-bounded and must not be rewritten into a positive claim that SC080 must precede SC087.

Interpretation `SC079-SC080` belongs here because it is the remaining opening bridge that the closing region needs for explicit prose. The report remains modest on purpose. SC079 is the stronger hinge with the broader historical load; SC080 is the one-sided final-geminate follower. Together they justify a short adjacent report, but not a larger closing chapter.

This makes the closing sequence much clearer in book form: SC078 on the left, `SC079-SC080` as the opening bridge, `SC081-SC083` as the technical middle bridge, then the `SC085-SC086` core and the SC087 terminal singleton.

Remaining cautions The report should stay narrow. It should not merge backward into the SC078 singleton report just because the two units are adjacent. It should not merge forward into the `SC081-SC083` bridge just because that sequence follows it. SC079's broad/far left edge to SC055 should remain only a cross-reference, and SC080's runner-bounded later side must not be turned into a positive historical boundary.

J-strengthening, vocalization, and ei-contraction bridge

Sound-change report

Historical formulation This is a cautious technical middle closing bridge centred on SC081 *OEJStrengtheningAfterFrontDiphthong*, SC082 *OEIntervocalicJVocalization*, and SC083 *OEUnstressedEIContraction*. The evidence shows that the chain is historically real and locally coherent, but also more technical and less book-style than the earlier `SC085-SC086` h-loss/contraction core. That makes SC082 the true center, with SC081 as the left flank and SC083 as the right follower.

The report therefore stays compact and explicit. It discusses only `SC081-SC083`. It does not merge into the `SC085-SC086` report, does not stretch backward into SC078, and does not try to turn one technically coherent bridge into a broader closing chapter than the evidence supports.

Source tradition The source tradition supports the bridge, but unevenly. Campbell's vocalization material and contracted-verb discussion give the clearest direct footing for the strew-type sequence behind SC081 and SC082 (Campbell 1959, sects 266–274, 731). Fulk's discussion of loss of *j* between unstressed vowels and contracted verbs similarly reinforces the same technical zone, especially for the strew, beon, and -ian type material (Fulk 2018, sect. 6.11, §12.21). Ringe and Taylor add the clearest local chronology support for the bridge itself, especially in the `SC081 < SC082 < SC083` chain (Ringe and Taylor 2014, sects 6.9.1–6.9.3).

What the sources do not give is a major handbook-style chapter of equal co-headliners. SC082 is the clearest technical center. SC081 is historically legible but broad/far on the left, and SC083 is historically real but one-sided on the right. The honest shape is therefore a cautious technical bridge, not a broad closing core.

CAPR implementation CAPR sharpens this technical closing sequence into three adjacent steps. SC081 *OEJStrengtheningAfterFrontDiphthong* preserves the strengthened consonantal stage after a front diphthong. SC082 *OEIntervocalicJVocalization* then vocalizes the intervocalic *j*, and SC083 *OEUnstressedEIContraction* removes the extra *ei*-like sequence in the resulting unstressed environment. That segmentation is more explicit than most handbook prose, but the chronology cards show why all three steps matter: the real local center of gravity is the chain `SC081 < SC082 < SC083`, with SC082 as the strongest internal anchor.

Place in the cascade This report belongs immediately after the `SC079-SC080` **Final-j loss and final-geminate simplification bridge** and immediately before the `SC085-SC086` **H-loss and contraction core**. Its meaningful outward chronology links point far left to `SC055-SC056` through SC081 and right only indirectly toward SC085, but neither relation should become a reason to pull this bridge out of strict local order.

SC078 remains a singleton left context only. It should not be treated as a co-member of this report, and this bridge should not be stretched forward into `SC085-SC086` just because the later core is more book-style.

Order evidence The local order evidence gives the report a real internal spine: `SC081 < SC082 < SC083`.

For SC081, the earlier boundary is broad/far to SC055. If OE J Strengthening After Front Diphthong is moved before SC055, PGmc stráwjaną yields strēaġan rather than expected OE strīeġan. That left edge is historically real, but it should not be narrated as a tight local adjacency claim. SC081 must also precede SC082: if it is moved later than SC082, the same strew derivation yields strīeian instead of strīeġan.

SC082 is the center of gravity because it has real local chronology on both sides. If OE Intervocalic J Vocalization is moved earlier than SC081, PGmc stráwjaną yields strīeian rather than strīeġan. If it is moved later than SC083, forms such as bore, handle, and make yield boreian, handleian, and maceian instead of expected OE borian, handlian, and macian. That makes SC082 the strongest internal anchor of the report.

SC083 must follow SC082. If OE Unstressed EI Contraction is moved before that stage, forms such as bore, learn, and lick preserve extra *ei*-like sequences: boreian, liorneian, and licceian instead of borian, liornian, and liccian. Its later side is different: the current runner finds no real later break before the SC087 boundary. That later no-break result is runner-bounded and must not be rewritten as an SC087 boundary claim.

Interpretation `SC081-SC083` belongs here because it is coherent enough for final prose once the stronger closing anchors around it are already explicit. The report does not pretend to be the main closing core. Instead, it shows how the book can give honest prose to more technical material without inflating it into a major handbook chapter: SC081 as the broad/far left flank, SC082 as the real center, SC083 as the right follower.

That is exactly what this middle bridge should do. It fills the technical gap between the modest opening `SC079-SC080` bridge and the `SC085-SC086` core while keeping the whole region strictly chronological.

Remaining cautions The cautions are mainly architectural. SC081's leftward SC055 relation is real but broad/far and should remain a cross-reference only. SC083 should stay with this bridge rather than being treated as a right-edge feeder into `SC085-SC086`. SC078 must remain outside the report as a singleton left context. And SC083's runner-bounded later side must not be turned into a positive claim that it must precede SC087.

H-loss and contraction core

Sound-change report

Historical formulation This is a compact closing report centred on SC085 *OEHLoss* and SC086 *OEContraction*. The evidence shows that this adjacent pair is the strongest compact core in the closing region: it has the clearest handbook support, the cleanest local chronology, and the best-defined right-edge seam from SC078. It remains more book-legible than the more technical `SC081-SC083` bridge immediately to its left.

That stronger reading is also narrower than the old grouped version. The report discusses only SC085 and SC086. It does not expand backward into the more technical `SC081-SC083` bridge, outward into the `SC072-SC073` late-tail core, forward into SC087, or leftward into SC078 itself.

Source tradition The source tradition strongly supports a focused h-loss/contraction report. Campbell treats contraction and related vocalization after the loss of *h* or glide material as a real late Old English zone and gives textbook material such as téon, sléan, séon, and téo that matches the core of this pair (Campbell 1959, sects 234–239, 266–274, 731). Ringe and Taylor sharpen the same picture with their explicit section on loss of intervocalic *h* and contraction, using forms such as flēon, slēan, téon, and tā (Ringe and Taylor 2014, sects 6.9.1–6.9.3). Hogg likewise treats contracted verbs after loss of intervocalic *h* as one coherent late pattern (Hogg 1992).

Luick and Sievers-Brunner reinforce the same point from the older grammar tradition: later contraction follows hiatus created by *h*-loss, while metathesis belongs to a different terminal zone (Luick 1914--1940, sect. 249; Brunner 1965, sects 127–135, §179). Fulk is useful here as support for the contracted-verb tradition, but it should not be used to blur this compact core with the more technical *j*-material that remains in scaffolded `SC081-SC083` (Fulk 2018, sect. 12.21).

CAPR implementation CAPR sharpens this late closing history into two adjacent rules. SC085 *OEHLoss* removes *h* in the relevant late Old English environments, and SC086 *OEContraction* then contracts the resulting hiatus. That model-level segmentation is more explicit than most handbook prose, but it is exactly why this pair now deserves a focused report: the order evidence shows that the closing region needs both steps stated explicitly without inflating them into a larger mixed chapter.

Place in the cascade This report belongs immediately after the `SC081-SC083` **J-strengthening, vocalization, and ei-contraction bridge** and immediately before the SC087 **R-metathesis terminal note**. Its meaningful outward chronology links point left to the `SC072-SC073` **Unstressed long-vowel shortening and ae-merger core** through SC085 and farther left to the SC078 **Weak-tail reduction right-edge note** through SC086.

Those links matter, but they remain cross-references only. `SC073 < SC085` should not pull the late-tail core into this chapter, and `SC078 < SC086` should not pull SC078 forward into a non-contiguous closing report.

Order evidence The local center of the report is `SC085 < SC086`.

SC085 must follow SC073 and precede SC086. If OE H Loss is moved earlier than SC073, PGmc táixōn yields tāæ rather than expected OE tā, so the leftward SC073 relation is real but should remain a cross-reference to the `SC072-SC073` core. If SC085 is moved later than SC086, forms such as flee, slay, ten, and toe retain over-long vocalic sequences: PGmc fléuxaną yields flēoan instead of flēon, PGmc sláxaną yields sleaan instead of slēan, PGmc téxun yields teoon instead of tēon, and PGmc táixōn yields tāe instead of tā.

SC086 reciprocates the left edge directly. If OE Contraction is moved earlier than SC085, the same four-row contraction set develops over-long vocalic outputs. That makes SC086 the right-hand member of the strongest compact closing pair. Its later side is different: the current runner finds no real later break before the SC087 boundary, so this report must not turn the runner-bounded later side into a positive claim that SC086 must precede SC087.

The already explicit SC078 seam also lands here. SC078 must precede SC086, but that relation remains a leftward cross-reference from the SC078 report into this report through SC086 only. It is not a reason to treat SC078 as part of the same adjacent chapter.

Interpretation `SC085-SC086` belongs here because it is the strongest compact closing report now available. It gives the closing region one clear production center without overstating the more technical `SC081-SC083` bridge or the broader, more terminal SC087 material. This is the same editorial move that has worked well elsewhere in the half: keep the adjacent source-backed core together, keep real but non-local relations as cross-references, and leave weaker neighboring material separate.

Remaining cautions The report should stay focused. It should not expand backward into the `SC081-SC083` bridge just because the more technical sequence precedes it. It should not expand forward into SC087 just because SC087 is the next closing note. It should not pull SC078 into the chapter merely because `SC078 < SC086` is real, and it should not turn `SC073 < SC085` into a reason to absorb the `SC072-SC073` core.

Most importantly, the later side of SC086 remains runner-bounded. This chapter may state the real local center `SC085 < SC086`, but it must not invent a positive `SC086 < SC087` boundary that the card does not show.

R-metathesis terminal note

Sound-change report

Historical formulation SC087 *OERMetathesis* appears here as a **short terminal singleton full report**. The evidence shows that r-metathesis is source-backed and book-style enough to deserve explicit prose, but the modest reading is still the right one: this is an independent terminal note, not an extension of the `SC085-SC086` h-loss/contraction core and not a reason to pull in SC078, `SC079-SC080`, or `SC081-SC083`.

Source tradition The source tradition for this note is real and distinct. Hogg treats r-metathesis as a genuine Old English process with variable chronology and makes clear that it belongs to a different late consonant/vowel history from contracted verbs and final-geminate simplification

(Hogg 1992). Ringe and Taylor likewise treat metathesis of *r* with short vowels as a real Old English development and use beornan / berstan type material to show that the process is historically interpretable without making it a compact local pair (Ringe and Taylor 2014). Luick and Sievers-Brunner reinforce the same point from the older grammar tradition, treating metathesis as its own late rearrangement history and using the familiar berstan, frost, and cerse family as core examples (Luick 1914--1940; Brunner 1965, sect. 179).

That is enough support for a short terminal production note. It is also a reason to keep the note separate from the new h-loss/contraction report: the historical process is real, but it is not just the last step of the `SC085-SC086` core under another name.

CAPR implementation CAPR isolates this late process as one explicit step: SC087 *OERMetathesis*. That formalization is narrower than the broader historical discussions in Hogg, Ringe and Taylor, Luick, and Sievers-Brunner, but it is useful in the book because it gives the terminal *r*-rearrangement material a precise chronological place instead of leaving it buried in a generic closing cluster.

Place in the cascade This report belongs immediately after the `SC085-SC086` **H-loss and contraction core** and closes the current sound-change half. Its meaningful positive chronology relation points far left to the `SC044-SC045` **Breaking and velar-fricative palatalization** report through SC044. That broad/far relation should remain a distant leftward cross-reference rather than a reason to build a non-contiguous chapter.

This is also why SC087 remains separate from `SC085-SC086` even though it follows that report directly. Adjacency in the assembled order is real, but the source tradition and the chronology card do not make SC087 the right-hand half of a tighter local pair.

Order evidence SC087 has one positive chronology relation, and it is broad/far: `SC044 < SC087`.

If OE R Metathesis is moved before the much earlier SC044 OE Breaking stage, PGmc bréstaną yields beorstan rather than expected OE berstan. That left edge is historically real, but it is not local and should be narrated as a distant cross-reference to the `SC044-SC045` report rather than as a compact local window.

The later side is different. The current runner finds no real break beyond the current order before the runner limit. That is a boundary-limited negative result, not a positive historical boundary. The report therefore must not claim a positive `SC086 < SC087` relation, and it must not turn the end of the runner into a must-follow or must-precede statement.

Interpretation SC087 belongs here because the terminal singleton reading is now the best one. The historical process is real and book-style enough to merit explicit prose, but its chronology is broad/far on the left and runner-bounded on the right. A short terminal note therefore fits the evidence better than either a larger closing cluster or a forced merger with `SC085-SC086`.

That makes this an architectural endpoint. It gives the closing region an explicit terminal note while leaving the but still modest `SC079-SC080` and more technical `SC081-SC083` bridges in place and keeping SC078 outside the closing report structure.

Remaining cautions The cautions are mostly structural. SC087 should not be merged with the `SC085-SC086` report merely because it follows it in the cascade. Its broad/far `SC044 < SC087` relation should remain only a distant cross-reference to the `SC044-SC045` report. Its later side must remain a runner-bounded limit rather than a positive boundary. And the report should stay focused: it is a short terminal singleton note, not a reason to reopen SC078, `SC079-SC080`, or `SC081-SC083` inside the same chapter.

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